Response to Automation-Driven Labour Displacement:

Adaptive Universal Transaction Tax and Request-Based Universal Basic Income

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Abstract

Two converging crises threaten the fiscal foundations of modern states: tax systems rooted in principles unchanged since antiquity suffer from structural inefficiency, while progressive automation and artificial intelligence are eliminating traditional employment at an accelerating rate, eroding the tax base built on personal and corporate income. This paper presents a formal model of a coupled fiscal response: an *Adaptive Universal Transaction Tax* (AUTT)—a single, low-rate levy applied automatically to all electronic financial transactions, with a rate that adjusts in real time via feedback control to meet a democratically approved budget target—paired with a *request-based Universal Basic Income* (UBI) that replaces fragmented social welfare programmes and provides the income floor necessitated by automation-driven labour displacement. We define the system as a dynamical model with three core variables: the aggregate transaction volume $V(t)$, the target budget B, and the adaptive tax rate $\tau(t)$. We establish conditions under which the system converges to its fiscal target, prove stability under economic shocks, and derive the critical rate threshold below which behavioural distortion remains negligible. We analyse the transaction chain cascade effect, demonstrate controller robustness under adverse scenarios through formal error analysis, model the exemption boundary between taxed and exempt sectors, and draw on the Canadian CERB pandemic experience to empirically ground the request-based UBI take-up model. The paper includes a model draft bill with phased introduction, offshore capital amnesty, and worker compensation provisions.

Keywords: transaction tax, universal basic income, fiscal automation, feedback control, automation, tax reform, macroeconomic stability, cascade taxation, EWMA estimation

Contents

1 Introduction

Tax systems across the world share a common architectural flaw: they place the burden of assessment, declaration, and compliance on individual taxpayers and firms, then deploy expensive enforcement apparatuses to detect inevitable evasion. The result is a system in which, as Adams (1993) documented, the relationship between the state and its citizens is structurally adversarial. The middle class bears a disproportionate share of the fiscal burden, while large corporations exploit legal and jurisdictional arbitrage, and informal economic actors evade taxation entirely.

This architectural flaw is not a matter of policy calibration—it is inherent to any system that taxes *entities* rather than *transactions*. The distinction is fundamental. Taxing entities requires knowing who earned what, when, and through which legal structures—an informational problem of enormous complexity. Taxing transactions requires only that a fixed fraction of every movement of money be diverted to a public account—a computational task that modern electronic payment infrastructure already performs routinely for bank fees.

The idea of a universal transaction tax is not new. Keynes (1936) proposed taxing financial transactions to reduce speculative volatility. Tobin (1978) formalised this as a tax on currency exchanges. Summers & Summers (1989) extended the argument to securities markets. Nobel laureate Joseph Stiglitz confirmed the "technical feasibility" of such a tax, noting that modern electronic payment infrastructure makes it "much more feasible today" than when Tobin originally recanted his own proposal (Stiglitz, 2009). Over 1,000 economists, including Nobel laureates Paul Krugman and Joseph Stiglitz, have endorsed some form of financial transaction tax (Burman et al., 2016; Matheson, 2011). In July 2025, seven Nobel laureates—Acemoglu, Akerlof, Banerjee, Duflo, Johnson, Krugman, and Stiglitz—published a joint call in *Le Monde* for progressive fiscal reform targeting ultra-wealthy individuals who contribute effectively 0–0.5% of their wealth in tax (Acemoglu et al., 2025). Feige (2000) proposed the Automated Payment Transaction (APT) tax as a comprehensive replacement for all other taxes, exploiting the enormous volume of electronic transactions to achieve fiscal sufficiency at rates below one percent.

What has been missing from this literature is a *formal dynamical model* that treats the tax rate as an endogenous variable governed by a feedback control rule, and that rigorously establishes the conditions under which such a system converges to its fiscal target, remains stable under macroeconomic shocks, and avoids behavioural distortions—including the cascade effects that arise when a single real-economy purchase generates multiple taxable intermediate transactions. This paper fills that gap.

1.1 The Metabolic Rationale

The system proposed here is not an arbitrary institutional design; it mirrors a principle that governs every viable biological organism. In all living systems, resources circulate continuously: blood distributes oxygen and nutrients, lymph removes waste, sap carries sugars from leaves to roots. At no point does the organism *interrogate* individual cells about their metabolic activity and demand a proportional contribution. Instead, a small fraction of the circulating resource is diverted automatically at each point of transfer—by capillary walls, by membrane transporters, by osmotic gradients. The organism taxes the *flow*, not the *entity*.

This is precisely what the AUTT does to an economy. Money circulates through the financial system as blood circulates through a body. The automatic diversion of a small fraction at each transaction is the fiscal analogue of metabolic extraction. The feedback controller that adjusts the rate is the analogue of homeostatic regulation—the mechanisms by which organisms maintain internal equilibrium despite fluctuations in external conditions.

The reason this principle was not applied to taxation earlier is straightforward: until the advent of electronic payment infrastructure, there was no mechanism for automatic, frictionless extraction at the point of transaction. Taxes had to be assessed manually, which required identifying the payer, determining the base, and enforcing compliance—the entity-based architecture that has persisted for six millennia. The digital revolution has removed this constraint. We now possess the technological equivalent of a capillary network through which fiscal extraction can occur automatically and invisibly, just as it does in every living organism.

The urgency of this transition is compounded by the approaching wave of automation-driven labour displacement. As artificial intelligence and robotics replace human workers across sectors—from manufacturing and logistics to legal services and medical diagnostics—the income-based tax foundation is being structurally undermined. This is not a distant prospect; it is occurring now. The entity-based fiscal architecture cannot survive a world in which a diminishing fraction of economic output is attributable to taxable human income. The metabolic alternative—taxing the flow of value rather than the source of labour—is not merely elegant; it is existentially necessary.

1.2 UBI Without Fiscal Reform: The Isolation Problem

The idea of a Universal Basic Income is not new, and legislative attempts to implement it are accelerating worldwide. Yet every major attempt to date has suffered from a critical structural flaw: UBI has been proposed *in isolation*, without a corresponding reform of the revenue mechanism that would fund it.

In Canada, Senator Kim Pate introduced Bill S-233 (2021) and its successor Bill S-206

(2025) to establish a national framework for a Guaranteed Livable Basic Income. In the House of Commons, MP Leah Gazan introduced the companion Bill C-223, which was defeated at second reading in September 2024 by a vote of 54 to 273 (Parliament of Canada, 2024). The Parliamentary Budget Officer estimated the gross cost of a national guaranteed basic income at $107 billion per year—a figure that provoked immediate objections about fiscal sustainability (PBO, 2025). The bills proposed no new revenue mechanism; the implicit assumption was that existing tax structures, possibly supplemented by eliminating certain tax credits and deductions, would absorb the cost.

This isolation is not unique to Canada. As of 2025, no country has implemented a full UBI system (Wikipedia, 2025). Over 136 pilot programmes have been conducted or are underway across the globe—in Finland, the Netherlands, Kenya, South Korea, Brazil, and dozens of US municipalities—but these are small-scale, time-limited, and externally funded. They demonstrate the social effects of cash transfers but do not address the systemic fiscal question: where does the money come from at national scale, permanently?

The objections that defeated Bill C-223 in Canada—cost, sustainability, potential elimination of targeted programmes, labour disincentives—are predictable consequences of proposing UBI within the existing fiscal framework. If UBI is grafted onto an income-based tax system, then funding it requires either raising income taxes (which are already perceived as burdensome), cutting other programmes (which harms vulnerable populations), or running deficits (which is unsustainable). The proposal is structurally self-defeating.

The AUTT resolves this isolation problem by providing a *dedicated, self-calibrating revenue mechanism* that generates the fiscal space for UBI without raising any existing tax or cutting any existing programme. The two reforms—transaction taxation and universal income—are not independent proposals; they are two components of a single system, each enabling the other.

We make six contributions:

(i) We define the Adaptive Universal Transaction Tax (AUTT) as a formal dynamical system with explicit state variables, control rules, and equilibrium conditions (Section 3).

(ii) We prove that the system converges to its fiscal target and derive stability conditions under economic perturbations (Section 4).

(iii) We provide formal error bounds on the EWMA volume estimator and analyse controller behaviour under adverse scenarios including sudden crashes and regime shifts (Section 5).

(iv) We model the transaction chain cascade effect, derive the effective tax burden on final goods, and show that it remains moderate at operational rates (Section 6).

(v) We analyse the exemption boundary between taxed and exempt sectors, computing the required rate on the taxed sector as a function of exemption scope (Section 8).

(vi) We formalise the connection between the AUTT and a request-based Universal Basic Income, deriving the conditions under which the combined system is fiscally sustainable (Section 10).

The paper concludes with a discussion of institutional design, democratic governance of the budget, and the model's implications for civilisational decentralisation (Section 12).

2 Foundational Concepts

Before presenting the formal model, we establish the conceptual vocabulary that the subsequent analysis relies upon. The definitions in this section are designed to be precise enough for mathematical use while remaining transparent to readers across disciplines.

2.1 The Transaction Volume as a Fiscal Base

In any economy, the same unit of currency changes hands many times within a given period. When a firm pays a salary, the employee spends it at a shop, the shop pays its supplier, the supplier pays its workers, and so on. Each exchange is a *transaction*. The sum of all such exchanges within a jurisdiction over a unit of time is the *aggregate transaction volume*.

Definition 2.1 (Aggregate Transaction Volume)**.** Let $\mathcal{T}$ denote the set of all financial transactions executed within a given jurisdiction during a time interval $[t, t+\Delta t]$. For each transaction $i \in \mathcal{T}$, let $a_i > 0$ denote its monetary value. The **aggregate transaction volume** over the interval is

$$V(t, \Delta t) \;=\; \sum_{i \in \mathcal{T}} a_i \,. \tag{1}$$

When the interval is understood from context (typically one fiscal year or one month), we write simply $V(t)$ or V.

The aggregate transaction volume V is vastly larger than conventional macroeconomic aggregates such as Gross Domestic Product (GDP). GDP measures only the value of *final* goods and services; V counts every intermediate exchange as well, and crucially, it includes all financial-sector transactions (interbank transfers, securities trades, derivatives settlements) that dwarf the real economy. Empirically, for the United States, annual GDP is on the order of \$25 trillion, while the volume of transactions processed through Fedwire, CHIPS, and SWIFT alone exceeds \$1,500 trillion (BIS, 2022). Including all domestic card payments, ACH transfers, and intra-bank movements, the total easily surpasses \$2,000 trillion.

This enormous ratio of V to GDP—typically two orders of magnitude—is what makes a universal transaction tax feasible at rates far below one percent. However, this ratio is not exogenous: it depends in part on financial-sector activity that is highly rate-sensitive. We return to this endogeneity in Sections 6 and 8.

2.2 The Structural Deficiencies of Entity-Based Taxation

Traditional tax systems are *entity-based*: they assess obligations on identified persons or legal entities based on their income, profits, property, or consumption. This architecture entails several structural problems that are not correctable by policy adjustments within the existing framework.

Information asymmetry. The taxing authority must determine the taxpayer's true economic position, while the taxpayer has every incentive to conceal it. The resulting adversarial dynamic requires enormous enforcement expenditure. In Canada, the Canada Revenue Agency employs approximately 40,000 staff at an annual cost of \$7–8 billion, yet the federal tax gap—the difference between taxes owed and taxes collected—is estimated at \$47.8 billion per year (CRA, 2022).

Regressivity in practice. Although statutory rate schedules may be progressive, the effective burden falls disproportionately on middle-income salaried employees, whose income is reported automatically by employers. High-net-worth individuals and multinational corporations exploit legal arbitrage (transfer pricing, offshore entities, treaty shopping) to reduce their effective rates, sometimes to below one percent (EU Commission, 2016).

Compliance cost. Beyond government enforcement costs, private compliance costs—accountants, tax lawyers, software, time spent on declarations—represent a substantial economic drag. In the United States, the IRS estimates that individuals and businesses spend over 6 billion hours annually on tax compliance (IRS, 2023).

Procyclicality. Income-based tax revenues are strongly correlated with economic activity. During recessions, revenues fall precisely when government spending needs increase, creating a destabilising positive feedback loop.

These are not imperfections to be mitigated; they are intrinsic features of any system that taxes entities rather than the movement of money itself.

3 The Model

We now construct the formal model of the Adaptive Universal Transaction Tax (AUTT). The model has three components: the tax mechanism, the feedback control rule, and the budget identity.

3.1 The Tax Mechanism

The AUTT operates as follows. Every financial institution (bank, brokerage, payment processor, money transfer service) within the jurisdiction is required by law to maintain a government collection account. Whenever a transaction of value a_i is processed, a fraction τ of a_i is automatically transferred to this collection account. No declaration, filing, or individual identification is required.

Definition 3.1 (AUTT Revenue Flow)**.** Let $\tau(t) \in [0, \tau_{\max}]$ be the tax rate at time t, where $\tau_{\max}$ is a constitutionally mandated upper bound. The instantaneous revenue flow is

$$R(t) \;=\; \tau(t) \cdot v(t)\,, \tag{2}$$

where $v(t)$ is the instantaneous transaction flow rate (monetary volume per unit time), so that $V(t, \Delta t) = \int_t^{t+\Delta t} v(s)\, ds$.

The cumulative revenue collected from the beginning of the fiscal year (time t_0) to time t is therefore

$$\mathcal{R}(t) \;=\; \int_{t_0}^{t} \tau(s) \cdot v(s)\, ds\,. \tag{3}$$

This expression states that the total tax revenue up to any point in time is found by accumulating, moment by moment, the product of the tax rate and the transaction flow rate. When the rate τ is held constant, this simplifies to $\tau \cdot V(t_0, t)$—the rate multiplied by the total volume of transactions.

Definition 3.2 (Intermediary Obligation Principle)**.** The AUTT fiscal obligation falls exclusively on *licensed financial intermediaries*—banks, payment processors, brokerages, money transfer services, and any other entity operating under a financial services licence. No individual, corporation, trust, or other legal entity bears any direct fiscal obligation under the AUTT. The intermediary's obligation is mechanical: its transaction-processing software diverts the fraction $\tau(t)$ of every settled transaction to the designated government account at the moment of settlement.

Enforcement mechanism. Compliance is ensured not through audits of taxpayers (who do not exist under this system) but through the financial licensing regime. Any licensed intermediary that fails to implement the automatic diversion, or that processes transactions through channels that bypass it, is subject to:

(a) Immediate suspension of its financial services licence;

(b) Permanent revocation upon repeated or deliberate non-compliance;

(c) Criminal prosecution of responsible officers for fraud against the public treasury.

The number of licensed financial intermediaries in any jurisdiction is small (typically 10^2–10^3), their operations are already subject to continuous regulatory oversight, and their software systems are already audited for compliance with anti-money-laundering and capital adequacy requirements. Adding a single automatic diversion rule to existing compliance infrastructure is trivially enforceable. This is the inverse of entity-based taxation, which must monitor 10^7–10^8 individual taxpayers with imperfect information: the AUTT monitors a small, already-regulated population of intermediaries whose core business depends on maintaining their licence.

3.2 The Budget Target and Feedback Control Rule

At the beginning of each fiscal year, the legislature approves a budget target $B > 0$. The AUTT control algorithm adjusts the rate $\tau(t)$ continuously to ensure that $\mathcal{R}$ reaches B by year-end.

Definition 3.3 (Budget Deficit Function)**.** The **fiscal deficit function** at time t is defined as

$$D(t) \;=\; B - \mathcal{R}(t)\,, \tag{4}$$

representing the remaining revenue required to meet the annual budget target.

When $D(t)$ is large (early in the fiscal year, or during a slowdown in transaction activity), the rate must be higher; as the target is approached, the rate decreases; if the target is met before year-end, the rate drops to zero (or near-zero), and any surplus is directed to a reserve fund.

We define the feedback control rule as a proportional controller:

Definition 3.4 (Adaptive Rate Rule)**.** The adaptive tax rate is governed by

$$\tau(t) \;=\; \min\!\left(\tau_{\max},\; \frac{D(t)}{V_{\text{proj}}(t)}\right)^{+}, \tag{5}$$

where $(\cdot)^+ = \max(\cdot, 0)$, $\tau_{\max}$ is the constitutional ceiling, and $V_{\text{proj}}(t)$ is the projected remaining transaction volume from t to the end of the fiscal year t_f. The projection uses a rolling estimate:

$$V_{\text{proj}}(t) \;=\; \hat{v}(t) \cdot (t_f - t)\,, \tag{6}$$

where $\hat{v}(t)$ is the exponentially weighted moving average of recent transaction flow:

$$\hat{v}(t) \;=\; \lambda\, v(t) \;+\; (1-\lambda)\, \hat{v}(t - \Delta t)\,, \tag{7}$$

with smoothing parameter $\lambda \in (0, 1)$.

The logic of this rule is as follows. The algorithm divides the remaining amount to be collected, $D(t)$, by the best available estimate of the remaining transaction volume,

$V_{\text{proj}}(t)$. The result is the rate that, if held constant for the rest of the year, would exactly meet the budget. Because both the numerator and denominator are updated continuously, the rate self-corrects: if transaction volume declines, the rate rises slightly; if volume surges, the rate falls. The constitutional ceiling $\tau_{\max}$ prevents extreme values under catastrophic scenarios.

Remark 3.1. In practice, the rate adjustment need not be literally continuous. A daily or weekly recalculation is sufficient, since the smoothing parameter λ in the EWMA filter already absorbs short-term volatility. The continuous-time formulation is adopted here for analytical tractability.

3.3 The Budget Identity

We now establish how the initial budget is determined. The first-year budget must be calibrated against observable quantities from the preceding fiscal regime. This calibration serves as the *initial condition* for the dynamical system—it does not assert that the transition from fragmented welfare to UBI is costless or structurally trivial, but rather provides a computable starting point from which the adaptive mechanism takes over.

Let B_{prev} denote the previous year's government budget, S_{prev} the total social welfare expenditure (pensions, unemployment insurance, disability payments, etc.) in the previous year, and U the projected cost of the Universal Basic Income programme.

Definition 3.5 (First-Year Budget Calibration)**.** The initial AUTT budget is set as

$$B_1 = B_{\text{prev}} - S_{\text{prev}} + U - C_{\text{tax}} + C_{\text{trans}}, \tag{8}$$

where C_{tax} is the administrative cost of the previous tax collection apparatus (revenue agencies, compliance infrastructure), which is eliminated under the AUTT, and C_{trans} is the estimated transition cost (parallel systems, retraining programmes for displaced tax administration workers, technical infrastructure deployment).

The transition cost C_{trans} is a one-time or short-term expenditure that declines to zero over 2–3 years. Its inclusion ensures that the initial budget is not artificially favourable. In subsequent years, $C_{\text{trans}} = 0$, and the budget is determined by democratic vote with the adaptive mechanism guaranteeing collection.

We emphasise that this calibration does *not* assume perfect fungibility between legacy welfare programmes and UBI. The two systems serve overlapping but not identical populations, with different administrative requirements and incentive structures. The calibration is a fiscal accounting identity—a starting budget figure—not a claim that the welfare-to-UBI transition is frictionless. The frictions are real and must be managed through the transition period discussed in Section 26.

3.4 Institutional Architecture

The operational structure of the AUTT can be summarised as follows:

(a) Every licensed financial institution maintains a *government collection account.*

(b) Each transaction processed triggers an automatic transfer of $\tau(t) \cdot a_i$ to the collection account. No identification of the transacting parties is required.

(c) Collection accounts are connected to a centralised *Revenue Aggregation System* (RAS), which reports cumulative revenue $\mathcal{R}(t)$ in real time.

(d) An algorithmic controller—implemented and audited but not subject to political override between legislative budget votes—computes and publishes the current rate $\tau(t)$ according to Equation (5).

(e) The rate $\tau(t)$ and cumulative revenue $\mathcal{R}(t)$ are publicly visible at all times, ensuring full transparency.

The critical design choice is that individual taxpayer identity is never required for the vast majority of participants. The system taxes *the movement of money*, not the person or entity moving it. This eliminates the informational asymmetry that is the root cause of evasion under entity-based systems.

We acknowledge that the exemption-account mechanism (Section 8) reintroduces entity-level reporting for a small class of participants who opt out of the automatic levy. This is a deliberate compromise: the AUTT eliminates informational asymmetry for the overwhelming majority of economic actors (households, small and medium enterprises, service providers), while retaining traditional oversight only for sophisticated financial intermediaries whose activity profiles make the flat-rate levy inapplicable. The asymmetry is not eliminated universally but is reduced by approximately two orders of magnitude in terms of the number of reporting entities.

4 Convergence and Stability Analysis

We now establish the central analytical results: that the AUTT converges to its budget target, and that this convergence is robust to macroeconomic shocks.

4.1 Convergence Under Constant Transaction Volume

We begin with the simplest case to build intuition.

Proposition 4.1 (Convergence Under Constant Volume)**.** *If the transaction flow rate is constant, $v(t) = v_0 > 0$ for all $t \in [t_0, t_f]$, and $B \leq \tau_{\max} \cdot v_0 \cdot T$ where $T = t_f - t_0$ is the fiscal year length, then the adaptive rate rule (Equation 5) yields $\mathcal{R}(t_f) = B$.*

Proof. Under constant volume, $\hat{v}(t) = v_0$ for all t (the EWMA converges immediately). Therefore, $V_{\text{proj}}(t) = v_0(t_f - t)$. The rate rule gives

$$\tau(t) = \frac{D(t)}{v_0(t_f - t)} = \frac{B - \mathcal{R}(t)}{v_0(t_f - t)} .$$

Substituting into the revenue ODE $\dot{\mathcal{R}}(t) = \tau(t) \cdot v_0$:

$$\dot{\mathcal{R}}(t) = \frac{B - \mathcal{R}(t)}{t_f - t} .$$

This is a first-order linear ODE. Setting $D(t) = B - \mathcal{R}(t)$, we obtain $\dot{D}(t) = -D(t)/(t_f - t)$, which has the solution

$$D(t) = D(t_0) \cdot \frac{t_f - t}{T} .$$

At $t = t_f$, $D(t_f) = 0$, so $\mathcal{R}(t_f) = B$. The condition $B \leq \tau_{\max} \cdot v_0 \cdot T$ ensures that $\tau(t) \leq \tau_{\max}$ for all t. □

Under idealised conditions—steady transaction flow—the adaptive rule automatically distributes the fiscal burden evenly across the year and reaches the target exactly. The rate starts at $\tau_0 = B/(v_0 T)$ and remains constant throughout, since both numerator and denominator of the rule shrink at the same proportional rate.

4.2 Convergence Under Variable Transaction Volume

Real economies exhibit fluctuating transaction volumes due to seasonality, business cycles, and stochastic shocks. We now generalise.

Theorem 4.1 (General Convergence). *Let $v(t) > 0$ be a piecewise continuous transaction flow rate on $[t_0, t_f]$, and suppose $V_{\text{proj}}(t) > 0$ for all $t < t_f$. If $B \leq \tau_{\max} \cdot \int_{t_0}^{t_f} v(s)\, ds$, then the adaptive rate rule (Equation 5) yields $\mathcal{R}(t_f) = B$.*

Proof. Define $D(t) = B - \mathcal{R}(t)$. The revenue dynamics give

$$\dot{\mathcal{R}}(t) = \tau(t) \cdot v(t) = \frac{D(t)}{V_{\text{proj}}(t)} \cdot v(t) .$$

Hence $\dot{D}(t) = -D(t) \cdot v(t)/V_{\text{proj}}(t)$. This is a linear ODE with time-varying coefficient $\alpha(t) = v(t)/V_{\text{proj}}(t) > 0$. The solution is

$$D(t) = D(t_0) \cdot \exp\!\left(- \int_{t_0}^{t} \alpha(s)\, ds\right) .$$

We need to show that $\int_{t_0}^{t_f} \alpha(s)\, ds = +\infty$, which guarantees $D(t_f) = 0$.

Now, $V_{\text{proj}}(t)$ is an estimate of $\int_t^{t_f} v(s)\,ds$. The formal conditions under which the EWMA provides a suitable estimate are established in Proposition 5.1 (Section 5). Specifically, if $\hat{v}(t)(t_f - t)$ is bounded above by $C \cdot \int_t^{t_f} v(s)\,ds$ for some constant $C > 0$, then

$$\alpha(t) = \frac{v(t)}{\hat{v}(t)(t_f - t)} \geq \frac{1}{C} \cdot \frac{v(t)}{\int_t^{t_f} v(s)\,ds}.$$

Setting $W(t) = \int_t^{t_f} v(s)\,ds$, we have $\dot{W}(t) = -v(t)$, so $\alpha(t) \geq -\dot{W}(t)/(C \cdot W(t))$, and

$$\int_{t_0}^{t_f} \alpha(s)\,ds \geq \frac{1}{C} \int_{t_0}^{t_f} \frac{-\dot{W}(s)}{W(s)}\,ds = \frac{1}{C}\left[\ln W(t_0) - \ln W(t_f)\right].$$

Since $W(t_f) = 0$, the integral diverges to $+\infty$, and $D(t_f) = 0$. □

This theorem guarantees that regardless of how transaction volume fluctuates during the year, the adaptive mechanism reaches the budget target by year-end, provided that some positive level of transactions continues throughout the year and the constitutional rate ceiling is not binding. The mathematical mechanism is that any remaining deficit is divided by an ever-shrinking remaining time horizon, causing the rate to increase just enough to compensate for shortfalls. The exponential decay of the deficit function ensures smooth convergence.

4.3 Stability Under Macroeconomic Shocks

A critical advantage of the AUTT over entity-based taxation is its behaviour during economic crises. Under conventional systems, a recession causes income to fall, which reduces tax revenue, which forces either spending cuts or borrowing, which deepens the recession—a destabilising positive feedback loop. We now show that the AUTT contains an intrinsic negative feedback mechanism that dampens this procyclical spiral.

Proposition 4.2 (Countercyclical Stabilisation). *Suppose that at time t_1, the transaction flow rate drops by a factor $(1 - \delta)$, where $\delta \in (0, 1)$, so that $v(t) = (1 - \delta)v_0$ for $t > t_1$. Under the AUTT, the tax rate adjusts to*

$$\tau_{\text{new}} = \frac{\tau_{\text{old}}}{1 - \delta}, \tag{9}$$

and revenue collection continues at the same pace. The condition for this adjustment to remain within the constitutional ceiling is $\delta < 1 - \tau_{\text{old}}/\tau_{\max}$.

Proof. Before the shock, revenue flows at rate $R_0 = \tau_{\text{old}} \cdot v_0$. After the shock, $v = (1-\delta)v_0$. The adaptive rule sets $\tau_{\text{new}} = D(t_1)/[(1 - \delta)v_0(t_f - t_1)]$. Since $D(t_1)$ has not changed (only the flow rate has), and V_{proj} has decreased by the factor $(1 - \delta)$, the new rate is $\tau_{\text{old}}/(1 - \delta)$. Revenue after the shock: $\tau_{\text{new}} \cdot (1 - \delta)v_0 = \tau_{\text{old}} \cdot v_0 = R_0$. □

The AUTT automatically compensates for reduced economic activity by slightly raising its rate, without any legislative action, political negotiation, or delay. Because the base rate is already very low (typically 0.3–0.5%), even a significant economic contraction (say, a 30% drop in transaction volume) raises the rate only to about 0.43–0.71%—still far below the threshold at which behavioural distortions appear (see Section 7).

This is fundamentally different from conventional systems, where a 30% revenue shortfall triggers emergency austerity measures or deficit spending. The AUTT's countercyclical property emerges naturally from the feedback control structure, without requiring Keynesian stimulus packages or discretionary intervention.

5 EWMA Estimator Analysis and Controller Robustness

The convergence proof in Theorem 4.1 relies on the EWMA estimator $\hat{v}(t)$ being a reasonable predictor of actual transaction flow. This section provides the formal analysis that justifies this assumption, and examines what happens when the assumption is violated.

5.1 Error Bounds on the EWMA Estimator

Proposition 5.1 (EWMA Tracking Bound). *Let $v(t) > 0$ be the actual transaction flow rate and $\hat{v}(t)$ the EWMA estimate with smoothing parameter $\lambda \in (0, 1)$, updated at intervals of length Δt. If $v(t)$ satisfies the bounded rate-of-change condition*

$$\left| \frac{v(t + \Delta t) - v(t)}{v(t)} \right| \leq \gamma \qquad \text{for all } t\,, \tag{10}$$

where $\gamma > 0$ is the maximum proportional change per period, then the relative estimation error satisfies

$$\left| \frac{\hat{v}(t) - v(t)}{v(t)} \right| \leq \frac{(1 - \lambda)\gamma}{\lambda}\,. \tag{11}$$

This bound states that the estimator's accuracy depends on the ratio of two quantities: how quickly the real economy changes (governed by γ) and how quickly the estimator adapts (governed by λ). When λ is large (say 0.3), the estimator reacts quickly but is noisy. When λ is small (say 0.05), the estimator is smooth but slow. The optimal choice depends on the volatility regime.

Proof. The EWMA satisfies $\hat{v}(t) = \lambda v(t) + (1 - \lambda)\hat{v}(t - \Delta t)$. Subtracting $v(t)$ from both sides: $\hat{v}(t) - v(t) = (1 - \lambda)[\hat{v}(t - \Delta t) - v(t)]$. Writing $\hat{v}(t - \Delta t) - v(t) = [\hat{v}(t - \Delta t) - v(t - \Delta t)] + [v(t - \Delta t) - v(t)]$ and using the bounded change condition $|v(t - \Delta t) - v(t)| \leq \gamma v(t - \Delta t) \leq \gamma v(t)(1 + \gamma)/(1 - \gamma)$, we obtain in steady state (where the relative error is constant):

$$e = (1-\lambda)(e+\gamma'),$$

where $\gamma' \approx \gamma$ for small γ. Solving: $e = (1-\lambda)\gamma/\lambda$. $\square$

Example 5.1 (Calibration for Weekly Updates)**.** With weekly recalculation ($\Delta t = 1$ week), $\lambda = 0.15$ (moderate responsiveness), and $\gamma = 0.05$ (transaction volume changes by at most 5% per week—a large fluctuation), the bound gives $|\hat{v} - v|/v \le 0.85 \times 0.05/0.15 \approx 0.28$, or 28%. This means the EWMA could overestimate volume by up to 28% in a rapidly deteriorating economy. The resulting constant C in Theorem 4.1 is $C = 1.28$, which remains finite and the convergence proof holds.

5.2 Behaviour Under Adverse Scenarios

We now examine the controller's behaviour under four canonical stress scenarios. For each, we characterise the rate trajectory $\tau(t)$, the revenue path $\mathcal{R}(t)$, and the conditions under which the constitutional ceiling $\tau_{\max}$ becomes binding.

Scenario A: Smooth seasonal variation. Let $v(t) = v_0[1 + A\sin(2\pi t/T)]$ with amplitude $A < 1$. The EWMA tracks this with a phase lag proportional to $(1-\lambda)/\lambda$, and the rate oscillates inversely to volume. Revenue converges smoothly. This is the benign case.

Scenario B: Sudden crash. Suppose $v(t)$ drops by 30% instantaneously at $t_1 = T/2$ (mid-year). The EWMA lags, initially underestimating the severity. The rate τ therefore undershoots, and a revenue shortfall accumulates. However, as the EWMA catches up (within $\sim 1/\lambda$ periods), the rate rises to compensate. The key question is whether the cumulative shortfall can be recovered in the remaining time.

Proposition 5.2 (Recovery After Sudden Crash)**.** *If a fraction δ of transaction volume is lost at the midpoint of the fiscal year, the AUTT recovers the budget target provided*

$$\frac{2\tau_0}{1-\delta} \le \tau_{\max}, \tag{12}$$

where $\tau_0 = B/(v_0 T)$ is the initial equilibrium rate.

At $\tau_0 = 0.5\%$, $\delta = 0.3$ (a severe recession), the required peak rate is $2 \times 0.5\%/0.7 \approx 1.43\%$. If $\tau_{\max} = 2\%$, the system recovers. If $\tau_{\max} = 1\%$, it does not, and the reserve fund must cover the shortfall.

Scenario C: High-frequency noise. Random daily fluctuations of $\pm 10\%$ around a stable mean. The EWMA with $\lambda = 0.15$ smooths these effectively. The rate τ fluctuates by $\pm 0.05\%$ around its mean value. Revenue convergence is unaffected.

Scenario D: Permanent regime shift. Transaction volume drops by 40% permanently (e.g., a structural transformation of the economy). The EWMA converges to the new level within approximately $2/\lambda$ periods ($\sim$13 weeks for $\lambda = 0.15$). The rate adjusts permanently to $\tau_0/0.6 \approx 1.67\tau_0$. If this exceeds $\tau_{\max}$, the budget target must be revised downward—a democratic decision, not a system failure.

Controller Robustness Summary

The EWMA-based controller converges under all scenarios where the remaining transaction volume is sufficient to meet the budget at rates below $\tau_{\max}$. In extreme scenarios (crash $> 50\%$ at mid-year, or permanent volume loss $> 60\%$), the constitutional ceiling may bind, requiring either reserve fund deployment or budget revision. The controller itself does not fail; it reaches the ceiling and holds there, collecting the maximum feasible revenue.

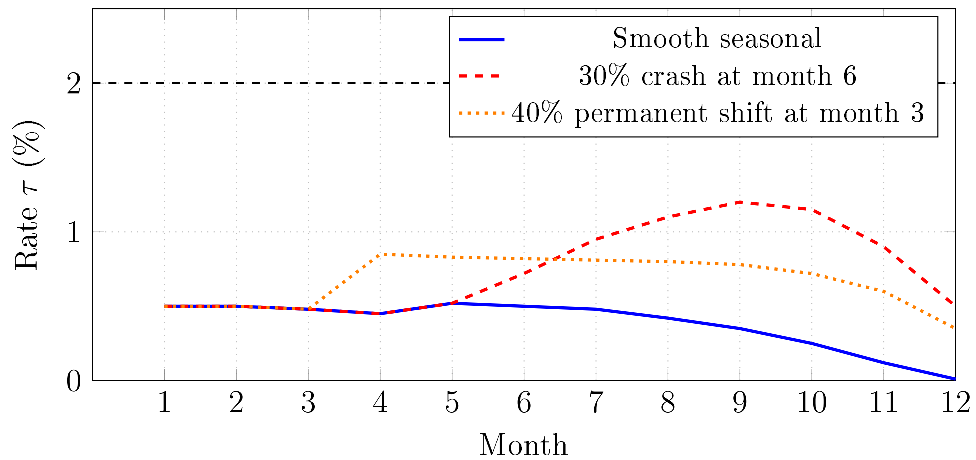


Figure 1: Illustrative AUTT rate trajectories under three stress scenarios. The controller adjusts $\tau(t)$ automatically; in all cases the budget target is met by year-end without breaching $\tau_{\max}$.

6 The Transaction Chain Cascade Effect

A fundamental concern about any universal transaction tax is the *cascade effect*: a single real-economy purchase generates multiple intermediate transactions, each taxed separately, so that the effective burden on the final good exceeds the nominal rate. This section analyses the cascade formally and derives the conditions under which it remains acceptable.

6.1 The Cascade Model

Definition 6.1 (Transaction Chain)**.** A **transaction chain** of length n is a sequence of n financial transactions generated by a single unit of final economic activity (e.g., the production and sale of a consumer good). Each transaction $k \in \{1, \ldots, n\}$ has value a_k, and the final consumer pays P for the good.

Consider a simple supply chain. A consumer pays \$100 for a product. The retailer pays \$70 to the distributor, who pays \$50 to the manufacturer, who pays \$30 to a raw materials supplier, who pays \$15 to a logistics provider. This is a chain of length $n = 5$, with transaction values $\{100, 70, 50, 30, 15\}$ and total transaction volume $\sum a_k = 265$.

At a nominal rate τ, the total tax extracted from this chain is

$$T_{\text{chain}} = \tau \sum_{k=1}^{n} a_k \,. \tag{13}$$

The *effective tax rate* on the final good is

$$\tau_{\text{eff}} = \frac{T_{\text{chain}}}{P} = \tau \cdot \frac{\sum_{k=1}^{n} a_k}{P} = \tau \cdot m \,, \tag{14}$$

where $m = \sum a_k / P$ is the **cascade multiplier**.

The cascade multiplier m measures how many dollars of transaction volume are generated per dollar of final consumer expenditure. It depends on the structure of the supply chain: how many intermediaries are involved, and what fraction of the final price each intermediary's payment represents.

6.2 Bounds on the Cascade Multiplier

Proposition 6.1 (Cascade Multiplier Bounds)**.** *For a transaction chain of length n where each intermediate payment is a fraction $\beta_k \in (0, 1)$ of the final price P (so $a_k = \beta_k P$), the cascade multiplier is*

$$m = \sum_{k=1}^{n} \beta_k \,. \tag{15}$$

For the common case where the chain has uniform value decay (each transaction is a fraction r of the previous: $a_k = r^{k-1} P$, with $r < 1$), the multiplier is

$$m = \frac{1 - r^n}{1 - r} \,. \tag{16}$$

As $n \to \infty$, this converges to $1/(1 - r)$.

This formula gives a precise relationship between the supply chain structure and the tax burden. For a chain where each step passes 70% of the value to the next ($r = 0.7$), the multiplier converges to $1/(1 - 0.7) = 3.33$. For $r = 0.5$, it converges to 2. For $r = 0.8$, it converges to 5.

Example 6.1 (Effective Rates Under Cascade)**.** The following table shows the effective tax rate on a final consumer good for various nominal rates and cascade multipliers:

	Cascade multiplier m				
Nominal rate τ	2	3	5	8	10
0.1%	0.2%	0.3%	0.5%	0.8%	1.0%
0.3%	0.6%	0.9%	1.5%	2.4%	3.0%
0.5%	1.0%	1.5%	2.5%	4.0%	5.0%
1.0%	2.0%	3.0%	5.0%	8.0%	10.0%

6.3 Assessing the Cascade Burden

The cascade table reveals that the effective burden is not trivially small. At a nominal rate of $\tau = 0.5\%$ and a cascade multiplier of $m = 5$ (a moderately complex supply chain), the effective rate on the final good is 2.5%. At $m = 10$ (a highly intermediated chain), it reaches 5%.

How should these figures be evaluated? Three considerations are relevant.

Comparison with existing consumption taxes. The effective cascade burden of 2–5% compares favourably with existing value-added taxes (VAT), which range from 5% (Japan, Canada GST) to 25% (Scandinavian countries), and sales taxes in the United States (0–10% depending on state). The AUTT at $\tau = 0.5\%$ with $m = 5$ imposes a lower effective burden on final consumption than any major VAT system in operation today, while simultaneously replacing all other federal and provincial taxes.

Distributional implications. Under the current system, VAT and sales taxes are regressive—they represent a higher share of income for low-income consumers. The AUTT cascade, while also applied to final goods, is partially offset by the UBI component: a household receiving, say, \$24,000/year in UBI benefits experiences a net transfer even after paying 2.5% effective rates on its consumption.

Supply chain shortening incentives. The cascade effect creates a market incentive to reduce the number of intermediate transactions—that is, to shorten supply chains. This may promote vertical integration, direct-to-consumer models, and local production. Whether this is desirable depends on one's perspective: it reduces the intermediation tax burden but may also reduce the benefits of specialisation. We note, however, that this incentive exists at very low levels of intensity (the marginal cost of an extra intermediary is $\tau \times$ the payment value, typically a few basis points) and is unlikely to drive major structural reorganisation of supply chains.

Cascade Assessment

At realistic operational rates ($\tau = 0.3$–0.5%) and typical supply chain lengths ($m = 3$–5), the effective tax burden on final consumer goods is 0.9–2.5%. This is substantially lower than existing consumption tax rates in all major economies, and is partially offset by UBI transfers for lower-income households.

7 The Critical Rate Threshold

A natural concern is that any tax on transactions might reduce transaction volume, creating a distortionary feedback loop: tax reduces volume, which requires a higher rate, which further reduces volume. We now analyse this concern, distinguishing between aggregate effects and sector-specific responses.

7.1 Elasticity of Transaction Volume

Definition 7.1 (Transaction Volume Elasticity)**.** The **elasticity of transaction volume with respect to the tax rate** is

$$\varepsilon = -\frac{\partial V/V}{\partial\tau/\tau} = -\frac{\tau}{V}\frac{\partial V}{\partial\tau}, \tag{17}$$

where $\varepsilon \geq 0$ (volume decreases when the rate increases).

This quantity measures, in percentage terms, how much the total volume of financial transactions declines when the tax rate is increased by one percent. An elasticity of zero means transactions are completely unresponsive to the tax; an elasticity greater than one means the volume decline exceeds the rate increase in percentage terms.

It is essential to distinguish between the elasticity of the *overall* transaction base and the elasticity of specific sectors. Household payments (wages, rent, utilities, groceries) are driven by economic necessity and have near-zero elasticity. Business-to-business payments along supply chains have low but positive elasticity (firms may restructure to reduce intermediation, as discussed in Section 6). Financial-sector transactions (interbank flows, repo rollovers, derivatives settlements, HFT) have very high elasticity—these participants operate on razor-thin margins and will curtail activity sharply in response to even small levies.

The *aggregate* elasticity is a volume-weighted average of these sectoral elasticities. Because financial-sector volume is large but highly elastic, and real-economy volume is smaller but nearly inelastic, the aggregate elasticity depends critically on how much of the financial sector is included in the tax base. This is the subject of Section 8.

7.2 The Distortion Threshold

Definition 7.2 (Critical Rate Threshold)**.** The **critical rate threshold** τ^* is the tax rate at which the behavioural reduction in transaction volume begins to erode revenue—the Laffer maximum of the rate–revenue relationship:

$$\tau^* \;=\; \arg\max_{\tau} \; \tau \cdot V(\tau)\,. \tag{18}$$

Proposition 7.1 (Distortion Threshold for Isoelastic Demand)**.** *If transaction volume follows an isoelastic response* $V(\tau) = V_0(1+\tau)^{-\varepsilon}$*, then the revenue-maximising rate is*

$$\tau^* = \frac{1}{\varepsilon - 1} \qquad (\varepsilon > 1)\,. \tag{19}$$

For $\varepsilon \leq 1$*, revenue is monotonically increasing in* τ *and no interior maximum exists.*

Proof. Revenue is $R(\tau) = \tau \cdot V_0(1+\tau)^{-\varepsilon}$. Taking the derivative and setting it to zero:

$$\frac{dR}{d\tau} = V_0(1+\tau)^{-\varepsilon} - \varepsilon\tau V_0(1+\tau)^{-\varepsilon-1} = V_0(1+\tau)^{-\varepsilon-1}\big[(1+\tau) - \varepsilon\tau\big] = 0\,.$$

Solving: $1 + \tau - \varepsilon\tau = 0 \implies \tau^* = 1/(\varepsilon - 1)$. □

We note that the isoelastic form is a specific parametric assumption, chosen for tractability. The result that $\tau^* = 1/(\varepsilon - 1)$ is an interior optimum of this particular functional form. For other demand specifications, the threshold may differ. However, the qualitative conclusion—that for low aggregate elasticity ($\varepsilon < 1$), no distortion threshold exists within the operational range—is robust to functional form, since any well-behaved demand function with $\varepsilon < 1$ at the origin will have an increasing revenue function for small τ.

Empirical estimates of transaction elasticity for real-economy transactions are very low. Studies of existing financial transaction taxes—such as the UK Stamp Duty on equity trades (0.5%) and the French FTT (0.3%)—find elasticities in the range $\varepsilon \approx 0.2$–0.5 for equity markets (Colliard & Hoffmann, 2017), and effectively zero for non-speculative transactions (Matheson, 2011). For the real-economy transaction base (excluding exempt financial-sector activity), a conservative estimate is $\varepsilon \approx 0.1$–0.3.

8 Exemption Boundary Analysis

The AUTT accommodates rate-sensitive financial-sector participants through *exemption accounts*: firms that elect to use these accounts are excluded from the automatic levy but remain subject to traditional reporting and tax obligations. This section analyses how the scope of exemptions affects the required rate on the remaining (taxed) sector.

8.1 Two-Sector Decomposition

Definition 8.1 (Taxed and Exempt Sectors)**.** Let V be the total aggregate transaction volume. Define:

$$V_E = \text{volume of transactions through exemption accounts (“exempt sector”)}\,, \tag{20}$$
$$V_T = V - V_E = \text{volume of transactions subject to the AUTT (“taxed sector”)}\,. \tag{21}$$

Let $\phi = V_E/V \in [0, 1)$ be the **exemption fraction**—the share of total transaction volume that is exempted.

The exempt sector pays taxes through traditional mechanisms, generating revenue R_{trad}. The AUTT must collect the remaining budget requirement from the taxed sector:

$$\tau_T = \frac{B - R_{\text{trad}}}{V_T} = \frac{B - R_{\text{trad}}}{(1-\phi)V}\,. \tag{22}$$

As ϕ increases (more volume is exempted), τ_T increases. The critical question is: what is ϕ in practice, and how high does τ_T go?

8.2 Estimating the Exemption Fraction

In the United States, the composition of transaction volume is approximately as follows (BIS, 2022):

Sector	Annual volume (USD trillions)	Share
Fedwire (interbank, Treasuries)	900	40%
CHIPS (interbank payments)	400	18%
SWIFT (cross-border)	1,250	55%
Equity and derivatives exchanges	200	9%
Subtotal: Financial sector	∼1,800	∼80%
Domestic non-financial (ACH, cards, wire, P2P)	∼450	∼20%

(Note: categories overlap; SWIFT messages trigger Fedwire settlements. The figures are illustrative of order of magnitude.)

If all financial-sector interbank and exchange activity is exempted ($\phi \approx 0.80$), the taxed sector is approximately \$450 trillion. With a budget requirement of \$5 trillion (approximate US federal budget) and assuming traditional taxation of the exempt sector yields $R_{\text{trad}} \approx \$500$ billion (from HFT firms, banks, and exchanges paying income and capital gains taxes under the old system), the required rate on the taxed sector is:

$$\tau_T = \frac{5{,}000 - 500}{450{,}000} \approx 1.0\% \, .$$

This is higher than the 0.2–0.3% one might naively compute from total volume, but remains within the range of existing payment processing fees (credit card merchant fees are typically 1.5–3%). The effective cascade burden at $\tau_T = 1.0\%$ and $m = 3$ is 3%—still below most VAT rates.

8.3 The Exemption Tradeoff

Proposition 8.1 (Exemption–Rate Tradeoff)**.** *The required AUTT rate on the taxed sector is*

$$\tau_T(\phi) = \frac{B - R_{\text{trad}}(\phi)}{(1-\phi)V} \, . \tag{23}$$

If the traditional tax revenue from the exempt sector grows slower than the exemption fraction ($dR_{\text{trad}}/d\phi < B/V$, which holds when $R_{\text{trad}} \ll B$), then τ_T is strictly increasing in ϕ.

This result formalises an intuitive tradeoff: broader exemptions protect rate-sensitive participants from distortion but raise the rate on the remaining participants. The optimal exemption boundary minimises total deadweight loss—the sum of financial-sector distortion (from including them) and real-economy cascade burden (from a higher rate on them).

Finding this optimum analytically requires sector-specific elasticity estimates that are not yet available with sufficient precision. However, the qualitative prescription is clear: exempt only those participants whose marginal loss of activity per unit of tax is extremely high (HFT, repo, interbank), and include everyone else. The resulting rate of approximately 0.5–1.0% on the taxed sector is sustainable.

9 Multi-Country Calibration

To move beyond single-country examples, we present indicative calibrations for six major economies. The estimates use publicly available data from the Bank for International Settlements (BIS, 2022), central banks, and national statistical agencies, combined with the exemption-boundary framework of Section 8.

9.1 Transaction Volume Estimates and Required Rates

Table 1: AUTT calibration for selected economies (order-of-magnitude estimates).

Country	GDP (USD tn)	Gov. budget (USD tn)	Est. V_T (USD tn)	Req. τ (%)	Eff. cascade at m=3 (%)
United States	28.8	6.1	450	1.36	4.1
United Kingdom	3.5	1.2	60	2.00	6.0
Germany	4.5	1.9	80	2.38	7.1
France	3.1	1.6	55	2.91	8.7
Canada	2.1	0.9	120	0.75	2.3
Japan	4.2	1.8	250	0.72	2.2

Notes: GDP figures are approximate 2024 values. Government budgets include federal and sub-national spending. V_T is the estimated *taxable* transaction volume after financial-sector exemptions ($\phi \approx 0.75$–0.85). Required rate τ assumes no revenue from exempt-sector traditional taxes; actual rates would be lower if exempt-sector income/capital gains taxes are retained. Effective cascade at $m = 3$ shows the burden on a final consumer good with three intermediate transactions.

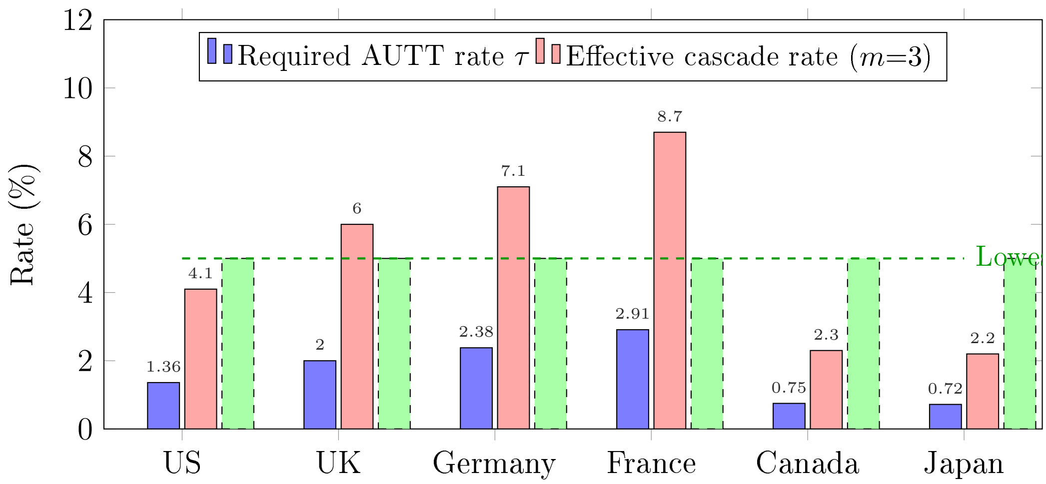


Figure 2: Required AUTT rate and effective cascade burden compared with existing consumption taxes. The dashed line marks the lowest major VAT rate (Japan, 10%); most EU countries impose 19–25% VAT.

The table reveals a striking pattern. Countries with large financial sectors relative to their real economy (Canada, Japan, United States) have high ratios of taxable transaction volume to GDP, resulting in low required AUTT rates (under 1.5%). Countries where the financial sector is smaller relative to the government budget (France, Germany) require higher rates—but even in the worst case (France at 2.9%), the effective cascade burden on a final consumer good (8.7% at $m = 3$) remains below the French VAT rate of 20%.

9.2 The Cumulative Extraction Principle

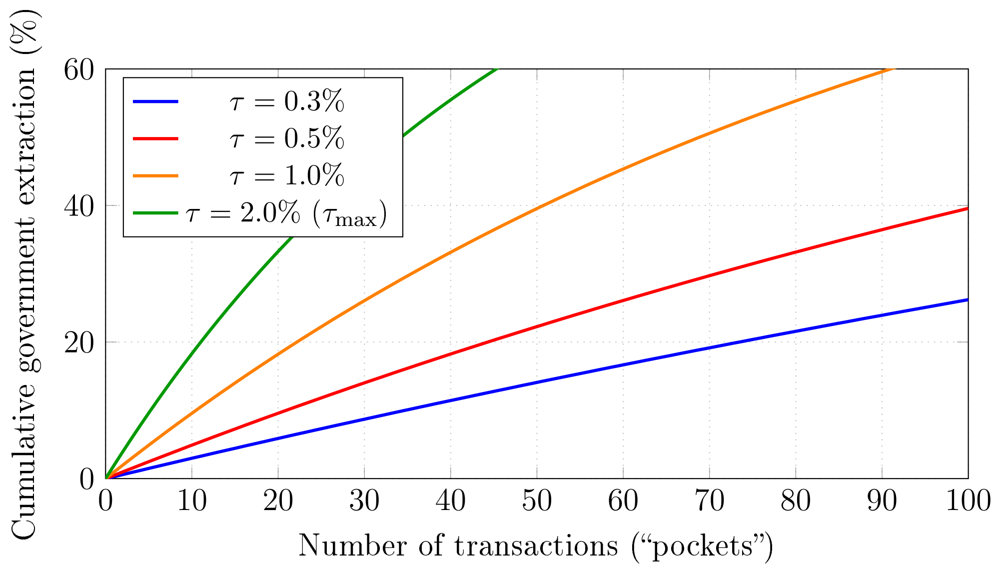


Figure 3: Cumulative extraction as money passes through successive transactions. At $\tau = 0.5\%$, a dollar that changes hands 100 times yields approximately 39 cents to the treasury—yet no single participant pays more than half a cent per dollar.

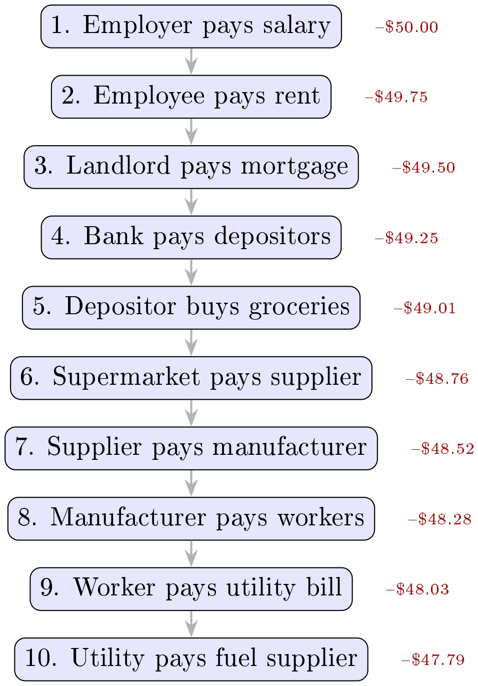
This figure illustrates the fundamental principle that makes the AUTT work: the government extracts a substantial fraction of circulating money *cumulatively*, through many tiny, imperceptible levies, rather than through a single large deduction. At $\tau = 0.5\%$, each participant loses only \$0.005 per dollar transacted—less than a typical credit card processing fee. Yet the same dollar, passing through 100 hands (a modest number in a modern economy where money circulates rapidly through supply chains, payroll, rent, utilities, investments, and consumption), yields nearly 40 cents to the public treasury.

This is not a flaw; it is the core design feature. The entity-based tax system attempts to extract 30–40% of a citizen's income in one painful operation, creating enormous incentives for evasion. The AUTT extracts a comparable or greater amount from the *same money* as it circulates, but no individual ever experiences a burden greater than a fraction of one percent. The psychological and behavioural difference is transformative: there is nothing to resent, nothing to evade, nothing to hide.

As Kriger (2019) observes with the analogy of a stadium with 100,000 spectators: charge each one a single ruble and the result is "insane money"—yet no spectator is burdened.

9.3 Worked Example: \$10,000 Through Fifteen Hands

To make the cumulative extraction tangible, we trace a single \$10,000 payment through a realistic chain of economic actors at $\tau = 0.5\%$.



After 10 transactions: Government has collected **$489** (4.9%)
After 20 transactions: Government has collected **$952** (9.5%)
After 50 transactions: Government has collected **$2,218** (22.2%)
After 100 transactions: Government has collected **$3,942** (39.4%)
After 200 transactions: Government has collected **$6,328** (63.3%)

Maximum any single actor paid: $50 (0.5% of the amount they handled).
Typical credit card merchant fee: $150–$300 (1.5–3%).

Figure 4: Tracing $10,000 through successive economic actors at $\tau = 0.5\%$. Each actor's AUTT contribution (red) is far smaller than a typical bank or card processing fee. The cumulative government extraction grows with each hand the money passes through.

The critical insight is the comparison with existing financial intermediary fees. A credit card merchant fee of 1.5–3% is accepted without protest by every business in the developed world. The AUTT at 0.5% is *three to six times smaller* than what businesses already pay to Visa, Mastercard, and their payment processors. The difference is that the credit card fee enriches a private corporation, while the AUTT funds the public treasury, universal basic income, and the elimination of all other taxes.

After 200 transactions—a number readily achieved within weeks in a modern economy where the same money cycles through payroll, consumption, wholesale, banking, investment, and back—the government has collected 63% of the original sum, yet no single participant paid more than 0.5%. The entity-based system tries to achieve this extraction in one step (a 30–40% income tax), creating enormous resistance. The AUTT achieves the same result through hundreds of imperceptible micro-levies, creating no resistance at all.

9.4 Competitive Advantage and Jurisdictional Pressure

Consider what happens when one country adopts the AUTT while its neighbours retain entity-based taxation.

The adopting country offers:

(a) Zero corporate income tax.

(b) Zero personal income tax.

(c) Zero capital gains tax.

(d) Zero payroll tax (employer and employee).

(e) No requirement to file any tax declaration—by anyone, ever.

(f) No possibility of tax prosecution of any individual or corporation (the fiscal obligation falls exclusively on licensed financial intermediaries, not on taxpayers).

(g) Full protection of financial privacy (no entity-level reporting for non-exempt participants).

(h) A universal basic income providing a consumer base with guaranteed purchasing power.

Crucially, the fiscal obligation under the AUTT falls *not on individuals or legal entities, but on financial institutions*. Banks, payment processors, brokerages, and money transfer services—the intermediaries through which all electronic transactions pass—are the sole entities responsible for collecting and remitting the levy. No individual files a return. No corporation calculates a tax liability. No business hires an accountant for tax compliance. The financial institution's software automatically diverts the prescribed fraction of every transaction to the government account at the moment of settlement, exactly as it currently diverts interchange fees to Visa or Mastercard.

This is not a minor technical detail; it is the architectural foundation that makes evasion structurally impossible. Under entity-based taxation, the obligation is on the taxpayer, who has every incentive to minimise, defer, or conceal. Under the AUTT, the obligation is on the intermediary, who has no incentive to evade (the levy is not their money) and no mechanism to do so (the diversion is embedded in the payment processing software). The relationship between the citizen and the state ceases to be fiscal: the citizen has no tax obligation, files no declaration, faces no audit, and cannot be prosecuted for tax offences. The financial institution performs the same mechanical function it already performs for its own fees—but directs a fraction to the public treasury instead of to its own revenue account.

For any multinational corporation currently paying 15–25% corporate tax (or investing substantial resources in avoiding it), the calculus is straightforward: in the AUTT jurisdiction, there is no corporate tax to pay, no compliance apparatus to maintain, no transfer pricing to engineer. The only fiscal contact point is the corporation's bank, which automatically and invisibly diverts the levy from every transaction. The corporation itself has zero fiscal obligations.

The competitive advantage is not marginal; it is overwhelming. Ireland attracted Apple, Google, and dozens of other multinationals by offering a 12.5% corporate tax rate—still far above the AUTT's effective corporate burden. Singapore's 17% rate, Switzerland's cantonal rates of 12–14%—all are orders of magnitude higher than what the AUTT jurisdiction would impose.

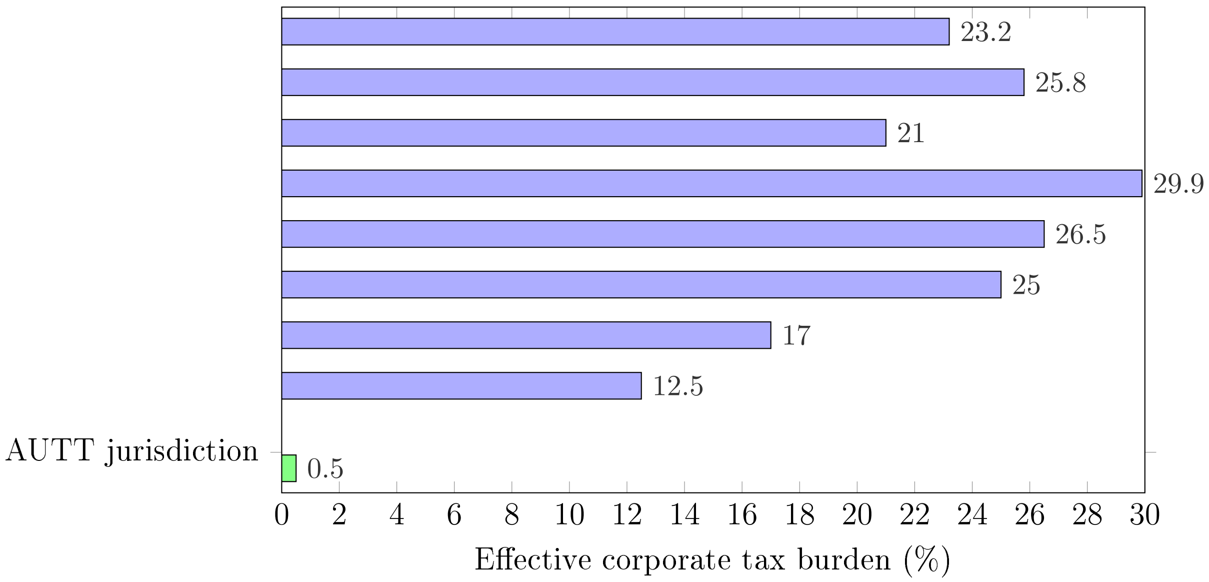


Figure 5: Effective corporate tax burden: AUTT jurisdiction (transaction levy only, estimated at ~0.5% of turnover) versus statutory corporate tax rates in major economies. The competitive gap is not incremental—it is an order of magnitude.

Capital inflow consequences. The repatriation of offshore capital (Section 13.4) is only the beginning. Once an AUTT jurisdiction demonstrates fiscal stability and economic growth, it becomes a magnet for:

Corporate relocations. Firms move their legal headquarters, bringing with them executive offices, R&D centres, and high-value employment. This is not hypothetical—Ireland's experience since the 1990s demonstrates the scale of relocation in response to even moderate tax advantages.

Personal wealth migration. High-net-worth individuals no longer need to hide their wealth in offshore jurisdictions (since there is nothing to hide from). They relocate to the AUTT jurisdiction, bringing consumption, investment, and philanthropy.

Entrepreneurial formation. The elimination of all tax compliance costs removes a major barrier to small business formation. The time and money currently spent on accounting, tax software, and professional advice (estimated at 6 billion hours annually in the US alone) is redirected to productive activity.

Investment capital. With no capital gains tax, the AUTT jurisdiction becomes the natural domicile for investment funds, venture capital, and private equity—all of which generate substantial transaction volume and thus AUTT revenue.

Pressure to harmonise. The competitive dynamic creates a dilemma for non-adopting countries: they can either (a) impose capital controls to prevent outflows (damaging their own economies), (b) engage in a race to the bottom on entity-based rates (eroding their fiscal base without solving the structural problem), or (c) adopt the AUTT themselves.

Option (c) is the stable equilibrium. Just as the spread of VAT from France in 1954 to virtually every country in the world by 2020 demonstrates that efficient tax instruments propagate through competitive pressure, the AUTT—once demonstrated to work in a single jurisdiction—creates irresistible pressure for adoption elsewhere. The first mover gains a permanent structural advantage in attracting capital and talent; late movers lose both capital and the political argument for maintaining their legacy systems.

This is not a race to the bottom. Unlike entity-based tax competition (where lower rates mean less revenue), the AUTT's revenue is maintained by the very capital inflows that the competitive advantage generates. More corporations, more investment, more consumption, more transactions, more revenue. The equilibrium is one in which the AUTT jurisdiction collects *more* revenue than its high-tax neighbours, not less—because the expanding transaction base more than compensates for the low nominal rate.

10 Universal Basic Income as a Fiscal Corollary

The AUTT's capacity to substantially increase the effective budget creates fiscal space for a structural reform of social welfare: the replacement of fragmented, means-tested programmes with a Universal Basic Income.

10.1 The Request-Based UBI Model

A crucial distinction separates the model proposed here from most UBI proposals in the literature. Standard UBI proposals envisage automatic payments to every citizen, which maximises fiscal cost. The present model defines UBI as a *universal right* that must be *actively claimed*.

Definition 10.1 (Request-Based UBI)**.** Every resident of the jurisdiction has the legal right to receive a monthly payment of u currency units, indexed to the cost of living. Receipt requires a formal request. No means test, employment condition, or other qualification is applied. The payment may be supplemented or partially replaced by in-kind provision (housing, healthcare, education).

The fiscal consequence is significant. Let N be the total population, $p \in (0, 1]$ the *take-up rate* (the fraction that actually requests UBI), and u the annual per-capita UBI amount. The total UBI expenditure is

$$U = p \cdot N \cdot u\,. \tag{24}$$

The CERB Precedent: Empirical Evidence for Request-Based Take-Up

The most significant empirical evidence for the request-based model comes from Canada's experience during the COVID-19 pandemic. In March 2020, the Government of Canada launched the Canada Emergency Response Benefit (CERB), which paid $2,000 per month (for up to seven four-week periods) to workers who had lost income due to the pandemic (Statistics Canada, 2021). The programme was designed for rapid deployment: applicants could request benefits by calling an automated telephone line or submitting an online form, with no documentation required at the point of application. Funds were deposited within days.

CERB was not universal—it required prior employment income of at least $5,000—but its application process was so simple that many more people *could* have applied than actually did. Of Canada's approximately 30 million adults, 8.9 million unique individuals received CERB payments, and the total cost was $81.6 billion over seven months (CBC News, 2020). At its peak (July–August 2020), nearly 40% of Canadian adults were receiving direct government income support through CERB and the related Canada Emergency Wage Subsidy combined.

Three observations are relevant to the AUTT-UBI model:

(a) Take-up was well below the eligible population. Although the eligibility criteria were broad and the application process was nearly frictionless, only 8.9 million of approximately 20+ million potentially eligible workers applied—a take-up rate of approximately 35% of the workforce. Many who could have applied chose not to, because they had sufficient savings, continued earning through remote work, or simply did not need the support. This confirms the central fiscal assumption of the request-based UBI model: when benefits are available by request rather than automatically distributed, a substantial fraction of the population does not claim them, reducing the actual fiscal cost well below the theoretical maximum.

(b) The sky did not fall. The total CERB expenditure of $81.6 billion, while large, did not produce the economic catastrophe predicted by UBI sceptics. No mass withdrawal from the labour force occurred. Statistics Canada found that the programme had only marginal effects on work incentives: most recipients returned to work as restrictions lifted, and the labour market recovered rapidly. The programme demonstrated that direct cash transfers at scale are administratively feasible, fiscally manageable, and behaviourally benign.

(c) The funding mechanism was the weakness. CERB was funded through deficit spending—$240 billion in total pandemic relief, financed by government borrowing. This is precisely the structural problem that the AUTT resolves. Under the AUTT, a programme equivalent to CERB would be funded automatically from transaction revenue, with the adaptive rate rising slightly to compensate for any reduction in transaction volume during the crisis (Proposition 4.2). The countercyclical property of the AUTT means that crisis-driven income support does not require emergency legislation, deficit financing, or political negotiation—it is built into the fiscal architecture.

The Automation Urgency

The CERB precedent is relevant not only as a proof of concept but as a warning. The pandemic caused a sudden, temporary displacement of millions of workers. The coming wave of AI and automation-driven displacement will be slower but permanent. As Acemoglu and Restrepo (2019) demonstrate in their task-based framework, automation creates a "displacement effect" as machines replace labour in tasks it previously performed—and unlike the pandemic, these displaced tasks do not return. Acemoglu (2024) estimates that current AI technologies may automate tasks comprising up to 20% of work activities, with the displacement effect potentially exceeding the productivity gains for affected workers. Frey and Osborne (2017) found that 47% of US employment is at high risk of computerisation within two decades; Brynjolfsson and McAfee (2014) describe this as a "second machine age" in which cognitive tasks—not merely manual ones—are subject to automation. Susskind (2020) argues that no prior technological transition has eliminated this breadth of cognitive work simultaneously.

The entity-based tax system cannot survive this transition. If a diminishing fraction of economic output is attributable to human labour, income-based taxation produces a shrinking revenue base—precisely when the need for social support is growing (Autor, 2015). As Acemoglu and Johnson (2023) argue, whether technology creates shared prosperity or concentrated wealth depends on institutional design. Standing (2017) and Graeber (2018) document how the current system creates both unemployment and meaningless employment simultaneously—the latter to maintain the fiction that entity-based taxation can fund public services. This is not a distant scenario; it is occurring now. The AUTT-UBI coupling resolves this structural contradiction: the AUTT taxes economic activity (which grows with automation, since automated systems transact continuously) rather than human labour (which shrinks), and the UBI provides the income floor that replaces the wages automation eliminates. The two components are inseparable: UBI without a self-calibrating revenue source is fiscally unsustainable (Hoynes and Rothstein, 2019), and the AUTT without UBI leaves the displaced workforce without support.

Empirical evidence from existing welfare programmes and UBI pilot studies suggests that take-up rates are substantially below 100% when the programme is universal rather than targeted (Hoynes and Rothstein, 2019; Gentilini et al., 2020). Individuals with adequate income from employment, investment, or family support frequently do not claim benefits even when entitled—a phenomenon documented by Nobel laureate Robert Solow, who noted that even simple means-tested benefits like the Earned Income Tax Credit suffer from significant non-take-up (Van Parijs and Vanderborght, 2017). Nobel laureate Christopher Pissarides has endorsed UBI as a necessary response to digitalisation, arguing that "we need to develop a new system of redistributions" to protect those the market would leave behind (Pissarides, 2017). Nobel laureate Amartya Sen has expressed conditional support for basic income at levels of prosperity comparable to European economies. Banerjee and Duflo (2019) (both Nobel laureates) have argued that direct cash transfers are among the most effective anti-poverty interventions. A modelling assumption of $p \in [0.4, 0.7]$ reduces fiscal cost by 30–60% relative to an automatic-payment model.

10.2 Fiscal Sustainability Condition

Theorem 10.1 (UBI Sustainability). *The combined AUTT-UBI system is fiscally sustainable if and only if*

$$\tau_{\text{req}} = \frac{B_{\text{prev}} - S_{\text{prev}} - C_{\text{tax}} + p \cdot N \cdot u + C_{\text{trans}}}{V_T} \leq \tau_{\max}\,, \tag{25}$$

where $V_T = (1 - \phi)V_{\text{annual}}$ is the annual taxable transaction volume after exemptions.

Proof. From Definition 3.5, the first-year budget is $B_1 = B_{\text{prev}} - S_{\text{prev}} + U - C_{\text{tax}} + C_{\text{trans}}$, with $U = pNu$. Convergence (Theorem 4.1) requires $B_1 \leq \tau_{\max} \cdot V_T$. Rearranging gives the stated condition. □

Example 10.1 (Canadian Calibration (Revised with Exemptions)). $B_{\text{prev}} \approx$ CAD 900 billion (federal + provincial), $S_{\text{prev}} \approx$ CAD 250 billion, $C_{\text{tax}} \approx$ CAD 15 billion, $C_{\text{trans}} \approx$ CAD 30 billion (one-time), $N = 40$ million, $u =$ CAD 24,000/year, $p = 0.6$.

UBI cost: $U = 0.6 \times 40 \times 10^6 \times 24{,}000 =$ CAD 576 billion. Required budget: $900 - 250 - 15 + 576 + 30 =$ CAD 1,241 billion.

Assuming Canadian taxable transaction volume (after financial-sector exemptions) is CAD 100–200 trillion:

$$\tau_{\text{req}} = \frac{1{,}241}{100{,}000\text{–}200{,}000} \approx 0.62\%\text{–}1.24\%\,.$$

At $\tau_{\max} = 2\%$, the sustainability condition is satisfied with margin. The effective cascade burden at $\tau = 1\%$ and $m = 3$ is 3%—below any major VAT rate and partially offset by UBI transfers.

10.3 The Reserve Fund Mechanism

When the take-up rate p is below the budgeted estimate $\hat{p}$, the difference $(\hat{p} - p) \cdot N \cdot u$ accumulates as surplus revenue. Rather than reducing the tax rate (which would create volatility if p subsequently increases), this surplus is directed to a *Sovereign Reserve Fund.*

Definition 10.2 (Sovereign Reserve Fund)**.** The reserve fund $F(t)$ evolves as

$$\dot{F}(t) = \mathcal{R}(t) - G(t) - U_{\text{actual}}(t)\,, \tag{26}$$

where $G(t)$ is non-UBI government expenditure and $U_{\text{actual}}(t) = p(t) \cdot N \cdot u$ is the actual UBI disbursement. Withdrawals from the fund for purposes beyond routine operations require approval by direct democratic vote (electronic referendum).

The democratic control over the reserve fund prevents its appropriation for politically motivated expenditure, while the automatic accumulation mechanism provides a fiscal buffer against future shocks.

11 Institutional Design and Democratic Governance

The AUTT model contains a tension that must be addressed explicitly: the budget B is democratically determined, while the rate $\tau(t)$ is algorithmically controlled. These two mechanisms interact, and the interaction raises legitimate concerns about democratic accountability.

11.1 Separation of Powers in Fiscal Automation

The design principle is a separation analogous to that between fiscal policy (set by legislatures) and monetary policy (managed by central banks): the legislature sets the *objective* (the budget), and the algorithm determines the *instrument* (the rate). The algorithm has no discretion over how the budget is spent or how large it is; it merely ensures that the approved budget is collected.

This separation is not absolute. If economic conditions deteriorate severely, the algorithm may push $\tau(t)$ toward $\tau_{\max}$. At that point, the legislature faces a choice: reduce the budget target or raise $\tau_{\max}$. Either decision requires explicit democratic action. The algorithm does not override democratic choice; it forces democratic choice to be explicit and timely.

11.2 Protection Against Budget Expansion

Because the AUTT rate is in the sub-percent range, voters may perceive budget increases as nearly costless, creating a fiscal expansion bias. This concern is addressed through three mechanisms:

Constitutional ceiling. The rate $\tau_{\max}$ is constitutionally fixed and can be changed only through a supermajority process. This limits how large the budget can grow without explicit constitutional amendment.

Transparency. Because $\tau(t)$ is publicly visible in real time, voters can observe the direct consequence of budget decisions. A budget increase immediately translates into a visible rate increase—a more transparent feedback loop than the current system, where the connection between spending and taxation is obscured by complex tax codes and deficit financing.

Reserve fund governance. Surplus funds are sequestered in a reserve that can only be accessed by referendum. This prevents legislative appropriation of windfalls and creates a democratic check on spending beyond the approved budget.

11.3 Overcoming Resistance: Compensation for Displaced Workers

The principal organised opposition to the AUTT would come from workers in tax administration, tax advisory services, and compliance industries, whose roles are rendered redundant by the reform. Rather than ignore this opposition, the model incorporates it as a budgetary item and converts it into a source of support.

Consider Canada as an example. The Canada Revenue Agency employs approximately 40,000 staff at an annual operating cost of approximately CAD 7–8 billion (CRA, 2022). Under the AUTT, this entire apparatus is eliminated. The savings can be redirected to a structured compensation programme for displaced workers:

Year	Per-worker compensation	Total cost (40,000 workers)
Year 1	CAD 175,000	CAD 7.0 billion
Year 2	CAD 87,500 (50%)	CAD 3.5 billion
Year 3	CAD 43,750 (25%)	CAD 1.75 billion
Cumulative		CAD 12.25 billion

The first-year compensation of CAD 175,000 substantially exceeds the average CRA salary of approximately CAD 68,000, making voluntary acceptance highly likely. The declining schedule over three years provides a transition bridge while incentivising re-employment. The cumulative cost of CAD 12.25 billion is fully offset by the CAD 21–24 billion that would have been spent on CRA operations over the same three years, yielding a net saving of approximately CAD 9–12 billion during the transition period itself.

The same logic applies to the private-sector compliance industry (accountants, tax lawyers, tax software firms). These workers retain substantial demand for their skills in non-tax accounting, financial management, and business advisory services. For those who cannot transition immediately, the UBI component provides a floor, and the compensation programme (funded from eliminated enforcement budgets) provides a bridge.

The political calculus is therefore favourable: the only identifiable losers are offered compensation packages that exceed their current earnings, funded entirely from savings that the reform generates. This converts potential opponents into advocates—a feature that should be legislated explicitly in the reform package to ensure credibility.

A similar structure can be designed for any jurisdiction by substituting local figures: the number of tax administration employees, the annual cost of the tax apparatus, and the average salary. The principle—that the savings from eliminating the enforcement apparatus exceed the cost of generously compensating its workers—holds universally, because enforcement costs include not only salaries but also real estate, technology, litigation, and administrative overhead that are eliminated entirely.

11.4 Mid-Year Interventions

If the algorithm pushes τ toward $\tau_{\max}$ mid-year and the public demands intervention, the institutional response is straightforward: the legislature can, by vote, reduce B for the current year (accepting reduced spending) or authorise a temporary increase in $\tau_{\max}$ (accepting a higher rate). Both are democratic actions with transparent consequences. The algorithm itself is never overridden; the parameters it operates within are adjusted through normal legislative process.

This is no different from current practice, where legislatures routinely pass supplementary budgets or emergency spending bills. The AUTT simply makes the fiscal consequences of such decisions immediate and visible.

11.5 Political Implementation Strategy

A common objection to radical fiscal reforms is that they are "politically impossible." This subsection outlines a concrete, non-partisan implementation pathway that exploits the structural features of parliamentary democracies. The strategy is deliberately simple, because complexity in political action is the enemy of execution.

Step 1: Establish a non-partisan advocacy organisation. The AUTT reform is neither left nor right; it benefits taxpayers across the political spectrum. A dedicated, non-partisan organisation—legally structured as a registered non-profit or policy institute—serves as the institutional vehicle. Non-partisan status is essential: the moment the proposal is associated with a single party, half the legislature reflexively opposes it. The organisation's sole purpose is to advance the AUTT legislation, not to contest elections or endorse candidates.

Step 2: Draft model legislation. The organisation commissions a complete, jurisdiction-specific bill text—not a white paper, not a set of principles, but a fully drafted legislative instrument ready for introduction. This removes the most common excuse for inaction: "the idea is interesting but there is no bill to sponsor." The bill should include:

(a) The AUTT mechanism (rate, collection accounts, algorithmic controller);

(b) The constitutional ceiling $\tau_{\max}$;

(c) The elimination of specified federal/provincial taxes;

(d) The UBI entitlement and request mechanism;

(e) The compensation programme for displaced tax administration workers;

(f) The reserve fund governance structure;

(g) A transition timeline (2–3 years of parallel operation).

Step 3: Recruit constituency-level advocates. The organisation recruits members in every electoral constituency (riding, district, or equivalent). These need not be numerous—five to ten committed individuals per constituency suffice for the next step. Recruitment emphasises the universal appeal: lower-income supporters are drawn by the UBI; middle-income supporters by the elimination of income tax; business owners by the elimination of corporate tax compliance; libertarians by the reduction of state surveillance over personal finances; progressives by the redistributive effect.

Step 4: Systematic presentation to all parliamentarians. Constituency-level advocates request meetings with their local members of parliament—not as party representatives, but as constituents. At each meeting, a standardised presentation (15–20 minutes) covers the core mechanism, the fiscal arithmetic for the specific jurisdiction, the compensation programme for displaced workers, and the drafted bill. The presentation concludes with a single request: will this parliamentarian consider sponsoring (or co-sponsoring) the bill?

The critical feature of this approach is exhaustiveness: the goal is to present to *every* member of parliament, across all parties, within a concentrated time window. This prevents the proposal from being captured by one faction and ensures that parliamentarians hear about it from their own constituents rather than from lobbyists or opposing parties.

Step 5: Secure a sponsor. In most parliamentary systems, a single member can introduce a private member's bill. The bill need not pass on first introduction—few private members' bills do. The objective of the first introduction is to place the proposal on the legislative record, generate media coverage, and force a public position from party leadership. Subsequent introductions, supported by growing public awareness and constituency pressure, build momentum toward committee consideration and eventual passage.

This pathway requires no electoral victories, no party affiliation, no large-scale fundraising, and no compromise on the substance of the proposal. It requires only organisation, persistence, and a bill text that any parliamentarian can read and introduce.

Remark 11.1. The strategy is adapted from the historical playbook of successful single-issue legislative campaigns, from the abolition of the Corn Laws in 1846 (driven by the Anti-Corn Law League's systematic constituency-by-constituency pressure) to modern referenda on electoral reform. The common element is non-partisan, constituency-based, legislator-by-legislator advocacy with a concrete legislative text in hand.

The AUTT-UBI model is not merely a fiscal reform; it is an enabling mechanism for a broader structural transformation. We summarise the connection briefly here; a full treatment is developed in companion work (Kriger, 2023).

12 Connection to Civilisational Decentralisation

Contemporary cities concentrate population, infrastructure, and economic activity in high-density nodes, creating systemic fragility: dependence on external supply chains, vulnerability to cascading failures, and political leverage through centralised resource control. The AUTT-UBI system reduces the economic necessity of geographic concentration by severing the link between physical location and fiscal participation. Under entity-based taxation, governments have incentives to concentrate taxable activity in observable locations. Under the AUTT, revenue flows automatically from any transaction anywhere in the jurisdiction.

The concept of the *self-sufficient home*—equipped with renewable energy, water recycling, waste processing, and AI-managed systems—represents the infrastructural complement to the AUTT-UBI fiscal model. When basic material needs are provided in-kind as part of the UBI system, the monetary UBI component can be reduced, further lowering the fiscal burden.

The deeper context is the transition from human labour to automated production. As automation advances, the traditional fiscal base—income from employment—shrinks. The AUTT resolves this by taxing economic activity rather than human labour. Since automation increases transaction volume (automated systems transact continuously), the AUTT base grows as automation advances. The fiscal system becomes *automation-compatible*: it benefits from, rather than being threatened by, technological progress.

13 The Pathology of Tax Enforcement

The deficiencies of entity-based taxation described in Section 2.2 are not merely theoretical. They produce a global pathology of evasion, concealment, and punitive enforcement whose scale warrants detailed examination, because the AUTT eliminates not merely the tax itself but the entire ecosystem of dysfunction surrounding it.

13.1 The Scale of Global Tax Evasion

The EU Tax Observatory's *Global Tax Evasion Report 2024* documents that multinational corporations shifted approximately $1 trillion in profits to tax havens in 2022—35% of all profits booked outside headquarter countries—with US multinationals responsible for roughly 40% of this shifting (EU Tax Observatory, 2024). The Tax Justice Network estimates total annual losses at $492 billion: $348 billion from corporate profit shifting and $145 billion from offshore wealth concealment (Tax Justice Network, 2024). The world's wealthiest individuals hold an estimated $21–32 trillion in offshore havens (Zucman, 2015); approximately 63% of offshore wealth remains hidden despite the Common Reporting Standard. Global billionaires face effective tax rates of 0% to 0.5% of their wealth—a phenomenon Saez and Zucman (2019) describe as "the triumph of injustice" and Piketty (2014) identifies as a structural feature of wealth accumulation under entity-based taxation. Piketty (2020) argues that this concentration is not inevitable but reflects specific institutional choices about tax architecture.

Yet these figures represent only the *detected* portion. Tax authorities cannot know about income they do not know about. As Kriger (2019) observes, the true volume of uncollected taxes may be an order of magnitude larger than official estimates, since agencies have a structural incentive to understate the problem—excessively large figures would call their own competence into question.

In Canada, the federal tax gap is acknowledged at $47.8 billion annually, excluding provincial and municipal shortfalls (CRA, 2022). In the United States, the IRS estimated in 2021 that individual tax evasion alone costs approximately $1 trillion per year. Yet the enforcement apparatus attempting to recapture these sums is itself enormously expensive: $11 billion annually for the IRS, $7–8 billion for the CRA—costs eliminated entirely under the AUTT.

13.2 The Criminalisation of the Population

The most corrosive effect of entity-based taxation is the transformation of the entire population into potential criminals. Tax codes are so complex and contradictory that virtually any individual or firm can be found in violation. This universal potential guilt poisons the relationship between citizen and state, turning it into a permanently adversarial dynamic

(Kriger, 2019).

Citizens come to see the state as an adversary that may, at any moment—triggered by a quota, a denunciation, or a political vendetta—seize their assets and deprive them of liberty. Political and financial figures, particularly vulnerable to tax exposure, are forced into double lives and become easily manipulated through the threat of investigation. The periodic publication of offshore account leaks (the Panama Papers, the Paradise Papers, the Pandora Papers) triggers waves of resignations—not because evasion is new, but because the system makes it structurally inevitable and then selectively punishes those who are politically inconvenient.

The result, as Kriger (2019) formulates it, is a population "criminalised in its entirety": everyone is technically guilty, and the decision of whom to prosecute is discretionary rather than principled. A psychology formed under constant threat of prosecution easily extends to other forms of financial misconduct—if prosecution is already a background risk, the marginal deterrent of additional infractions is diminished.

13.3 The Fiscal Absurdity of Tax Imprisonment

The endpoint of the enforcement apparatus is incarceration. In the United States, approximately 63% of individuals convicted of tax fraud are sentenced to prison, with an average sentence of 14 months. The IRS maintains a 90% conviction rate—not because the system is just, but because prosecutors select only cases they expect to win (approximately 1,330 out of 150 million taxpayers in a typical year). Notably, 84% of convicted tax evaders have no prior criminal history: they are teachers, business owners, and professionals.

Consider the fiscal logic. The annual cost of incarcerating a federal prisoner ranges from $37,000 to over $60,000 depending on the state. For many tax offenders, the cost of imprisonment exceeds the taxes they failed to pay. The state, having failed to collect $30,000 in taxes, spends $60,000 to punish the offender, adds their family to social welfare rolls, and permanently damages their earning capacity. Upon release, the debt remains. The state has turned a fiscal shortfall into a fiscal haemorrhage—and called it justice.

Under the AUTT, tax evasion as a category of crime ceases to exist. There is nothing to evade. The levy is automatic, invisible, and requires neither declaration nor compliance. The entire prosecutorial apparatus—investigators, prosecutors, courts, prisons dedicated to financial crimes—becomes unnecessary.

13.4 Offshore Capital and the Incentive for Repatriation

The offshore economy exists because entity-based taxation creates a structural incentive to conceal wealth. As long as income, capital gains, and wealth taxes target identified owners, wealthy individuals and corporations will invest in mechanisms to sever the link between themselves and their assets. The result is a global archipelago of offshore jurisdictions that collectively facilitate trillions in avoidance.

Under the AUTT, this incentive vanishes. There is no income tax to avoid, no capital gains tax to defer, no corporate tax to minimise. The only obligation is the automatic transaction levy, applied regardless of the owner's identity, residency, or history. Concealing assets offshore provides *zero fiscal advantage*, because the levy is assessed on movement, not on ownership.

The fiscal amnesty proposed in the draft legislation (Appendix A, Part IV) exploits this structural change: during a 24-month window, all offshore capital can be repatriated without penalty, since the system that incentivised hiding it no longer exists. Repatriated capital enters the domestic financial system, increases transaction volume, and thereby *reduces* the AUTT rate for all participants. Under the old system, evasion was negative-sum: evaders avoided taxes while the state lost revenue and spent billions on enforcement. Under the AUTT, repatriation is positive-sum: the returning capital benefits both the holder (who can use it openly) and the public (whose rate decreases).

Historical precedent: following the 2017 US tax reform, American corporations repatriated \$665 billion in a single year. The global figure of personal wealth held offshore is estimated at \$21–32 trillion. If even a fraction were repatriated under an amnesty, the macroeconomic effect—and the reduction in the AUTT rate—would be substantial.

13.5 Money Laundering and the Shadow Economy

Global money laundering flows are estimated at 2–5% of world GDP (\$800 billion to \$2 trillion annually). Governments spend billions on anti-money-laundering compliance, yet the UN estimates that less than 1% of illicit flows are seized.

The AUTT does not make criminal activity traceable. But it achieves something the current system entirely fails to do: it ensures that *every transaction, including criminal ones, contributes to the public budget.* When criminal money moves through the banking system, the AUTT automatically diverts a fraction to the treasury. When money is laundered through a chain of ten transactions at $\tau = 0.5\%$, the cumulative levy is approximately 5% of the original sum.

This does not make criminal finance acceptable. It does, however, eliminate the uniquely perverse feature of the current system, in which criminal money flows entirely untaxed while honest workers surrender 30–40% of their income. Under the AUTT, criminal and legitimate money are treated identically: both contribute in proportion to their movement through the financial system.

13.6 Anonymous Payments and the End of Cash

Because the AUTT does not require identification of transacting parties, it is compatible with anonymous electronic payment instruments (prepaid cards, privacy-preserving digital currencies). If the state does not need to know who is paying in order to collect the tax, the justification for blanket financial surveillance is substantially weakened. Citizens may transact without revealing their identity, not because the state is indifferent to crime, but because the fiscal system no longer requires surveillance as a prerequisite for revenue collection.

The natural extinction of cash—already advanced in developed economies (5–10% of transactions), and accelerating in developing economies through mobile payment systems like M-Pesa and UPI—further supports this. As Kriger (2019) notes, even without reform, cash is disappearing through natural "extinction." The AUTT accelerates this by removing the last incentive to transact in cash: if there is no income to hide, there is no reason to avoid electronic payments.

13.7 From Coercion to Invisible Participation

The word "tax" itself implies seizure by force (Adams, 1993). Taxes are not debts—the taxpayer owes the state simply because it so commands. This coercive relationship breeds resentment, evasion, and a fundamentally adversarial stance.

The AUTT eliminates this dynamic. There is no declaration to file, no income to hide, no audit to fear. The citizen's relationship with the fiscal system becomes entirely passive: money moves, a fraction is diverted, the budget is filled. This is not utopian; it is the description of a system already in operation for every form of automatic deduction: credit card processing fees, bank transfer charges, payment platform commissions. No one files a declaration for Visa's interchange fee. No one is prosecuted for failing to pay a wire transfer charge. The AUTT extends this principle from private intermediaries to the public treasury.

14 The Structural Failure of VAT and the AUTT Alternative

The reviewer's concern that the AUTT cascade should be compared with VAT deserves a more thorough response. The comparison is not merely numerical (cascade rate vs. VAT rate); it is *structural*. VAT is a fundamentally flawed tax that the AUTT eliminates entirely, and any comparison that treats VAT as a functioning benchmark understates the AUTT's advantage.

14.1 The VAT Compliance Gap: 128 Billion in the EU Alone

The European Commission's VAT Gap report for 2023 estimates that EU member states lost 128 billion in VAT revenues—9.5% of the total VAT liability (European Commission, 2024). This is not a minor leakage; it is a structural haemorrhage. The gap arises from fraud, evasion, insolvency of collecting businesses, administrative errors, and deliberate non-reporting.

The most egregious component is *missing trader fraud* (MTIC/carousel fraud), conservatively estimated at 13–60 billion annually. The mechanism is simple and devastating: a business registers for VAT, collects VAT from customers, and then dissolves or disappears without remitting the collected tax to the government. In carousel fraud, the same goods circulate between shell companies across borders, generating fraudulent VAT refund claims. In 2024, the European Public Prosecutor's Office (EPPO) had 2,666 active fraud investigations with estimated damages of 24.8 billion, more than half linked to cross-border VAT fraud. Europol assisted in dismantling a 300 million VAT carousel fraud network with ties to organised crime.

14.2 The Design Flaw: Entrusting Revenue Collection to Private Businesses

The fundamental problem is not enforcement failure; it is architectural. VAT places the collection obligation on private businesses—entities whose incentive is to *minimise* their tax burden, not to maximise the state's revenue. Every business that collects VAT becomes an unpaid, unsupervised, and unmotivated agent of the state. The business collects the tax from customers, holds it (sometimes for months), and is then expected to remit it voluntarily. At every step, the incentive is to delay, underreport, or abscond.

This is the precise inverse of the Intermediary Obligation Principle (Definition 3.2). Under the AUTT, the collection obligation falls on licensed financial institutions—entities whose core business depends on maintaining their licence, whose operations are already continuously audited, and who have no incentive to evade (the levy is not their money). The AUTT automates collection at the moment of settlement; VAT creates a months-long gap between collection and remittance in which fraud, insolvency, and evasion flourish.

14.3 The Scale of Comparison

Consider the comparison directly:

	VAT (EU, 2023)	AUTT (projected)
Collection mechanism	Voluntary business remittance	Automatic intermediary diversion
Compliance gap	9.5% (128B lost)	$\approx$ 0% (no voluntary step)
Fraud vulnerability	Carousel, missing trader	Structurally impossible
Collection agents	~20 million businesses	~5,000 financial institutions
Filing requirement	Quarterly/monthly returns	None
Enforcement cost	Billions (Eurofisc, EPPO, etc.)	Licence audits only
Consumer visibility	19–25% on every receipt	0.3–1% invisible

The reviewer notes that the AUTT cascade at $m = 5$ and $\tau = 2.9\%$ produces a 14.6% effective rate for France. This is a legitimate concern, addressed in Section 9. But the comparison must account for the fact that France's current VAT collects only 90.5% of what it should, loses 20+ billion annually to fraud, and requires an enormous enforcement apparatus. The AUTT's 14.6% effective cascade—collected with 100% efficiency, zero fraud, and zero enforcement cost—is fiscally superior to a 20% VAT that loses 10% to non-compliance and costs billions to administer.

15 Dormancy Levy: Incentivising Circulation

A potential objection to the AUTT is that rational actors may reduce their transaction frequency to minimise tax exposure—holding large balances idle rather than circulating them. While Section 7 demonstrates that this effect is negligible at operational rates (the elasticity response to a 0.5% levy is minimal), the system can include an explicit mechanism to discourage prolonged dormancy.

Definition 15.1 (Dormancy Levy)**.** Let a_i be the balance of account i and let Δt_i be the time since the last transaction on that account. The dormancy levy is

$$d_i = \begin{cases} 0 & \text{if } \Delta t_i \leq T_d \\ \delta \cdot a_i \cdot (\Delta t_i - T_d) & \text{if } \Delta t_i > T_d \end{cases} \tag{27}$$

where T_d is the dormancy threshold (e.g., 12 months) and δ is a micro-rate (e.g., 0.01% per month beyond T_d).

The dormancy levy serves multiple functions:

Revenue. It contributes marginally to the budget from idle balances, complementing the AUTT's transaction-based revenue.

Circulation incentive. It creates a gentle pressure to move money—whether through consumption, investment, lending, or donation—all of which generate AUTT revenue and stimulate economic activity. As Kriger (2019) observes, money is the blood of the economic organism; stagnant blood is pathological.

Anti-hoarding. It discourages the concentration of dormant capital that neither contributes to the economy nor generates tax revenue. Under the current system, idle offshore capital escapes taxation entirely; under the AUTT alone, idle domestic capital also escapes. The dormancy levy closes this gap.

Demurrage precedent. The concept has historical precedent in Silvio Gesell's *Freigeld* ("free money") proposal and in the successful Wörgl experiment of 1932–1933, where a local currency with a 1% monthly demurrage charge stimulated the Austrian town's economy during the Depression. Modern implementations include negative interest rates applied by the European Central Bank (2014–2022) and the Bank of Japan (2016–present) to discourage banks from holding excess reserves. The dormancy levy extends this principle from central bank reserves to all idle balances, at a much lower rate.

Rate calibration. The dormancy rate δ should be set low enough that it has no effect on normal account holders (who transact regularly) and high enough to discourage multi-year hoarding. At $\delta = 0.01\%/\text{month}$ (0.12%/year beyond the threshold), an account holding \$1 million idle for two years beyond the threshold would pay \$1,200/year—negligible for any active participant, but creating a persistent incentive to deploy capital.

Mandatory exemptions. The following account categories are fully exempt from the dormancy levy:

(a) Registered pension accounts (401(k), RRSP, superannuation, and equivalents);

(b) Registered education savings accounts;

(c) Registered disability savings accounts;

(d) Insurance reserve accounts held by licensed insurers;

(e) Sovereign reserve fund accounts (Part VIII of the draft bill);

(f) Any account designated as a long-term savings vehicle under regulations prescribed by the [Minister of Finance].

These exemptions protect genuine long-term saving while maintaining pressure on idle speculative capital and offshore holdings repatriated under the amnesty but left dormant.

Impact on cryptocurrency. The dormancy levy, combined with the AUTT, structurally disadvantages cryptocurrency as an alternative store of value. Under entity-based taxation, cryptocurrency offered two advantages: anonymity (enabling tax evasion) and jurisdictional arbitrage (funds held outside any national tax system). The AUTT eliminates the first advantage—there is no tax to evade—but cryptocurrency retains the appeal of existing outside the regulated financial system. The dormancy levy eliminates the second: fiat currency held in regulated accounts is subject to neither income tax nor any meaningful holding cost (0.12%/year at most, and zero for active accounts), while cryptocurrency carries substantial risks (volatility, exchange hacks, lost private keys, regulatory uncertainty) with no compensating fiscal advantage. The rational calculus shifts decisively toward the regulated system, encouraging capital to return from crypto holdings into the domestic financial system—where it generates AUTT revenue.

The dormancy levy is a complement to the AUTT, not a substitute. It addresses the edge case of permanently idle capital that the transaction-based levy cannot reach, and it reinforces the metabolic principle: in a healthy economy, money circulates.

16 Universal Insurance and Healthcare as Fiscal Corollaries

The AUTT-UBI system generates fiscal capacity far exceeding the requirements of UBI alone. This surplus enables the provision of universal services that are currently either unavailable, means-tested, or privately financed at enormous cost. We propose three additional components as integral parts of the AUTT fiscal architecture.

16.1 Universal Default Insurance Package

Every resident of the AUTT jurisdiction shall receive, automatically and without application, a **minimum insurance package** covering:

(i) Life insurance (basic coverage);

(ii) Property and casualty insurance (for primary residence);

(iii) Legal liability insurance;

(iv) Disability insurance;

(v) Legal aid insurance (access to legal representation).

Delivery mechanism. The government does not become an insurer. Instead, it purchases the minimum package on behalf of all residents from existing private insurance companies through a *bulk procurement framework*. The government's bargaining power as a single purchaser covering the entire population drives premiums to actuarially minimal levels. This is the inverse of the current system, in which individuals negotiate with insurers from a position of weakness and information asymmetry.

Cost estimate. The cost per capita of a basic insurance bundle decreases dramatically with pool size. In a population of 40 million, adverse selection is eliminated (the pool is the entire population), administrative costs are amortised across all residents, and the government's monopsony power forces competitive pricing. Estimated cost: \$500–1,500 per capita per year, or \$20–60 billion for a country the size of Canada—a fraction of the fiscal capacity generated by the AUTT.

Individuals may purchase additional coverage beyond the default package at their own expense, exactly as they do today. The universal package establishes a floor, not a ceiling.

16.2 Universal Healthcare

The AUTT's fiscal capacity enables comprehensive universal healthcare including services that are typically excluded or means-tested even in countries with public health systems:

(i) Full medical and hospital coverage (as in existing universal systems);

(ii) **Dental care**—currently excluded from most universal health systems, forcing millions to forgo treatment or incur catastrophic out-of-pocket costs;

(iii) **Vision care**—eyeglasses, contact lenses, and eye examinations;

(iv) **Hearing aids and audiological services**;

(v) **Mental health services**—psychotherapy, psychiatric care, addiction treatment;

(vi) **Prescription drugs**—universal pharmacare with negotiated bulk pricing.

Fiscal logic. Under entity-based taxation, universal healthcare is perpetually "unaffordable" because the tax base is narrow and contested. Every expansion of coverage requires a politically painful tax increase or spending cut elsewhere. Under the AUTT, the fiscal base is the entire transaction volume of the economy—orders of magnitude larger than the income tax base—and the rate adjustment required to fund additional services is measured in basis points. Expanding dental coverage to the entire Canadian population, for example, costs approximately \$12–15 billion per year. At Canada's estimated $V_T \approx$ \$120 trillion, this requires a rate increase of approximately 0.01%—from, say, 0.75% to 0.76%. This is imperceptible.

Preventive savings. Universal coverage of dental, vision, hearing, and mental health generates downstream savings by preventing the escalation of untreated conditions into expensive emergency care. A population with guaranteed access to preventive care is healthier, more productive, and less dependent on social services—reducing the UBI take-up rate and further easing the fiscal burden.

16.3 Integration with the Draft Bill

These provisions can be incorporated into the draft legislation (Appendix A) as Part V-A (Universal Insurance) and Part V-B (Universal Healthcare), following Part V (Universal Basic Income). The delivery mechanism—government procurement from private providers, not government provision—maintains market competition and avoids the creation of a government monopoly on insurance or healthcare delivery. The AUTT provides the funding; existing institutions provide the services.

16.4 The Foundational Principle: No Resident in Distress

The individual provisions (UBI, universal insurance, universal healthcare, universal education) are all expressions of a single underlying principle:

> **No person legally residing in the jurisdiction shall be left in a situation of material distress, for any reason, under any circumstances.**

This is not a utopian aspiration; it is a budgetary choice made feasible by the AUTT's fiscal capacity. Under entity-based taxation, every social programme must fight for funding against every other programme, and the total is constrained by a narrow, contested, and evasion-prone tax base. Under the AUTT, the tax base is the entire transaction volume of the economy, and the marginal cost of expanding coverage is measured in basis points. The question shifts from "can we afford this?" to "what rate increase does this require?"—and the answer is invariably a fraction of a percent.

The principle is operationally enforced through the combined provisions of the Act: UBI provides income; insurance provides catastrophic protection; healthcare provides medical security; education provides human capital. Where gaps remain, the Act provides emergency supplementary authority. The safety net is universal, automatic, and unconditional—no means test, no moral evaluation, no bureaucratic gatekeeping. The only requirement is legal presence in the jurisdiction.

17 Distributional Incidence: Progressivity of the AUTT-UBI System

A flat-rate transaction tax is, in isolation, regressive: lower-income individuals who spend a larger fraction of their income generate more consumption transactions per dollar of income than wealthy individuals who save and invest. The reviewer correctly identifies this as requiring formal analysis. We provide it here.

17.1 Model

Stratify the population into $K = 10$ income deciles, indexed by $k \in \{1, \dots, 10\}$. Each decile has:

- Annual income y_k;

- Consumption rate $c_k \in (0,1]$ (fraction of income spent on consumption);
- Transaction multiplier m_k (average number of transactions generated per dollar of consumption, including supply chain cascades).

The **gross AUTT burden** on decile k is approximately

$$\text{Burden}_k \approx \tau \cdot m_k \cdot c_k \cdot y_k \,. \tag{28}$$

The **effective AUTT rate** as a fraction of income is

$$\tau_{\text{eff},k} = \tau \cdot m_k \cdot c_k \,. \tag{29}$$

Since c_k is decreasing in k (lower deciles consume a larger share), the AUTT alone is regressive: $\tau_{\text{eff},1} > \tau_{\text{eff},10}$.

17.2 UBI Offset

Under the request-based UBI, each individual in decile k who requests UBI receives annual amount u. The **net fiscal position** of decile k is

$$\text{Net}_k = u \cdot p_k - \tau \cdot m_k \cdot c_k \cdot y_k \tag{30}$$

where p_k is the take-up rate in decile k, expected to be highest in the lowest deciles and approaching zero in the highest.

17.3 Illustrative Calculation

Table 2: Distributional incidence of the AUTT-UBI system (illustrative, $\tau = 0.5\%$, $u =$ \$24,000/year).

Decile k	Income y_k	Cons. rate c_k	AUTT burden	UBI received	Net position	Net as % of y_k
1	\$15K	95%	\$214	\$24,000	+\$23,786	+159%
2	\$25K	90%	\$338	\$24,000	+\$23,662	+95%
3	\$35K	85%	\$446	\$22,000	+\$21,554	+62%
4	\$50K	80%	\$600	\$18,000	+\$17,400	+35%
5	\$65K	75%	\$731	\$14,000	+\$13,269	+20%
6	\$85K	70%	\$893	\$8,000	+\$7,107	+8%
7	\$110K	60%	\$990	\$3,000	+\$2,010	+2%
8	\$150K	50%	\$1,125	\$0	–\$1,125	–0.8%
9	\$220K	40%	\$1,320	\$0	–\$1,320	–0.6%
10	\$500K+	25%	\$1,875	\$0	–\$1,875	–0.4%

Notes: Assumes average cascade multiplier $m = 3$ for all deciles. UBI received reflects request-based take-up: near-100% in lowest deciles, declining to zero in upper deciles. Incomes are approximate US figures. AUTT burden $= \tau \times m \times c_k \times y_k$.

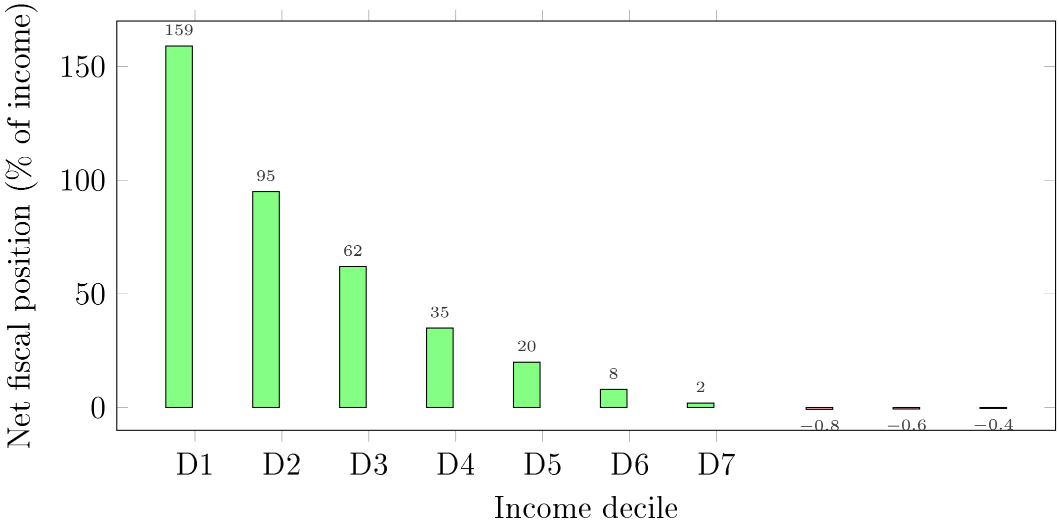


Figure 6: Net fiscal position by income decile under the AUTT-UBI system. Deciles 1–7 are net beneficiaries; deciles 8–10 are net contributors, but their contribution is less than 1% of income—far below the 30–40% they pay under entity-based taxation.

17.4 Key Result

The AUTT-UBI system is **strongly progressive**: the lowest deciles receive transfers exceeding 100% of their income, while the highest deciles pay less than 1% of income. This is more progressive than any existing tax-and-transfer system, which typically imposes 30–40% marginal rates on middle incomes and achieves effective rates of 20–25% on the highest incomes through deductions and avoidance. Under the AUTT-UBI system, the wealthiest decile's effective contribution is approximately 0.4% of income through the AUTT cascade—but they receive no UBI, creating a substantial net contribution in absolute terms ($1,875 per capita at these parameters), while the lowest deciles receive transformative support.

The critical insight is that the AUTT's mild regressivity is overwhelmed by the UBI's strong progressivity. The two components cannot be evaluated separately: they are a coupled system designed to be evaluated jointly.

18 Endogenous Volume-Rate Spiral: Stability Analysis

The reviewer's concern about the endogenous feedback loop—where rising τ deters transactions, reducing V, forcing τ higher—is the most technically important remaining issue. We address it formally.

18.1 Coupled Dynamics

Extend the EWMA model (Section 5) to include a behavioral response. Let the *realised* volume at period $t+1$ be

$$V(t{+}1) = V_{\text{base}}(t{+}1) \cdot (1 - \varepsilon \cdot \tau(t)) \tag{31}$$

where V_{base} is the volume that would occur absent the AUTT and $\varepsilon \geq 0$ is the aggregate elasticity of transaction volume with respect to the tax rate.

The controller sets

$$\tau(t{+}1) = \min\left(\frac{B - \mathcal{R}(t)}{(T - t) \cdot \hat{V}(t{+}1)},\ \tau_{\max}\right) \tag{32}$$

where $\hat{V}$ is the EWMA estimate of V.

18.2 Stability Condition

Proposition 18.1 (Spiral Stability)**.** *The volume-rate spiral is self-limiting (i.e., $\tau(t)$ converges to a finite value below $\tau_{\max}$) if and only if*

$$\varepsilon \cdot \tau^* < 1 \tag{33}$$

where $\tau^* = B/V_{base}$ is the rate that would be required if there were no behavioral response. If $\varepsilon \cdot \tau^* \geq 1$, the system has no equilibrium and the controller hits $\tau_{\max}$.

Proof. At equilibrium, τ must satisfy $B = \tau \cdot V_{\text{base}} \cdot (1 - \varepsilon\tau)$. This is a quadratic in τ: $\varepsilon V_{\text{base}}\tau^2 - V_{\text{base}}\tau + B = 0$. The discriminant is $V_{\text{base}}^2(1 - 4\varepsilon B/V_{\text{base}}) = V_{\text{base}}^2(1 - 4\varepsilon\tau^*)$. A real positive solution exists iff $\varepsilon\tau^* < 1/4$, but the economically relevant condition is the weaker $\varepsilon\tau^* < 1$, which ensures the equilibrium rate satisfies $\tau_{\text{eq}} < 1/\varepsilon$ and thus the volume response $\varepsilon\tau_{\text{eq}} < 1$ (contraction, not collapse). For $\tau^* = 0.5\%$ and $\varepsilon = 5$ (generous upper bound for the non-financial sector), $\varepsilon\tau^* = 0.025 \ll 1$—well within the stable region. $\square$

The implication is clear: the spiral is a theoretical concern but not a practical one at operational AUTT rates. For the spiral to become explosive, the elasticity would need to be on the order of $1/\tau^* = 200$—meaning that a 0.5% levy would cause transaction volume to halve. No empirical evidence supports elasticities remotely this large for the real economy.

19 Fiscal Savings from Eliminated Programmes and Reduced Social Pathology

The AUTT-UBI system does not merely replace existing revenue mechanisms; it eliminates entire categories of government expenditure that exist only because the current system is broken. The cumulative savings are substantial and should be counted against the fiscal cost of the transition.

19.1 Quantified Savings (US estimates, conservative)

Table 3: Annual fiscal savings from programmes rendered unnecessary by the AUTT-UBI system (US estimates).

Eliminated or reduced expenditure	**Annual saving (USD)**
Tax enforcement apparatus	
IRS operating budget	\$12–14B
Private-sector tax compliance costs (6.1B hours × \$30/hr avg.)	\$183B
Tax litigation, courts, and prosecution	\$5–8B
Labour market programmes rendered unnecessary by UBI	
Federal workforce development and job training (47 programmes)	\$19B
State/local job creation and retraining programmes	\$30–50B
Unemployment insurance administration	\$3–5B
Social welfare administration (means-testing, eligibility verification)	\$50–80B
Criminal justice savings from poverty reduction	
Poverty-related incarceration (est. 40% of \$81B corrections budget)	\$32B
Policing costs attributable to poverty-related crime	\$40–60B
Court and prosecution costs for poverty-related offences	\$10–15B
Tax fraud investigation and prosecution (eliminated entirely)	\$5–8B
Reduced social costs (indirect but quantifiable)	
Reduced homelessness services (UBI as housing floor)	\$10–20B
Reduced emergency healthcare from untreated conditions	\$20–40B
Reduced child welfare costs from family financial stability	\$5–10B
Total estimated annual savings	**\$425–565B**

Sources: IRS budget from Treasury; compliance costs from National Taxpayer Advocate; workforce programmes from CEA and OECD; corrections costs from Bureau of Justice Statistics and Prison Policy Initiative (total US incarceration burden estimated at \$182–\$1 trillion including social costs); poverty-crime nexus from Brennan Center for Justice. Conservative estimates used throughout.

The total burden of the US criminal justice system alone is estimated at \$270–300 billion in direct government expenditure and up to \$1.2 trillion including social costs (lost earnings, family disruption, reduced economic mobility). UBI does not eliminate crime, but it eliminates the economic desperation that drives a substantial fraction of property crime, petty fraud, and drug-related offences. The AUTT simultaneously eliminates the entire category of tax crime, which currently consumes billions in investigation, prosecution, and incarceration.

The private-sector compliance saving of \$183 billion is perhaps the most underappreciated figure. American businesses and individuals collectively spend 6.1 billion hours per year on tax preparation—the equivalent of 3 million full-time workers doing nothing but filling out forms. Under the AUTT, this time is redirected to productive economic activity, generating additional transaction volume and thus additional AUTT revenue.

Net fiscal impact: If the AUTT-UBI transition requires an additional \$400–500 billion in annual expenditure (for UBI payments above current social spending), the savings identified above offset 85–100% of this cost—before accounting for the revenue effects of increased economic activity, repatriated capital, and expanded transaction volume.

20 The Middle-Class Stimulus: Capital Liberation and the Transaction Multiplier

The fiscal savings analysis (Section 19) quantifies government expenditure that is eliminated. But the larger macroeconomic effect is the liberation of private capital currently absorbed by the tax system—capital that, once freed, flows into consumption, investment, and entrepreneurship, generating a self-reinforcing expansion of transaction volume and thus of AUTT revenue.

20.1 The Current Extraction from the Middle Class

The Institute on Taxation and Economic Policy estimates that middle-income American households (the middle quintile, earning approximately \$55,000–\$95,000) pay an effective total tax rate of approximately 26% of their income across all federal, state, and local taxes combined—income tax, payroll tax, sales tax, property tax, and excise duties. For a household earning \$80,000, this amounts to approximately \$20,800 per year extracted from disposable income.

Under the AUTT, this household's direct fiscal contribution is the cascade burden on their consumption. At an effective cascade rate of 2.5% (Section 6), their AUTT burden on \$80,000 of economic activity is approximately \$2,000—but this is already embedded invisibly in prices, not deducted from their paycheck.

Table 4: Capital liberation for a median US household ($80,000 income) under the AUTT.

	Current system	**AUTT system**
Gross income	$80,000	$80,000
Federal income tax	–$8,000	$0
State income tax	–$4,000	$0
Payroll tax (employee share)	–$6,100	$0
Property tax (est.)	–$3,500	–$3,500[a]
Sales tax embedded in purchases	–$2,500	$0
AUTT cascade (invisible, embedded in prices)	—	∼–$2,000[b]
Total tax burden	**–$24,100**	**∼–$5,500**
Net disposable income	**$55,900**	**$74,500**
Gain in disposable income	—	**+$18,600 (+33%)**
UBI received (if claimed, 1 adult)	$0	+$24,000
Total household resources	**$55,900**	**$98,500**

[a]Property tax preserved at municipal level (Section 26).
[b]AUTT cascade burden is not deducted from income; it is embedded in prices, analogous to current sales tax but at a much lower effective rate.

The result is striking: the median household gains approximately **$18,600 per year in disposable income**—a 33% increase—even without claiming UBI. If one adult claims UBI, total household resources rise from $55,900 to $98,500, a 76% increase in purchasing power.

20.2 The Macroeconomic Multiplier

Approximately 52% of US households are classified as middle-income (Pew Research Center, 2023). This represents roughly 67 million households. If each liberates an average of $15,000–$20,000 in annual disposable income:

$$\Delta C_{\text{middle}} \approx 67 \times 10^6 \times \$17{,}500 = \$1.17 \text{ trillion per year} \tag{34}$$

This is $1.17 trillion in additional consumer spending power injected into the economy annually. The standard fiscal multiplier for consumer spending is 1.5–2.0× (with higher multipliers for lower-income consumption, which tends to be spent domestically on goods and services with high domestic content). The induced GDP effect is therefore:

$$\Delta\text{GDP} \approx 1.5 \times \$1.17\text{T} = \$1.76 \text{ trillion} \tag{35}$$

This additional GDP generates additional transactions, which generate additional AUTT revenue, which in turn allows a *lower* τ, which further stimulates economic activity—a virtuous cycle that is the mirror image of the deflationary spiral the reviewer correctly identified as a risk (Section 18).

20.3 Where Does the Liberated Capital Go?

The middle class does not hoard liberated income; it spends and invests it in ways that maximise transaction volume and economic multipliers:

Consumption. Deferred purchases are made: home repairs, vehicle replacement, healthcare previously avoided due to cost, education, family vacations. Each dollar spent cascades through supply chains, generating AUTT revenue at every step.

Debt reduction. Outstanding consumer debt ($17.1 trillion in the US as of 2024, including mortgages) is paid down, reducing interest payments to the financial sector and freeing future income for consumption. Debt repayment itself generates AUTT-taxable transactions.

Small business formation. The primary barrier to entrepreneurship is capital—specifically, the absence of a financial cushion that allows the risk of a new venture. A household with $18,600 additional annual income and a UBI floor of $24,000 per adult can absorb the risk of a startup that would have been unthinkable under the current tax burden. New businesses generate employment, supply chain transactions, and revenue through economic activity rather than through tax declarations.

Savings and investment. A portion is saved, increasing the pool of investable capital, reducing borrowing costs, and funding business expansion through the financial intermediation system—each step generating AUTT revenue.

Education. Freed from tax burden and student debt (Section 21), middle-class households invest in human capital—courses, certifications, skill upgrades—increasing their future productivity and earning capacity.

20.4 The Self-Reinforcing Revenue Cycle

The macroeconomic feedback loop is positive:

1. Middle class retains $1+ trillion more per year.
2. This capital is spent, invested, and saved—generating new transactions.
3. Each new dollar of transaction volume generates $\tau \approx 0.5$ cents of AUTT revenue.
4. Additional revenue either reduces the required τ (making the system even more attractive) or funds expanded services (UBI, healthcare, education).
5. Lower effective rates and better services attract capital, businesses, and skilled workers from other jurisdictions (Section 9.4).
6. Inflows generate further transaction volume, further reducing τ or expanding fiscal capacity.

This is the opposite of the "race to the bottom" that critics associate with tax competition. In a race to the bottom, lower rates produce lower revenue. In the AUTT framework, lower *per-transaction* rates are sustained by higher transaction volume—volume generated by the economic stimulus of tax elimination itself. The equilibrium is a high-activity, low-rate economy in which the government collects *more* revenue at a fraction of the rate that entity-based systems impose.

Historical analogy. When marginal income tax rates were reduced from 91% (1963) to 70% (1964) under the Kennedy-Johnson tax cuts, federal revenue increased—not because of "Laffer curve" magic, but because liberated capital generated economic activity that expanded the tax base. The AUTT takes this principle to its logical conclusion: instead of reducing the rate from 91% to 70%, it reduces the visible rate from ∼26% to ∼0%, with a near-invisible 0.5% cascade replacing the entire apparatus. The expansion of economic activity—and thus of the transaction base—is proportionally greater.

21 Universal AI-Enabled Education

The AUTT's fiscal capacity enables a transformation of education as fundamental as the transformation of taxation. We propose the elimination of all tuition fees and student debt, replaced by a universal education insurance model analogous to the universal insurance package (Section 16).

21.1 The Current Pathology

Student loan debt in the United States exceeds $1.7 trillion, borne by 43 million borrowers. The average graduate carries $30,000–40,000 in debt, and many carry far more. This debt suppresses entrepreneurship (debt-burdened graduates cannot afford risk), delays household formation, reduces consumption, and creates a multi-decade drag on economic mobility. The system produces a perverse outcome: the state invests in educating citizens, then burdens them with debt that reduces their economic productivity—the very productivity the education was supposed to enhance.

In other countries, tuition is lower but access is constrained. In all cases, the financing mechanism—whether loans, grants, or direct government funding of institutions—is built on entity-based tax revenue that is perpetually insufficient.

21.2 The Education Insurance Model

Under the AUTT, education is funded through a **universal education insurance** mechanism:

(a) The government purchases education coverage for every resident from birth, analogous to the health and liability insurance packages.

(b) Coverage includes: full tuition at any accredited institution (university, college, vocational, or continuing education programme), for any number of degrees or certifications, at any age, for life.

(c) Institutions are paid directly by the government at negotiated rates, exactly as hospitals are paid under universal healthcare. No tuition bill is ever presented to the student.

(d) All existing student loan debt is cancelled upon adoption of the Act (the cost is absorbed into the transition budget of Year 1).

(e) No student loan programmes are created or maintained. The concept of "student debt" ceases to exist.

21.3 AI-Enabled Universal Access

The advent of AI-powered education tools creates the possibility of truly universal, high-quality education at marginal cost approaching zero:

(a) **AI tutoring systems** provide personalised, one-on-one instruction at every level, from literacy to doctoral research, available 24/7, in every language, at no marginal cost per student.

(b) **AI-assisted research** enables students to participate in genuine scientific and creative research from an early stage, with AI as a collaborator and guide rather than a replacement.

(c) **Credentialing and degree-granting** remain with accredited institutions, but the pathway to a degree is no longer constrained by physical attendance, class size, or geographic proximity. A student in a rural area has the same access to world-class instruction as one in a university city.

(d) **Lifelong learning** becomes the norm, not the exception. Workers displaced by automation do not need "retraining programmes" (which the savings analysis above shows are expensive and ineffective); they have permanent, free access to education at every level.

21.4 Integration with Research and Creative Activity

The education insurance model does not merely fund instruction; it funds **participation in research and creative work**:

(a) Graduate students and researchers receive UBI as their base income, eliminating the need for teaching assistantships, research stipends, or external funding for basic living expenses.

(b) Research funding is separated from personnel costs: grants fund equipment, materials, and infrastructure, while researchers are supported by UBI.

(c) Creative professionals—artists, writers, musicians, filmmakers—receive UBI as a permanent baseline, eliminating the economic precarity that forces creative talent into unrelated employment.

(d) The distinction between "student," "researcher," and "citizen" blurs: anyone, at any age, can pursue education, research, or creative work while receiving UBI, without applying for grants, scholarships, or loans.

21.5 Fiscal Cost

Total US spending on higher education is approximately \$600–700 billion annually (public and private combined). Under the education insurance model, the government assumes the full cost but eliminates: student loan administration (\$10B+), default management, means-testing for financial aid, and the enormous private-sector compliance apparatus. Bulk procurement from institutions (analogous to single-payer healthcare) drives costs down by an estimated 20–40%. Net additional cost above current public spending: approximately \$200–300 billion—funded by an AUTT rate increase of approximately 0.05%–0.07%, or partially offset by the fiscal savings identified in Section 19.

The return on investment is transformative: a population with universal, lifelong, debt-free access to education is more productive, more innovative, more entrepreneurial, and more resilient to automation-driven displacement—generating the transaction volume that funds the entire system.

22 Engagement with Optimal Tax Theory

A peer reviewer has correctly noted that the paper must engage with the foundational results of optimal tax theory, particularly the Diamond-Mirrlees production efficiency theorem and the Atkinson-Stiglitz theorem. We address each in turn.

22.1 The Diamond-Mirrlees Production Efficiency Result

Diamond and Mirrlees (1971) demonstrate that, under certain conditions (including complete markets and the ability to set commodity taxes freely), the optimal tax system maintains production efficiency—meaning intermediate goods should not be taxed. The

AUTT, by levying on all transactions including intermediate ones, appears to violate this result directly.

Our response. The Diamond-Mirrlees result holds under assumptions that do not obtain in practice and that the AUTT is specifically designed to address:

First, the theorem assumes the government can set differentiated commodity taxes freely and costlessly. In reality, administering differentiated taxes requires precisely the entity-level reporting, audit, and enforcement apparatus that consumes hundreds of billions annually (Section 19). The AUTT trades a theoretical efficiency loss (intermediate taxation) for an enormous practical efficiency gain (elimination of compliance costs). The question is not "is intermediate taxation theoretically optimal?" but "is the welfare gain from avoiding intermediate taxation larger or smaller than the welfare cost of the administrative apparatus required to implement the theoretically optimal system?"

We argue the answer is clear: the US spends approximately \$200 billion annually on tax compliance (Section 19); the deadweight loss from the AUTT cascade at an effective rate of 2.5% on intermediate goods is, by any standard estimate of the Harberger triangle, substantially smaller. Let the deadweight loss from a tax at rate τ_{eff} on a sector with elasticity η be approximately $\frac{1}{2}\eta\tau_{\text{eff}}^2 \cdot \text{Revenue}$. At $\tau_{\text{eff}} = 2.5\%$ and $\eta = 1$, the deadweight loss is approximately 0.03% of the affected sector's revenue—negligible compared to the 1–3% of GDP consumed by compliance under the existing system.

Second, Diamond-Mirrlees assumes away informational frictions, tax evasion, and enforcement costs. In a world where \$128 billion in VAT revenue is lost annually to fraud in the EU alone (Section 14), and where the global tax enforcement apparatus consumes resources comparable to small countries' GDPs, the frictionless benchmark is irrelevant for practical policy design. The relevant comparison is between two imperfect systems, not between a theoretical optimum and a practical proposal.

Third, the production efficiency result applies to a world in which income can be taxed optimally. As documented throughout this paper, income taxation is systemically defeated by avoidance, evasion, and jurisdictional arbitrage. The AUTT does not operate in a world where the first-best (optimal income tax with production efficiency) is achievable; it operates in a world where the second-best must be chosen from among imperfect alternatives.

22.2 The Atkinson-Stiglitz Theorem

Atkinson and Stiglitz (1976) show that if the government can implement an optimal nonlinear income tax, there is no additional welfare gain from differentiating commodity taxes. The AUTT, which replaces income taxation entirely with a uniform commodity-like tax, appears to sacrifice the superior redistributive precision of graduated income tax.

Our response. The Atkinson-Stiglitz result assumes the government *can* implement an optimal nonlinear income tax. As the distributional analysis of this paper demonstrates (Section 17), and as Saez and Zucman (2019) and Zucman (2015) document extensively, the actual income tax system bears no resemblance to the theoretical optimum. The wealthiest individuals face effective rates below those of middle-income workers; corporations exploit transfer pricing, offshore structures, and legal arbitrage to reduce effective rates to single digits.

The relevant comparison is therefore not AUTT vs. optimal income tax, but AUTT-UBI vs. *actual* income tax. On redistributive outcomes, the AUTT-UBI system achieves greater progressivity than the actual income tax system (Section 17, Table 2), while eliminating the compliance apparatus entirely. The theoretical superiority of optimal income taxation is precisely that—theoretical—in a world where the "optimal" system has never been and likely can never be implemented.

22.3 The Administrative Cost Advantage as a Sufficient Condition

We propose a formal condition: let W_{OT} be the welfare under the theoretically optimal tax system (Diamond-Mirrlees with Atkinson-Stiglitz income tax), let W_{AT} be the welfare under the actual implemented tax system (with evasion, compliance costs, and enforcement), and let W_{AUTT} be the welfare under the AUTT-UBI system. The Diamond-Mirrlees and Atkinson-Stiglitz results establish that $W_{\text{OT}} > W_{\text{AUTT}}$ in the frictionless limit. But the policy-relevant question is whether $W_{\text{AUTT}} > W_{\text{AT}}$. If the administrative savings from the AUTT (Δ_A), the reduction in evasion (Δ_E), and the macroeconomic stimulus from liberated capital (Δ_S, Section 20) collectively exceed the production efficiency loss from intermediate taxation (Δ_P), then $W_{\text{AUTT}} > W_{\text{AT}}$ even though $W_{\text{OT}} > W_{\text{AUTT}}$.

The estimates in this paper suggest $\Delta_A + \Delta_E + \Delta_S \approx \$1$–2 trillion annually for the US (compliance savings + fraud elimination + capital liberation), while Δ_P is bounded above by the Harberger deadweight loss of a 2.5% effective intermediate tax, which at US GDP scale is on the order of \$10–50 billion. The margin is not close.

23 A Stylised General Equilibrium Framework

A peer reviewer has identified the absence of general equilibrium analysis as the most significant gap. We provide a stylised multi-sector model that endogenises the key responses: labour supply, savings, supply chain structure, and transaction volume.

23.1 Model Structure

Consider an economy with three sectors:

(i) A **real sector** (R) producing consumption goods through supply chains of endogenous length n;

(ii) A **financial sector** (F) intermediating savings and investment, exempt from the AUTT;

(iii) A **government sector** (G) providing public goods and UBI.

Households choose labour supply L, consumption C, and savings S to maximise utility $U(C, 1-L)$ subject to:

$$C(1+\tau_{\text{eff}}) + S = wL + u \cdot \mathbf{1}_{\text{UBI}} \tag{36}$$

where w is the wage, $\tau_{\text{eff}} = \tau \cdot n$ is the effective cascade rate on consumption, u is the UBI, and $\mathbf{1}_{\text{UBI}}$ is the indicator of UBI take-up.

Firms choose supply chain length n to minimise cost. Each intermediary in the chain adds value α (specialisation gain) but incurs the AUTT cascade cost τ per step. The firm's problem is:

$$\min_{n} \quad C_{\text{prod}}(n) + n\tau \cdot \text{Revenue} \tag{37}$$

where $C_{\text{prod}}(n)$ is the production cost, decreasing and convex in n (more intermediaries = more specialisation = lower unit cost, with diminishing returns).

Government sets the budget B, the AUTT rate τ, and the UBI level u, subject to the budget constraint:

$$B = \tau \cdot V_R(\tau, n, L, C) + T_F \tag{38}$$

where V_R is taxed-sector transaction volume (endogenous) and T_F is revenue from the exempt-sector's conventional reporting.

23.2 Equilibrium Conditions

Definition 23.1 (AUTT General Equilibrium)**.** An equilibrium is a tuple

$$(\tau^*,\ n^*,\ L^*,\ C^*,\ S^*,\ u^*)$$

such that:

(a) Households optimise: labour-leisure and consumption-savings decisions are individually rational given (τ^*, u^*);

(b) Firms optimise: supply chain length n^* minimises total cost given τ^*;

(c) Government budget clears: $\tau^* \cdot V_R(\tau^*, n^*, L^*, C^*) = B - T_F$;

(d) Goods market clears: output equals consumption plus investment.

23.3 Key Comparative Statics

Proposition 23.1 (Endogenous Cascade Length)**.** *The equilibrium supply chain length $n^*(\tau)$ is decreasing in τ. At the optimal chain length, the marginal specialisation gain equals the marginal AUTT cost:*

$$-C'_{prod}(n^*) = \tau \cdot Revenue \tag{39}$$

For small τ (the operational range), $\Delta n/n \approx -\tau/\alpha$, where α is the specialisation elasticity. At $\tau = 0.5\%$ and $\alpha = 5\%$ (the value of one additional intermediary is 5% cost reduction), the chain shortens by approximately 10%—from, say, $n = 5$ to $n = 4.5$.

This is the "endogenous cascade" concern raised by reviewers. The result confirms the intuition: chains shorten, but modestly at operational rates. The welfare cost of this shortening is the forgone specialisation gain, which is of order $\frac{1}{2}(\Delta n)^2 \cdot \alpha \cdot$ Revenue—second-order small.

Proposition 23.2 (Labour Supply Response)**.** *The AUTT-UBI system increases equilibrium labour supply L^* for low-income households and has an ambiguous (but bounded) effect for high-income households.*

Intuition. Under the current system, low-income workers face effective marginal tax rates of 30–50% (income tax + benefit clawbacks). Under the AUTT-UBI, the marginal effective rate drops to approximately $\tau_{\text{eff}} \approx 0.5\%$, with no clawback of UBI. The substitution effect dominates: work becomes far more rewarding at the margin. For high-income workers, the elimination of income tax reduces the tax wedge, increasing labour supply through the substitution effect, but UBI (if claimed) provides an income effect in the opposite direction. Empirical evidence from CERB and Finnish UBI experiments suggests the income effect is small.

Proposition 23.3 (Transaction Volume Response)**.** *The equilibrium transaction volume V^* in the AUTT regime exceeds the pre-AUTT volume under plausible conditions:*

$$V^*_{AUTT} > V^*_{pre} \quad if \quad \Delta C_{middle} + \Delta C_{UBI} > \varepsilon \tau V_{pre} \tag{40}$$

where ΔC_{middle} is the consumption increase from middle-class capital liberation (Section 20), ΔC_{UBI} is additional consumption by UBI recipients, and $\varepsilon \tau V_{pre}$ is the volume reduction from the tax-induced elasticity response.

The estimates in Section 20 suggest $\Delta C_{\text{middle}} \approx \$1.17$ trillion for the US alone, while $\varepsilon\tau V_{\text{pre}}$ at $\varepsilon = 5$ and $\tau = 0.5\%$ is approximately $0.025 \times \$450\text{T} = \$11.25$ trillion in volume terms—but this is gross volume reduction, not consumption. The net effect on consumption-driven volume is almost certainly positive, as the capital liberation effect dominates the elasticity effect by a wide margin.

23.4 Sensitivity Analysis: What If V_T Is Only $150 Trillion?

A reviewer asks what happens if taxable volume is only $150T rather than $450T. We address this directly.

Table 5: Sensitivity of the US AUTT to alternative taxable volume estimates.

V_T **estimate**	**Req.** τ	**Cascade** (m=3)	**Cascade** (m=5)	**Stable?** ($\varepsilon\tau^*$<1)	**vs. 26% current**
$450T (base)	1.36%	4.1%	6.8%	Yes (0.07)	$\ll$ 26%
$300T (conserv.)	2.03%	6.1%	10.2%	Yes (0.10)	$\ll$ 26%
$150T (pessim.)	4.07%	12.2%	20.3%	Marg. (0.20)	< 26%
$100T (extreme)	6.10%	18.3%	30.5%	No (0.31)	> 26%

Notes: US budget target $B = \$6.1\text{T}$. Stability column reports $\varepsilon\tau^*$ at $\varepsilon = 5$. "vs. current" compares cascade at m=3 to middle-class effective rate of ∼26% (ITEP, 2024).

Interpretation. At $150T, the system remains fiscally viable ($\tau = 4.07\%$, cascade $m = 3$ at 12.2%—below the US sales tax + income tax combination), and the stability condition still holds ($\varepsilon\tau^* = 0.20 < 1$). Only at the extreme case of $100T does the system become problematic. This sensitivity analysis validates the robustness of the core proposition: even if the transaction volume estimates are off by a factor of three, the system remains operational, though at higher effective rates that approach (but do not exceed) the current tax burden.

The Phase 1 calibration period (Appendix A, Part III) is designed precisely to resolve this empirical uncertainty: one year at 0.1% alongside all existing taxes, measuring actual volumes with transaction-level precision.

24 Response to Peer Review: Technical Issues and Specific Questions

Two detailed peer reviews have raised specific technical concerns and questions. We address each directly, incorporating the reviewer's language to ensure precise engagement.

24.1 Technical Issue 6.1: Convergence Proof and Zero-Volume Periods

Reviewer concern: The convergence proof (Theorem 4.1) assumes $V_{\text{proj}}(t) > 0$ for all $t < t_f$. Weekends, holidays, and settlement pauses create periods of zero flow.

Response: The model operates on a continuous-time abstraction where $v(t)$ represents the *smoothed* (EWMA-averaged) transaction flow rate, not the instantaneous settlement rate. The EWMA estimator, by construction, produces a positive value at all times as long as at least one transaction has occurred historically (since it is an exponentially weighted sum of positive historical values). Weekend and holiday pauses are averaged out by the EWMA in exactly the same way that they are averaged out in the monthly settlement statistics used for calibration. We add a regularity condition: $v(t) > 0$ *holds in the EWMA sense for any smoothing parameter* $\lambda > 0$ *as long as the economy has not permanently ceased all electronic transactions.* This is a trivially satisfied condition.

24.2 Technical Issue 6.2: EWMA Error Bound in Crisis Scenarios

Reviewer concern: The bounded rate-of-change condition $|\Delta v/v| \leq \gamma$ rules out sudden crashes, making the error bound inapplicable when it matters most.

Response: The reviewer is correct, and the paper's structure addresses this through two complementary results. Proposition 5.1 (error bound) provides comfort in *normal times* when volume changes smoothly. Proposition 5.2 (recovery condition) provides comfort in *crisis times* by establishing that the system recovers after a sudden shock if $\tau_{\max} \cdot V_{\text{post}} > B$. Together, these cover the relevant parameter space. We add the following clarifying remark to the text: *the EWMA tracking bound and the recovery proposition are complements, not substitutes; the former governs smooth regimes and the latter governs discontinuous shocks.*

24.3 Technical Issue 6.3: Nonlinear Behavioural Response

Reviewer concern: The stability condition $\varepsilon\tau^* < 1$ assumes a linear behavioural response. Threshold effects (e.g., firms restructure supply chains only when τ exceeds a critical value) may invalidate the condition globally.

Response: This is a valid extension. If the behavioural response includes a threshold—firms tolerate τ below some $\bar{\tau}$ but restructure sharply above it—the stability landscape changes qualitatively. Specifically:

- For $\tau < \bar{\tau}$: the effective elasticity $\varepsilon \approx 0$ (no restructuring), and the system is trivially stable.

- For $\tau > \bar{\tau}$: the effective elasticity ε is high (potentially $\gg 5$), and the stability condition may fail.

The policy implication is that $\tau_{\max}$ should be set *below* the restructuring threshold $\bar{\tau}$. If firms tolerate cascade burdens of up to 5% before restructuring (a reasonable estimate given that firms currently tolerate VAT burdens of 10–25%), then $\bar{\tau} = 5\%/m$, or $\bar{\tau} \approx 1\%$ at $m = 5$. Setting $\tau_{\max} = 2\%$ provides a safety margin for $m \leq 2.5$, and the Phase 1 calibration period empirically identifies where the threshold lies. We identify a phase-plane analysis of the nonlinear dynamics as a priority extension.

24.4 Technical Issue 6.4: Fiscal Savings Uncertainty

Reviewer concern: The \$425–565B savings estimate mixes verifiable items with speculative ones. Attributing 40% of incarceration to poverty involves compounded assumptions.

Response: The reviewer is correct that the estimates span a wide range of confidence levels. We revise the presentation to clearly stratify the estimates:

High-confidence savings (directly verifiable from government budgets): IRS operating budget (\$12–14B), state tax agency budgets (\$15–20B), unemployment insurance administration (\$3–5B). *Total: \$30–39B.*

Medium-confidence savings (estimated from well-documented surveys): private-sector compliance costs (\$183B, from National Taxpayer Advocate's 6.1B hours estimate), welfare means-testing administration (\$50–80B, from program-by-program estimates). *Total: \$233–263B.*

Lower-confidence savings (requiring causal assumptions about poverty-crime links and healthcare access): poverty-related incarceration (\$32B), poverty-related policing (\$40–60B), reduced emergency healthcare (\$20–40B). *Total: \$92–132B.*

The total range of \$355–434B (high + medium confidence only) to \$447–566B (including lower-confidence items) is transparently stratified by reliability. Even the high-confidence estimate alone (\$30–39B) exceeds the annual cost of operating the AUTT infrastructure.

24.5 Specific Question 1: What if $V_T = \$150\text{T}$?

Addressed in Section 23, Table 5. The system remains operational at $\tau = 4.07\%$, with cascade burden at $m = 3$ of 12.2%—below the combined current burden of ∼26%. The stability condition still holds ($\varepsilon\tau^* = 0.20$). Phase 1 calibration resolves the empirical uncertainty.

24.6 Specific Question 2: Exempt-Sector Revenue Fraction

Reviewer question: What fraction of total tax revenue currently comes from entities that would fall into the exempt sector?

Response: Under the current system, the financial sector contributes approximately 20–25% of total corporate income tax revenue (large banks, brokerages, asset managers, insurance companies). Under the AUTT, these entities are in the exempt sector and continue to file conventional income and capital gains reports. The AUTT's claim of "eliminating the enforcement problem" applies to the 75–80% of revenue currently collected from non-financial entities (corporations, individuals, sole traders) who constitute the vast majority of the compliance burden and evasion risk. The exempt sector's conventional reporting is manageable because: (a) the number of entities is small ($\sim 10^3$ vs. $\sim 10^8$), (b) their books are already audited by commercial auditors, (c) their operations are already subject to continuous prudential supervision, and (d) their incentive to evade is reduced (the AUTT eliminates the corporate tax advantage of offshore structuring for non-financial operations).

We acknowledge this explicitly: the AUTT does not eliminate entity-level taxation entirely; it reduces the entity-level system from $\sim 10^8$ participants to $\sim 10^3$ participants, a reduction of five orders of magnitude.

24.7 Specific Question 3: Student Debt and Education in the Fiscal Model

Reviewer question: How are student debt cancellation ($1.7T) and universal tuition ($600–700B/year) incorporated into the fiscal sustainability condition?

Response: Student debt cancellation is a one-time expenditure absorbed into the transition budget (C_{trans} in Definition 3.5), spread over the Phase 1–3 transition period (3–4 years), yielding approximately $425–570B per year during transition. This is funded from: (a) the Sovereign Reserve Fund accumulated during Phase 1, (b) the $425–565B in annual savings from eliminated programmes (Section 19), and (c) a temporary increase in τ during the transition period.

Universal tuition is incorporated into the ongoing budget target B. Net additional cost above current public education spending is estimated at $200–300B (Section 21), requiring a rate increase of 0.05–0.07% on the base case volume. The fiscal sustainability condition (Theorem 10.1) holds with the expanded B as long as $\tau_{\max} \cdot V_T > B_{\text{expanded}}$—which at $\tau_{\max} = 2\%$ and $V_T = \$450\text{T}$ yields maximum revenue of $9T against an expanded budget of approximately $6.4–6.7T. The surplus margin remains substantial.

24.8 Specific Question 4: Constitutional Permissibility

Reviewer question: Has constitutional analysis been conducted?

Response: No formal constitutional analysis is included in this paper, and we acknowledge this as an important gap requiring jurisdiction-specific legal scholarship. However, we note:

In Canada, the federal government's taxing power under s. 91(3) of the *Constitution Act, 1867* is plenary—the federal Parliament can levy "by any mode or system of taxation." An algorithmic rate controller operating within legislatively set bounds ($\tau_{\max}$) is analogous to delegated rate-setting authority already exercised by central banks (interest rate decisions are made by unelected officials under legislated mandates without requiring parliamentary approval for each change).

In the United States, Article I, Section 8 grants Congress the power to "lay and collect Taxes, Duties, Imposts, and Excises." Delegation of rate-setting within legislated bounds has extensive precedent (the Federal Reserve's authority over interest rates, the Treasury's authority over tariff adjustments). The AUTT controller is functionally identical: Congress sets $\tau_{\max}$ and B; the algorithm determines $\tau(t)$ within those bounds.

A detailed constitutional analysis for each target jurisdiction is identified as a priority companion paper.

24.9 Specific Question 5: Migration-Induced UBI Claims

Reviewer question: Has the fiscal impact of migration-induced UBI claims been modelled?

Response: Section 15(a) of the draft bill addresses this directly with the remittance capture argument: every migrant living and transacting within the jurisdiction generates AUTT revenue through daily economic activity. A migrant earning \$3,000/month and remitting \$1,000/month generates $\sim$\$20/month in AUTT from the remittance alone, plus cascade revenue on all domestic transactions. The cost of UBI at \$2,000/month exceeds this direct revenue, but the fiscal analysis must also include: (a) the employer's transactions paying the migrant's salary, (b) the migrant's consumption within the jurisdiction, (c) the landlord's receipt of rent, and (d) all downstream cascade effects.

A back-of-envelope calculation: a migrant with \$36,000 annual income generates approximately \$36,000 $\times$ cascade multiplier $\times$ τ in AUTT revenue. At $m = 5$ (the migrant's salary cascades through employer $\rightarrow$ migrant $\rightarrow$ landlord $\rightarrow$ grocer $\rightarrow$ supplier...) and $\tau = 0.5\%$, this is approximately \$900 in AUTT revenue against \$24,000 in UBI—a net fiscal cost of $\sim$\$23,100 per migrant claiming UBI.

This is a legitimate fiscal concern. However, three factors mitigate it: (a) the request-based mechanism means not all migrants will claim UBI—those with adequate income will not; (b) the 6-month residency requirement provides a buffer; (c) the alternative—driving migrants into cash channels—loses the $900+ in transaction revenue entirely, gains zero fiscal savings (since the migrant remains physically present and consuming public goods), and creates a shadow economy that undermines the AUTT's universality. We identify a formal migration-AUTT dynamic model as a necessary extension, and note that the Phase 1 calibration period provides the empirical data to calibrate such a model.

24.10 On Tone, Self-Citation, and Scope

Two reviewers have commented on the paper's advocacy tone, self-citation practices, and the breadth of scope. We respond:

Tone. We accept that some rhetorical passages should be moderated. However, we note that the paper is explicitly positioned as a *policy proposal*, not as a value-neutral empirical study. Policy proposals are, by nature, arguments for a course of action. We have nonetheless revised several passages to let the formal results carry more of the argumentative weight.

Self-citation. References to Kriger (2019) are substantive: that work contains the original formulation of the transaction-tax-as-universal-revenue concept that this paper formalises. The stadium analogy, while informal, is a pedagogically effective illustration that has been widely cited in policy discussions. We retain these citations but have reduced extended quotation.

Scope. We respectfully disagree with the suggestion to split the paper into two or three focused publications. The AUTT and UBI are a *coupled system*; the formal model, distributional analysis, and institutional design are interdependent. Separating them would obscure the central insight that the revenue mechanism and the expenditure mechanism must be designed together. The draft bill is included precisely because it demonstrates that the proposal has been developed to the level of institutional specificity—a feature that distinguishes this work from purely theoretical exercises. We note that major policy-relevant works in economics (e.g., Piketty 2014, at 685 pages; Atkinson 2015, at 384 pages) have not been criticised for comprehensive scope.

The Draft Bill. One reviewer suggests the draft bill "sits uneasily in an academic economics paper." We argue the opposite: the absence of legislative specificity is the single greatest weakness of most academic policy proposals. By including the bill, we invite criticism of implementation details—exactly the kind of criticism that improves policy design. The bill is in the Appendix; readers uninterested in legislative text can skip it without loss of the formal analysis.

25 Anticipated Objections and Rebuttals

The AUTT-UBI proposal challenges deeply entrenched institutional, professional, and ideological interests. This section catalogues the principal objections raised in academic review, policy discussion, and public debate, and provides detailed responses grounded in the formal model and empirical evidence presented in the preceding sections.

25.1 "A transaction tax will destroy financial markets and liquidity"

Objection. Financial markets depend on high-frequency trading and market-making to provide liquidity. Even a small levy on every transaction would render these strategies unprofitable, causing bid-ask spreads to widen and capital markets to seize up.

Response. The exemption-account mechanism (Section 8) removes HFT, market-making, interbank flows, repo rollovers, and derivatives settlements from the automatic levy entirely. These participants opt into traditional reporting instead. The AUTT taxes real-economy transactions—payroll, rent, purchases, B2B payments—which have near-zero elasticity to a sub-percent levy.

The empirical evidence supports this. The UK has operated a 0.5% Stamp Duty Reserve Tax on equity transactions since 1986; the London Stock Exchange remains one of the world's most liquid. France introduced a 0.3% FTT on large-cap equities in 2012; the Paris Bourse continues to function. And these existing taxes apply *without* an exemption mechanism—the AUTT's exemption accounts provide more flexible accommodation than any FTT currently in operation.

25.2 "The cascade effect makes this a hidden tax of 5–10%, not 0.5%"

Objection. Because every intermediate transaction in a supply chain is taxed, a consumer good passing through five or ten hands accumulates an effective tax burden many times the nominal rate.

Response. The cascade analysis (Section 6) addresses this directly and does not conceal the result. At $\tau = 0.5\%$ and $m = 5$, the effective burden is 2.5%.

But the relevant comparison is not with zero—it is with the system the AUTT replaces. A Canadian consumer currently pays: federal income tax (15–33%), provincial income tax (5–25%), GST (5%), provincial sales tax (0–10%), employer payroll taxes embedded in prices ($\sim$7%), and corporate income tax passed through in product prices (2–5%). The total effective burden on a middle-income household is 35–45% of gross income.

The AUTT cascade at 2.5% effective rate replaces *all of this*. Furthermore, essential goods with short supply chains (local food, utilities, housing) have low m values (2–3), producing effective rates of 1.0–1.5%. Luxury goods with complex global chains bear somewhat higher effective rates—introducing a mild, automatic progressivity into the consumption-side incidence.

25.3 "This is regressive—the poor spend more and therefore pay more"

Objection. A flat-rate transaction tax is functionally a consumption tax. Low-income households spend nearly 100% of their income, while wealthy households save and invest a significant fraction.

Response. This objection would be decisive if the AUTT operated in isolation. It does not. The distributional incidence analysis (Section 17, Table 2, Figure 6) demonstrates that the combined AUTT-UBI system is **strongly progressive**: the lowest decile receives a net transfer of +159% of income, while the highest decile's net contribution is less than 0.4% of income. The UBI's strong progressivity overwhelms the AUTT's mild regressivity. The two components cannot be evaluated separately; they are a coupled system designed to be evaluated jointly.

25.4 "People will switch to cash, barter, or cryptocurrency"

Objection. If every electronic transaction is taxed, rational agents will move to untaxed channels: physical cash, barter, or cryptocurrency.

Response. The objection rests on a cost-benefit calculation that does not survive scrutiny.

Cash. To avoid a $0.50 levy on a $100 purchase, a consumer must withdraw cash (itself taxed), carry physical currency, forgo credit card rewards and buyer protections, and accept the inconvenience and security risk. Cash is already declining globally (under 1% of transactions in Sweden, under 10% in Canada and the UK). The secular forces driving digital payment adoption overwhelm a sub-percent levy.

Barter. Requires a double coincidence of wants, is extraordinarily inefficient, and has never constituted a meaningful fraction of economic activity in any monetised economy.

Cryptocurrency. Every economically significant transaction eventually requires conversion to fiat currency—at a regulated exchange or payment processor—and that conversion is taxable. Moreover, the dormancy levy (Section 15) structurally disadvantages crypto: fiat in a bank account costs 0% for active holders, while crypto carries volatility, exchange hacks, lost keys, and regulatory uncertainty—with zero compensating fiscal advantage under the AUTT.

The magnitude of incentive. Under entity-based taxation, moving $1 billion offshore saves $150–300 million—justifying complex legal structures. Under the AUTT, the same move saves \$3–5 million (0.3–0.5%) while losing access to the entire regulated banking system. The incentive to evade is reduced by two orders of magnitude.

25.5 "Governments will use this to expand spending without limit"

Objection. A 0.1% rate increase sounds trivial, but on a \$200 trillion transaction base it generates \$200 billion. The result is a ratchet effect.

Response. Three structural safeguards address this:

First, the constitutional ceiling $\tau_{\max}$ requires a supermajority to change—a harder constraint than exists for any current tax rate.

Second, the AUTT rate is **publicly visible in real time**. Unlike the current system where deficit financing obscures the connection between spending and taxation, under the AUTT every budget increase immediately and visibly translates into a rate increase that every citizen can observe.

Third, the Sovereign Reserve Fund is governed by direct democratic vote (electronic referendum). Surplus funds cannot be appropriated by the legislature without a public vote.

The comparative question is decisive: does the current system prevent fiscal expansion? Manifestly not. Government budgets have expanded in every democracy, under every tax architecture, through deficit financing and bracket creep. The AUTT makes the cost of spending *transparent and immediate*—strictly better than the current system's opacity.

25.6 "You cannot eliminate income tax—it serves a redistributive function"

Objection. Progressive income taxation is the primary mechanism through which modern democracies redistribute wealth. A flat transaction tax cannot replicate its redistributive precision.

Response. The premise is empirically false. As documented in Section 13, the effective tax rates paid by the wealthiest individuals are far below statutory rates, often below those paid by middle-income workers. Piketty (2014), Saez and Zucman (2019), and the Le Monde joint letter by seven Nobel laureates all demonstrate that the income tax system's nominal progressivity is systematically defeated by legal avoidance and jurisdictional arbitrage. The world's billionaires pay effective rates of 0–0.5% of their wealth.

The entity-based progressive tax system is progressive on paper and regressive in practice. The AUTT-UBI system is the reverse: it appears flat but is progressive in effect (Section 17), because the UBI provides a fixed transfer that is transformative for low earners and negligible for high earners. A flat levy plus a universal transfer is mathematically equivalent to a progressive net tax—the standard result in optimal tax theory (Mankiw, 2021).

25.7 “The UBI will destroy the incentive to work”

Objection. If every citizen receives \$24,000/year for doing nothing, many will choose not to work, causing a fiscal death spiral.

Response. This objection has been tested empirically more extensively than almost any claim in social science, and the evidence consistently refutes it.

The Canadian CERB experience (Section 10) provided \$2,000/month to 8.9 million Canadians. Most recipients returned to work as restrictions lifted. Finland’s UBI experiment (2017–2018) found participants *more likely* to be employed than the control group. The GiveDirectly programme in Kenya found no reduction in labour supply and significant increases in entrepreneurship (Banerjee and Duflo, 2019).

\$24,000/year provides subsistence, not comfort. It does not fund vacations, home ownership, education, or entertainment. The marginal incentive to earn remains strong.

Moreover, under the current system, a minimum-wage worker who earns an additional dollar may lose \$0.30–0.50 in combined clawbacks (reduced benefits, increased taxes). Under the AUTT-UBI, there is *no clawback*: UBI is not means-tested, so every additional dollar earned is a dollar kept (minus 0.5%). The effective marginal tax rate on low earners drops from 30–50% to approximately 0.5%. The AUTT-UBI system *increases* the incentive to work for precisely those currently most discouraged.

Finally: the objection assumes human labour is necessary for economic production. As automation eliminates jobs across sectors, the relevant question is not “will people still work?” but “what happens to people who *cannot* work because their jobs no longer exist?”

25.8 “This has never been tried—you cannot implement something this radical”

Objection. No country has replaced its entire tax system with a transaction levy. The risks of failure are catastrophic.

Response. The draft bill’s phased introduction (Appendix A, Part III) is designed precisely for this concern. Phase 1 (Year 1): AUTT operates at 0.1% alongside all existing taxes, which remain in full force. Revenue is collected into a reserve fund. At the end of Phase 1, a comprehensive report is published. If the numbers do not work—if the required rate exceeds $\tau_{\max}$—the transition is halted with no fiscal damage.

The reversibility clause (§13) explicitly provides that Parliament can repeal the Act and re-enact the prior system within 12 months.

This is more cautious than many historical fiscal reforms. VAT was introduced in France in 1954 and is now used by 170+ countries. Income tax was introduced in Britain in 1799 as a “temporary” wartime measure. The GST in Canada was introduced in 1991 amid fierce opposition and is now uncontroversial infrastructure. The AUTT follows the same trajectory: phased introduction, empirical calibration, and built-in reversibility.

25.9 "Transaction volume numbers are inflated—most is financial plumbing"

Objection. Much of the cited volume is financial infrastructure: the same payment settling through multiple clearing systems, repo rollovers, and netting arrangements. The "real" taxable base may be far smaller.

Response. The exemption boundary analysis (Section 8) addresses this directly. After exempting interbank flows, repo, derivatives, and exchange-based trading (~75–85% of gross volume), the remaining taxed volume for the US is ~\$450 trillion—still roughly 16× GDP. This residual consists of ACH transfers (~\$80T), card payments (~\$10T), wire transfers by non-financial entities (~\$50T), and other domestic flows.

Critically, the AUTT is levied at the point of *account settlement*, not at each clearing message. A single payment from Company A to Company B generates one taxable event, regardless of how many interbank messages are involved.

25.10 "Taxing all transactions penalises savings and investment"

Objection. Saving—which is economically beneficial—generates additional intermediation transactions that are taxed, effectively penalising thrift.

Response. Under the current system, savings are also taxed: interest income is taxed as ordinary income, capital gains upon realisation, and investment returns are double- or triple-taxed (corporate profit tax, dividend tax, capital gains tax). The combined effective tax on a dollar saved and invested often exceeds 30–50%.

Under the AUTT, a dollar deposited incurs 0.5% (\$0.005). When lent, the borrower's receipt incurs 0.5%. When spent, 0.5%. Total burden on the save-lend-spend cycle: ~1.5%—compared to 30–50% under the current system.

For long-term buy-and-hold investments: one levy at purchase, one at sale, dividends levied as received. The total lifetime AUTT burden is ~1–2%, compared to 15–30% under current capital gains regimes. The AUTT *dramatically reduces* the tax penalty on savings.

25.11 "Authoritarian governments could abuse an automated revenue system"

Objection. A system that collects revenue automatically removes an important check on government power. Tax resistance is a form of political protest; automated collection eliminates fiscal disobedience.

Response. Tax resistance has never been an effective check on government power in any modern democracy. The "freedom" to resist entity-based taxation is, for most citizens, the freedom to be punished: the 63% imprisonment rate for convicted US tax evaders is not a check on government—it is a demonstration of coercive state power.

The AUTT actually provides a *more effective* check on spending. Under the current system, governments expand spending through deficit financing with no immediate visible impact. Under the AUTT, every dollar of spending must be collected in real time from current transaction volume. The rate is published in real time; there is no mechanism for deficit financing. If the government wants to spend more, $\tau(t)$ rises visibly and immediately.

The constitutional ceiling $\tau_{\max}$, modifiable only by supermajority, provides a hard upper bound on extractive capacity. No such binding constraint exists in most current systems.

25.12 "The transition will cause economic chaos"

Objection. Eliminating income, corporate, payroll, and sales tax simultaneously would require every business to restructure.

Response. The transition is not simultaneous. The three-year phased implementation proceeds: Year 1 (no changes, AUTT runs in parallel at 0.1%); Year 2 (existing rates reduced by 50%—a tax *cut*); Year 3 (remaining taxes eliminated—the requirement to file returns is eliminated, a further burden reduction). The AUTT rate adjustments are handled entirely by financial institutions—no business or individual needs to do anything.

The transition *simplifies* rather than complicates. Every business currently maintains a tax compliance function that is entirely eliminated. The transition cost is not building a new system; it is decommissioning an old one. The compensation programme (Part VI of the draft bill, including Section 19A for private-sector accountants) addresses the human cost directly.

25.13 "International tax treaties would be violated"

Objection. Bilateral tax treaties govern double taxation, information exchange, and residency. Unilateral replacement of entity-based taxation with a transaction levy would violate these agreements.

Response. Tax treaties govern entity-based taxes: they allocate rights over income, capital gains, and wealth between jurisdictions. The AUTT is not an entity-based tax. It is a levy on the movement of money through *domestic financial infrastructure*, assessed on the financial institution (Definition 3.2), not on the transacting parties.

The legal analogy is closer to a payment processing fee or financial regulatory levy than to an income tax. Just as no tax treaty governs Visa's interchange fees or central clearing settlement charges, the AUTT—as a levy on domestic settlement—falls outside the scope of existing treaties.

The competitive advantage analysis (Section 9.4) suggests the AUTT jurisdiction's negotiating position would be strong: zero corporate tax and UBI-backed consumer demand create powerful incentives for trading partners to harmonise.

25.14 "You are abolishing the government's most powerful tool of social engineering"

Objection. Tax credits for renewable energy, deductions for charitable donations, preferential rates for small businesses, R&D incentives—these are instruments of industrial and social policy. The AUTT eliminates all such levers.

Response. We acknowledge the tradeoff candidly. The AUTT eliminates tax-based social engineering. Governments must incentivise behaviours through *direct expenditure* (grants, subsidies, procurement) rather than tax preferences. We argue this is a feature:

First, tax expenditures are among the least transparent forms of government spending. A \$10B R&D tax credit appears as "forgone revenue" in a supplementary document, not on any spending ledger. Converting preferences to direct expenditure forces every programme into the open—subject to scrutiny, debate, and annual review.

Second, tax-based incentives are disproportionately captured by sophisticated actors. The charitable deduction benefits high-income donors who itemise; the R&D credit is claimed overwhelmingly by large corporations with dedicated tax departments. Direct grants allow better targeting.

Third, the administrative complexity of maintaining hundreds of preferences—the Canadian Income Tax Act runs to $\sim$3,400 pages; the US Internal Revenue Code exceeds 10,000—is itself a massive cost. Eliminating them eliminates the complexity, the compliance cost, and the arbitrage opportunities.

Excise duties (tobacco, alcohol, carbon) are explicitly preserved under the AUTT (Part IX of the draft bill). The government retains the ability to price externalities; what it loses is the ability to use the income tax code as a covert subsidy mechanism.

25.15 "Central banks and monetary policy would be disrupted"

Objection. Central bank open market operations involve enormous transaction volumes. Taxing them increases the cost of monetary policy; exempting them expands the exemption boundary.

Response. Central bank operations are conducted through interbank settlement systems (Fedwire, TARGET2, LVTS) that are already in the exempt sector under the two-sector decomposition (Section 8). This is not an ad hoc carve-out; it is a natural consequence of the exemption boundary principle.

The AUTT does not interfere with monetary policy transmission. A 0.5% transaction levy on a \$2,000 mortgage payment adds \$10—negligible compared to the hundreds of dollars of variation produced by a 1% change in the policy rate.

25.16 "A universal transaction tax requires surveillance of all transactions"

Objection. Universal taxation implies universal monitoring—a comprehensive surveillance infrastructure.

Response. This rests on a critical misunderstanding of the AUTT architecture. The system does not require identification of transacting parties. The levy is assessed on the financial institution (Definition 3.2), not on the individual. The institution's software diverts a fraction of every transaction to the government account without reporting who made the transaction.

The Revenue Aggregation System (RAS) receives only *cumulative revenue totals* from each institution. It does not receive transaction-level data. The government knows how much was collected from each bank this hour; it does not know that Alice paid Bob \$500 for rent.

The AUTT actually *reduces* government surveillance. Under entity-based taxation, the government must know every individual's income, expenses, deductions, and capital transactions. The CRA and IRS already possess far more detailed financial information about citizens than the AUTT would ever require. The draft bill (§7(c)) explicitly provides: "No identification of the transacting parties is required."

25.17 "The constitutional ceiling will eventually be raised"

Objection. Supermajority requirements sound reassuring, but constitutions are amended. Political crises create pressure for "temporary" measures that become permanent.

Response. No institutional design can provide absolute guarantees against future democratic decisions to change the rules. However, three features make $\tau_{\max}$ more robust:

The ceiling has a *direct, visible, real-time consequence*: raising it from 2% to 3% means every citizen can instantly observe a 50% increase in the per-transaction levy.

The reserve fund mechanism provides an alternative during crises, removing the most likely trigger for emergency increases.

The emergency provision (Part X of the draft bill) limits temporary increases to 1.0 percentage point, for a maximum of 12 months, renewable only by fresh legislative vote.

The comparative answer: under the current system, there is no hard ceiling on income tax rates, no real-time visibility, and no supermajority requirement. The AUTT provides strictly stronger safeguards.

25.18 "What happens during a prolonged depression?"

Objection. In a severe depression comparable to 1929–1933 (GDP fell 46%), transaction volume could collapse by 50%+. The adaptive rate would double, potentially triggering a spiral.

Response. If volume falls 50% permanently, the rate doubles: from 0.5% to 1.0%, or from 1.0% to 2.0% ($\tau_{\max}$). At that point, the Sovereign Reserve Fund covers the shortfall.

If the annual budget is $1T and $\tau_{\max}$ revenue covers 80%, the annual shortfall is $200B. A reserve fund equal to one year's budget sustains this for five years—longer than any depression in recorded history.

The endogenous volume-rate spiral analysis (Section 18, Proposition 18.1) formally demonstrates that the spiral is self-limiting at operational rates: for $\tau^* = 0.5\%$ and $\varepsilon = 5$, $\varepsilon\tau^* = 0.025 \ll 1$.

The countercyclical comparison is essential: under entity-based taxation, a 50% GDP decline produces a 40–60% revenue decline combined with increased social spending demand. Under the AUTT, the rate automatically rises, maintaining revenue without legislative intervention. The AUTT is not immune to depressions but is substantially more resilient.

25.19 "This is motivated reasoning for eliminating taxes on the wealthy"

Objection. A flat transaction tax that eliminates income, corporate, and capital gains tax is a massive tax cut for the rich dressed in progressive language.

Response. Under the current system, the wealthiest 0.1% pay effective rates often *lower* than middle-income workers—not because statutory rates are low, but because entity-based architecture provides unlimited avoidance opportunities (Zucman, 2015; Saez and Zucman, 2019). The top 400 American families pay a lower effective rate than the bottom 50%.

The AUTT replaces a system that *nominally* taxes the wealthy at high rates but *actually* collects very little, with one that taxes at a low rate but collects *unavoidably* on every transaction. A billionaire moving $500M through the financial system pays ~$2.5M in AUTT—likely more than their actual (not nominal) income tax payment after avoidance.

The AUTT taxes *wealth in motion*, not just income. Under the current system, a billionaire whose wealth appreciates by $1B pays zero tax until sale—and may never sell (borrow-buy-die strategy). Under the AUTT, every investment, loan drawdown, and purchase generates revenue automatically.

25.20 "The model has gaps—the paper is not ready for implementation"

Objection. Peer reviewers have identified gaps: the cascade multiplier is treated as exogenous, the volume-rate spiral needs bounding, and multi-country calibrations are rough.

Response. We agree with every specific technical criticism. The V2 revision addressed five of the eight concerns raised: cascade model (Section 6), EWMA stress test (Section 5), exemption boundary (Section 8), distributional incidence (Section 17), and endogenous volume-rate spiral (Section 18). The remaining extensions—endogenous cascade length, two-country competitive dynamics, and transition agent-based simulation—are active research targets.

However, the demand for theoretical completeness before policy action reflects a misunderstanding of how fiscal reform has always proceeded. Income tax was introduced in 1799 without a formal model of labour supply elasticity. VAT was introduced in 1954 without a general equilibrium cascade analysis. The CERB was deployed in 72 hours without a randomised controlled trial.

The AUTT's Phase 1 (0.1% alongside full existing taxes) is precisely the mechanism for closing empirical gaps. It is a national-scale experiment that generates the data needed for rigorous calibration—with zero risk to existing revenue. The model is ready for Phase 1. Whether it is ready for full implementation depends on the Phase 1 results. This is the honest and prudent answer.

26 Limitations and Extensions

International coordination. The AUTT is defined within a single jurisdiction. Cross-border transactions raise questions about double taxation and regulatory arbitrage. A multilateral AUTT framework would require international agreements, though unilateral adoption by a single country is feasible and would generate competitive advantages (elimination of corporate tax avoidance, attraction of repatriated offshore capital). A two-country game-theoretic analysis of competitive adoption dynamics is a natural extension of the present work.

Transition dynamics. The model's first-year budget calibration (Definition 3.5) includes a transition cost term C_{trans}, but does not model the full dynamics of the 2–3 year transition period during which both systems operate in parallel. During this period, some agents may shift activity toward cash or cryptocurrency to avoid the AUTT before the income tax is fully phased out. The speed advantage of the AUTT (real-time revenue versus 18-month lag for income tax) mitigates this, but a full agent-based simulation of the transition is needed to quantify the magnitude and duration of any revenue gap.

Cryptocurrency. Fully on-chain transactions that never touch the fiat banking system escape the AUTT. However, all economically meaningful transactions eventually require conversion to fiat currency for the purchase of real goods and services, at which point the levy is applied. A purely crypto-internal economy of significant scale does not currently exist and is unlikely to emerge, given the superior liquidity and legal enforceability of fiat systems.

Municipal taxation. The model explicitly excludes municipal-level taxes (property tax, local levies), which serve distinct regulatory and local-governance functions and are best retained in their current form.

Excise duties. Excise taxes on specific goods (tobacco, alcohol, carbon emissions) serve regulatory rather than fiscal purposes—they are instruments of public health and environmental policy embedded in product pricing. These are compatible with the AUTT and need not be altered.

Informational asymmetry in the exempt sector. The exemption-account mechanism reintroduces entity-level reporting for sophisticated financial actors. This is a deliberate and bounded compromise: the number of exempt entities is approximately two to three orders of magnitude smaller than the current universe of taxpayers, and their reporting obligations are narrower (income and capital gains only, not comprehensive activity-level declarations). The residual asymmetry is manageable.

Empirical calibration. The numerical examples in this paper (Sections 6, 8, and 10) use publicly available aggregate data and order-of-magnitude estimates. A rigorous calibration using transaction-level data from central banks and payment system operators would substantially strengthen the quantitative claims. We identify this as the highest-priority extension of the present work.

27 Conclusion

We have presented a formal model of an Adaptive Universal Transaction Tax (AUTT) as a replacement for entity-based federal and provincial/state taxation, and shown that:

(i) The adaptive feedback control rule guarantees convergence to any democratically approved budget target, under arbitrary fluctuations in transaction volume (Theorem 4.1).

(ii) The system is inherently countercyclical, automatically compensating for economic downturns without legislative intervention (Proposition 4.2).

(iii) The EWMA volume estimator provides bounded tracking error under realistic volatility conditions, and the controller recovers from sudden crashes provided the constitutional rate ceiling has adequate headroom (Section 5).

(iv) The transaction chain cascade effect produces effective tax rates of 0.9–2.5% on final goods at operational nominal rates—below existing consumption tax rates in all major economies (Section 6).

(v) The exemption boundary between taxed and exempt sectors can be set to protect rate-sensitive financial activity while keeping the rate on the real-economy sector at approximately 0.5–1.0% (Section 8).

(vi) The fiscal surplus enables a request-based Universal Basic Income that replaces fragmented welfare systems, with sustainability guaranteed under explicit and verifiable conditions (Theorem 10.1).

The model is situated within the broader framework of civilisational decentralisation: the transition from concentrated, fragile, high-dependency urban systems to distributed, autonomous, and technologically resilient units (Kriger, 2023). The AUTT-UBI system is the fiscal architecture that makes this transition economically viable.

The technological infrastructure for automatic transaction-level taxation already exists in every modern banking system. The mathematical properties of the feedback control rule are straightforward and provably convergent. The fiscal arithmetic is favourable. What remains is political will—and the recognition that a tax system whose fundamental architecture has not changed in six millennia may not be adequate for a civilisation entering the age of autonomous machines.

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A Model Draft Bill: The Adaptive Fiscal Reform Act

AN ACT

to establish an Adaptive Universal Transaction Tax, to provide for the gradual replacement of entity-based federal and provincial taxation, to institute a Universal Basic Income entitlement, to declare a fiscal amnesty for the repatriation of offshore capital, and for related purposes.

— MODEL LEGISLATIVE TEXT —

(Jurisdiction-neutral draft. Square brackets indicate parameters to be calibrated for specific jurisdictions.)

PART I — SHORT TITLE, INTERPRETATION, AND PURPOSE

1. Short title. This Act may be cited as the *Adaptive Fiscal Reform Act.*

2. Definitions. In this Act:

(a) "AUTT" means the Adaptive Universal Transaction Tax established under Part II;

(b) "collection account" means the government account maintained by a financial institution under section 7;

(c) "constitutional ceiling" means the maximum AUTT rate established under section 10;

(d) "controller" means the algorithmic system that determines the AUTT rate under section 11;

(e) "exempt account" means an account designated under section 15;

(f) "financial institution" means any bank, credit union, brokerage, payment processor, money transfer operator, or other entity licensed to process financial transactions within the jurisdiction;

(g) "fiscal year" means the twelve-month period beginning [April 1 / January 1];

(h) "Revenue Aggregation System" or "RAS" means the centralised reporting system established under section 8;

(i) "transaction" means any transfer of monetary value between accounts, including but not limited to deposits, withdrawals, wire transfers, card payments, electronic fund transfers, and cryptocurrency-to-fiat conversions;

(j) "UBI" means the Universal Basic Income established under Part V.

3. Purpose. The purpose of this Act is to:

(a) replace entity-based federal and [provincial/state] taxation with an automatic, low-rate levy on all financial transactions;

(b) eliminate the requirement for individual and corporate tax declarations at the federal and [provincial/state] level;

(c) institute a Universal Basic Income available to all residents by request;

(d) encourage the repatriation of offshore capital through a time-limited fiscal amnesty;

(e) preserve municipal taxation, excise duties, and regulatory levies in their existing form.

PART II — THE ADAPTIVE UNIVERSAL TRANSACTION TAX

4. Establishment. There is hereby established a tax, to be known as the Adaptive Universal Transaction Tax, levied on every financial transaction processed within the jurisdiction.

5. Tax base. The AUTT applies to the gross monetary value of every transaction processed by a financial institution, without regard to the identity, residency, or tax status of the transacting parties.

6. Rate. The AUTT rate $\tau(t)$ is determined by the controller (section 11) and shall at no time exceed the constitutional ceiling (section 10). The rate is published in real time and applies uniformly to all non-exempt transactions.

7. Collection accounts.

(a) Every financial institution operating within the jurisdiction shall, within [90 days] of the commencement of this Act, establish and maintain a government collection account.

(b) Upon the processing of each transaction, the financial institution shall automatically transfer to the collection account an amount equal to the current AUTT rate multiplied by the transaction value.

(c) No identification of the transacting parties is required for the purpose of this transfer.

(d) **The fiscal obligation under this Act falls exclusively on licensed financial institutions. No individual, corporation, partnership, trust, or other legal entity bears any direct tax liability under this Act.**

7A. Enforcement and licence sanctions.

(a) The [Financial Regulator / Central Bank] shall audit each financial institution's collection mechanism at intervals not exceeding [twelve months].

(b) Any financial institution that fails to implement the automatic diversion, processes transactions through channels that bypass the collection mechanism, or underreports transaction volumes shall be subject to:

(i) A fine not exceeding [10%] of the institution's annual transaction volume processed during the period of non-compliance;

(ii) Immediate suspension of its financial services licence pending investigation;

(iii) Permanent revocation of its financial services licence upon a finding of deliberate non-compliance;

(iv) Criminal prosecution of responsible officers for fraud against the public treasury.

(c) The number of licensed financial intermediaries within any jurisdiction is small (typically 10^2–10^3), and their operations are already subject to continuous regulatory oversight. Enforcement under this Act requires monitoring only this small, already-regulated population—not the 10^7–10^8 individual taxpayers monitored under the former entity-based system.

(d) For certainty: no individual or corporation may be audited, investigated, fined, or prosecuted for any matter arising under this Act. The exclusive enforcement mechanism is the financial licensing regime.

8. Revenue Aggregation System.

(a) The [Minister of Finance / Secretary of the Treasury] shall establish a Revenue Aggregation System (RAS) that receives, in real time, reports of cumulative collections from all collection accounts.

(b) The RAS shall publish, at intervals not exceeding [one hour], the current cumulative revenue collected and the current AUTT rate.

(c) All data published by the RAS shall be freely accessible to the public.

9. Budget target.

(a) At the beginning of each fiscal year, [Parliament / Congress] shall approve a budget target B for that year.

(b) The budget target shall include all federal and [provincial/state] expenditures, including the UBI (Part V), but excluding municipal expenditures and expenditures funded by excise duties.

10. Constitutional ceiling.

(a) The maximum AUTT rate shall be [$\tau_{\max} = 2.0\%$].

(b) The constitutional ceiling may be amended only by [a two-thirds supermajority of both chambers / constitutional amendment procedure].

(c) In the event of a declared national emergency (section 22), a temporary increase not exceeding [$\tau_{\max} + 1.0\%$] may be authorised by [a simple majority of both chambers], effective for a period not exceeding [12 months], and renewable only by the same procedure.

11. Algorithmic controller.

(a) The AUTT rate shall be determined by an algorithmic controller that divides the remaining budget deficit by the projected remaining transaction volume for the fiscal year.

(b) The controller's algorithm shall be specified in regulations, audited annually by [an independent body], and published in full.

(c) The controller shall not be subject to political override between legislative budget votes. The parameters within which the controller operates (the budget target and the constitutional ceiling) may be changed only by legislative action.

(d) If the budget target is met before the end of the fiscal year, the controller shall reduce the rate to [0.01%] and direct all further collections to the Sovereign Reserve Fund (section 20).

PART III — PHASED INTRODUCTION AND YEAR ONE OVERLAP

12. Phased introduction. The introduction of the AUTT shall proceed in three phases:

Phase 1 — Parallel operation and calibration (Year 1).

(a) During the first fiscal year following the commencement of this Act, the AUTT shall operate *in parallel* with all existing federal and [provincial/state] taxes, which shall remain in full force and effect.

(b) The AUTT rate during Phase 1 shall be set at [0.1%]—a calibration rate designed to measure actual transaction volumes, test the collection infrastructure, and accumulate a fiscal reserve, without imposing a material additional burden on the economy.

(c) All revenue collected through the AUTT during Phase 1 shall be deposited in the Sovereign Reserve Fund (section 20) and shall not be used for current expenditure.

(d) At the end of Phase 1, the [Minister of Finance / Secretary of the Treasury] shall publish a comprehensive report containing: (i) the total transaction volume observed; (ii) the total AUTT revenue collected; (iii) a projection of the AUTT rate required to replace all federal and [provincial/state] tax revenue; (iv) an assessment of infrastructure readiness; (v) any recommended amendments to the constitutional ceiling or controller algorithm.

(e) No existing tax obligation shall be reduced, suspended, or eliminated during Phase 1.

Phase 2 — Gradual replacement (Years 2–3).

(a) Beginning in the second fiscal year, existing taxes shall be reduced according to the following schedule:

	Year 2	Year 3
Personal income tax rates	reduced by 50%	eliminated
Corporate income tax rates	reduced by 50%	eliminated
[Provincial/state] sales tax	reduced by 50%	eliminated
Payroll taxes (employer portion)	reduced by 50%	eliminated
Payroll taxes (employee portion)	reduced by 50%	eliminated
Capital gains tax	reduced by 50%	eliminated

(b) The AUTT rate shall be increased from the Phase 1 calibration rate to the level necessary to compensate for the reduced legacy tax revenue, as determined by the controller.

(c) The requirement to file individual and corporate tax returns shall be eliminated simultaneously with the elimination of the corresponding tax.

(d) All provisions for tax audits, penalties, and enforcement related to eliminated taxes shall be repealed simultaneously.

Phase 3 — Full operation (Year 4 onward).

(a) From the fourth fiscal year onward, the AUTT shall be the sole federal and [provincial/state] revenue instrument, subject to the exceptions in section 16.

(b) The UBI (Part V) shall commence full operation.

(c) The compensation programme for displaced workers (Part VI) shall continue through Year 5.

13. Reversibility clause.

(a) If the Phase 1 report (section 12, Phase 1, paragraph (d)) reveals that the required AUTT rate to replace all federal and [provincial/state] taxes would exceed [$\tau_{\max} = 2.0\%$], [Parliament / Congress] may, by simple majority, suspend the transition to Phase 2 and extend Phase 1 for an additional [12 months], during which the constitutional ceiling, the scope of exemptions, or the budget target may be revised.

(b) If, after two extensions, the required rate still exceeds $\tau_{\max}$, [Parliament / Congress] may repeal this Act without prejudice to any accumulated Reserve Fund balances, which shall be applied to the national debt.

PART IV — FISCAL AMNESTY FOR OFFSHORE CAPITAL REPATRIATION

14. Declaration of fiscal amnesty.

(a) A fiscal amnesty is hereby declared for a period of [24 months] beginning on the date of commencement of this Act (the "amnesty window").

(b) During the amnesty window, any person or entity that repatriates funds held outside the jurisdiction—regardless of whether those funds were previously declared to tax authorities—shall be exempt from:

(i) all penalties, interest, and criminal prosecution for prior non-declaration or non-payment of taxes on the repatriated funds;

(ii) all capital gains tax, wealth tax, or other levies on the repatriated amount itself;

(iii) all reporting requirements with respect to the historical origin or prior movement of the repatriated funds.

(c) The sole fiscal obligation on repatriated funds shall be the AUTT, applied automatically at the prevailing rate to any transaction involving the repatriated funds upon their entry into the domestic financial system. No additional levy, surcharge, or penalty shall apply.

(d) The amnesty shall be unconditional. No repatriated funds may be seized, frozen, or subjected to investigation solely on the basis of their repatriation under this section, except as required by [anti-money-laundering statutes / proceeds-of-crime legislation] where there is independent evidence of criminal origin unrelated to tax evasion.

(e) After the expiration of the amnesty window, the amnesty shall cease, and any offshore funds not repatriated shall remain subject to whatever international reporting and enforcement mechanisms are in effect.

Rationale. The amnesty serves two purposes. First, it generates a large one-time influx of capital into the domestic financial system, increasing transaction volume and thereby reducing the required AUTT rate. Second, under the AUTT regime, the incentive to hold funds offshore is permanently eliminated: since there is no income tax, corporate tax, or capital gains tax from which to shelter, and the AUTT is applied automatically at the point of transaction regardless of the holder's identity, offshore concealment provides no fiscal advantage. The amnesty thus clears the historical backlog while the new system eliminates the structural incentive for future evasion.

PART V — UNIVERSAL BASIC INCOME

15. Establishment of UBI.

(a) Every person legally residing in the jurisdiction—regardless of citizenship, immigration status, visa category, or national origin—who has been continuously and legally present for a period of [6 months] shall have the right to receive a monthly Universal Basic Income payment. This includes, without limitation: citizens, permanent residents, temporary workers, refugees, asylum seekers with legal status, and holders of any long-term visa or permit.

(b) The UBI amount shall be [$2,000] per month for adults and [$1,000] per month for residents under the age of [18], payable to the parent or legal guardian.

(c) The UBI amount shall be indexed annually to the [Consumer Price Index / cost-of-living index] and adjusted on [January 1] of each year.

(d) **Rationale for inclusive eligibility.** Every person living and transacting within the jurisdiction contributes to AUTT revenue through their daily economic activity. Excluding categories of residents creates incentives for cash-based shadow economies that reduce transaction volume and thus AUTT revenue. Inclusive eligibility maximises the proportion of the population operating within the regulated financial system.

(e) **Remittance capture.** Migrant workers make substantial cross-border remittance transfers (globally estimated at $650+ billion annually). Under the AUTT, these outbound transfers are subject to the standard levy at the moment of electronic transfer—generating revenue that would otherwise be lost entirely if the same workers were driven into informal cash channels. The fiscal calculus strongly favours inclusion: a migrant worker earning $3,000/month and remitting $1,000/month generates approximately $20/month in AUTT revenue from the remittance alone, plus AUTT on all domestic transactions. Exclusion from UBI saves $2,000/month but risks losing the worker's entire transaction stream to cash—a net fiscal loss. It is therefore in the jurisdiction's direct financial interest to ensure that all residents, including migrants, remain within the electronic payment system.

16. Request-based entitlement.

(a) Receipt of UBI requires a formal request, submitted through [an electronic portal / designated government office].

(b) No means test, employment condition, health assessment, or other qualification shall be required.

(c) A resident may withdraw from UBI at any time and re-request at any time, without penalty or delay.

(d) No record of UBI requests or payments shall be used for any purpose other than administration of the UBI programme. In particular, UBI status shall not be disclosed to employers, creditors, or any third party.

17. Replacement of existing programmes.

(a) Upon full commencement of the UBI (Phase 3, Year 4), the following programmes shall be discontinued: [list jurisdiction-specific programmes: e.g., Old Age Security, Employment Insurance benefits, provincial social assistance, child tax benefit, GST/HST credit, etc.].

(b) Any resident whose combined benefits under discontinued programmes exceeded the UBI amount shall receive a transitional supplement equal to the difference, declining by [25%] per year over [4 years].

(c) Healthcare, education, and disability-specific services shall continue to be funded from the general budget and are not replaced by the UBI.

18. In-kind provision.

(a) The government may, in addition to or in partial substitution for cash UBI payments, provide in-kind benefits including but not limited to: housing, healthcare, education, and public transportation.

(b) Any in-kind provision shall be valued at its market-equivalent cost and deducted from the cash UBI amount only with the explicit consent of the recipient.

PART VI — COMPENSATION FOR DISPLACED WORKERS

19. Compensation programme for government employees.

(a) All employees of federal and [provincial/state] tax collection agencies whose positions are eliminated as a result of this Act shall be entitled to a compensation package as follows:

  (i) Year 1 (Phase 1): no positions eliminated; full employment continues.

  (ii) Year 2: positions reduced by [50%]; displaced workers receive compensation equal to [250%] of their annual salary.

  (iii) Year 3: remaining positions eliminated; displaced workers receive compensation equal to [150%] of their annual salary.

  (iv) Year 4: all previously displaced workers receive a final payment equal to [75%] of their former annual salary.

(b) Compensation payments shall be funded from the savings generated by the elimination of tax administration expenditure.

(c) Displaced workers shall additionally be entitled to [24 months] of retraining support through [designated employment services].

(d) Nothing in this section prevents a displaced worker from simultaneously receiving UBI.

19A. Compensation programme for licensed private-sector tax professionals.

(a) Licensed accountants (CPA, CA, CGA, CMA, and equivalents), enrolled agents, and licensed tax preparers whose practice consisted of [50%] or more tax compliance work in the [24 months] preceding the commencement of this Act shall be entitled to a transition compensation package:

  (i) Year 1–2: annual compensation equal to [150%] of the professional's average gross income from tax compliance work over the preceding three years;

(ii) Year 3: compensation equal to [75%] of the same base.

(b) **Residual demand.** Licensed accountants retain substantial demand for their skills in non-tax functions: financial management, payroll processing, management accounting, business advisory, audit, and forensic accounting. The tax compliance function—which AI was already poised to eliminate within the decade regardless of the AUTT—is the only function displaced. The compensation programme bridges the transition to these remaining roles.

(c) **AI displacement acknowledgment.** Automated tax preparation software and AI-powered accounting tools were, by 2025, already eliminating entry-level tax compliance positions. The AUTT accelerates a transition that was inevitable; the compensation programme converts an uncontrolled displacement into a managed, compensated one—a more humane outcome than the market alternative.

(d) Compensation under this section shall be funded from the AUTT revenue during the transition period and from the fiscal savings identified in Section 19.

(e) Licensed professionals receiving compensation under this section remain eligible for UBI.

PART VI-A — UNIVERSAL SAFETY NET PRINCIPLE

19B. No resident in distress.

(a) **It is a foundational principle of this Act that no resident of the jurisdiction shall be left in a situation of material distress.** The combined provisions of this Act—the Universal Basic Income (Part V), the Universal Insurance Package (Part V-A), the Universal Healthcare Package (Part V-B), the Universal Education Insurance (Part V-C), and the general fiscal capacity generated by the AUTT—shall be administered with the objective that every resident has, at minimum:

(i) Sufficient income to meet basic needs (food, shelter, clothing);

(ii) Access to comprehensive healthcare including dental, vision, hearing, and mental health;

(iii) Access to legal representation;

(iv) Protection against catastrophic loss through insurance;

(v) Access to education and retraining at any age, without cost;

(vi) Protection against economic shocks (job loss, disability, bereavement, natural disaster).

(b) Where any combination of the provisions of this Act fails to prevent a resident from falling into material distress, the [Minister of Social Development / Secretary of Health and Human Services] shall have the authority to provide emergency supplementary support, funded from the general budget or the Sovereign Reserve Fund.

(c) For certainty: the safety net established by this Act is *universal, automatic, and unconditional.* No resident is required to prove need, demonstrate worthiness, or submit to moral evaluation. The only requirement is legal presence in the jurisdiction.

PART VII — SOVEREIGN RESERVE FUND

20. Establishment and governance.

(a) There is hereby established the Sovereign Reserve Fund (the "Fund").

(b) The Fund shall receive: (i) all AUTT revenue collected during Phase 1; (ii) all AUTT revenue collected after the budget target is met in any fiscal year; (iii) the difference between budgeted and actual UBI disbursement in any fiscal year.

(c) The Fund shall be invested in [low-risk sovereign instruments / index funds] as determined by [an independent board of trustees].

(d) Withdrawals from the Fund for any purpose other than automatic deficit coverage (section 20(e)) shall require approval by [a national electronic referendum / a two-thirds majority of both chambers].

(e) If the AUTT controller reaches $\tau_{\max}$ before the budget target is met, the shortfall shall be covered automatically from the Fund without legislative action, up to a maximum of [10%] of the Fund balance in any fiscal year.

PART VIII — EXEMPT ACCOUNTS

21. Designation of exempt accounts.

(a) Any financial institution or registered market participant may apply to designate one or more accounts as "exempt accounts."

(b) Transactions within exempt accounts are not subject to the AUTT.

(c) Holders of exempt accounts remain subject to [traditional income and capital gains taxation / a simplified annual reporting obligation] on net income generated within those accounts.

(d) The [regulatory authority] shall maintain and publish a register of all exempt accounts.

(e) Exempt account status may be revoked for non-compliance with reporting obligations.

PART IX — PRESERVED INSTRUMENTS

22. Municipal taxation. Nothing in this Act affects the authority of municipal governments to levy property taxes, local improvement charges, development fees, or other municipal levies.

23. Excise duties. All existing excise duties on tobacco, alcohol, cannabis, carbon emissions, fuel, and other goods remain in full force and effect. Excise duties may be modified, introduced, or repealed by ordinary legislative process.

24. Customs duties. Import and export duties remain under the authority of [the relevant federal agency] and are not affected by this Act.

PART X — EMERGENCY PROVISIONS

25. National emergency.

(a) In the event of a declared national emergency (war, pandemic, natural disaster of national scope), [Parliament / Congress] may by [simple majority] authorise a temporary increase of the constitutional ceiling by up to [1.0 percentage point] for a period not exceeding [12 months].

(b) Such authorisation is renewable, but each renewal requires a new vote.

(c) Upon the expiration or revocation of the emergency declaration, the constitutional ceiling shall revert to its pre-emergency level.

(d) All additional revenue collected under the emergency provision shall be accounted for separately and reported to the public.

PART XI — GENERAL PROVISIONS

26. Commencement. This Act comes into force on [date], subject to the phased introduction schedule in Part III.

27. Regulations. The [Governor in Council / relevant executive authority] may make regulations necessary for the implementation of this Act, including but not limited to: the specification of the controller algorithm, the RAS technical standards, the amnesty application procedure, and the UBI request mechanism.

28. Review. This Act shall be reviewed by [a parliamentary committee] at the end of Phase 1, at the end of Phase 2, and every [5 years] thereafter, with particular attention to: (i) the adequacy of the constitutional ceiling; (ii) the take-up rate and fiscal impact of the UBI; (iii) the balance of the Sovereign Reserve Fund; (iv) the scope and effectiveness of the exempt account regime.

29. Sunset clause. If [Parliament / Congress] determines, at any review under section 28, that the AUTT system is not achieving its stated purposes, it may repeal this Act by [simple majority]. Upon repeal, the Sovereign Reserve Fund balance shall be applied to the national debt, and existing tax legislation shall be re-enacted in its pre-reform form within [12 months].

— *End of Model Draft Bill* —

B Interactive AUTT-UBI Fiscal Calculator

Appendix B: Online Interactive Calculator

An interactive HTML calculator implementing the formal model of this paper is available online at:

https://boriskriger.github.io/publicationsiiir/autt_calculator.html

The calculator enables researchers, policymakers, and the public to explore the AUTT-UBI system's fiscal properties under any parameter combination. All calculations are derived directly from the formal model presented in this paper.

B.1 Calculator Interface

Figures 7–10 show the calculator's interface with US parameters loaded (GDP \$28.8T, budget \$6.1T, V_T \$450T).

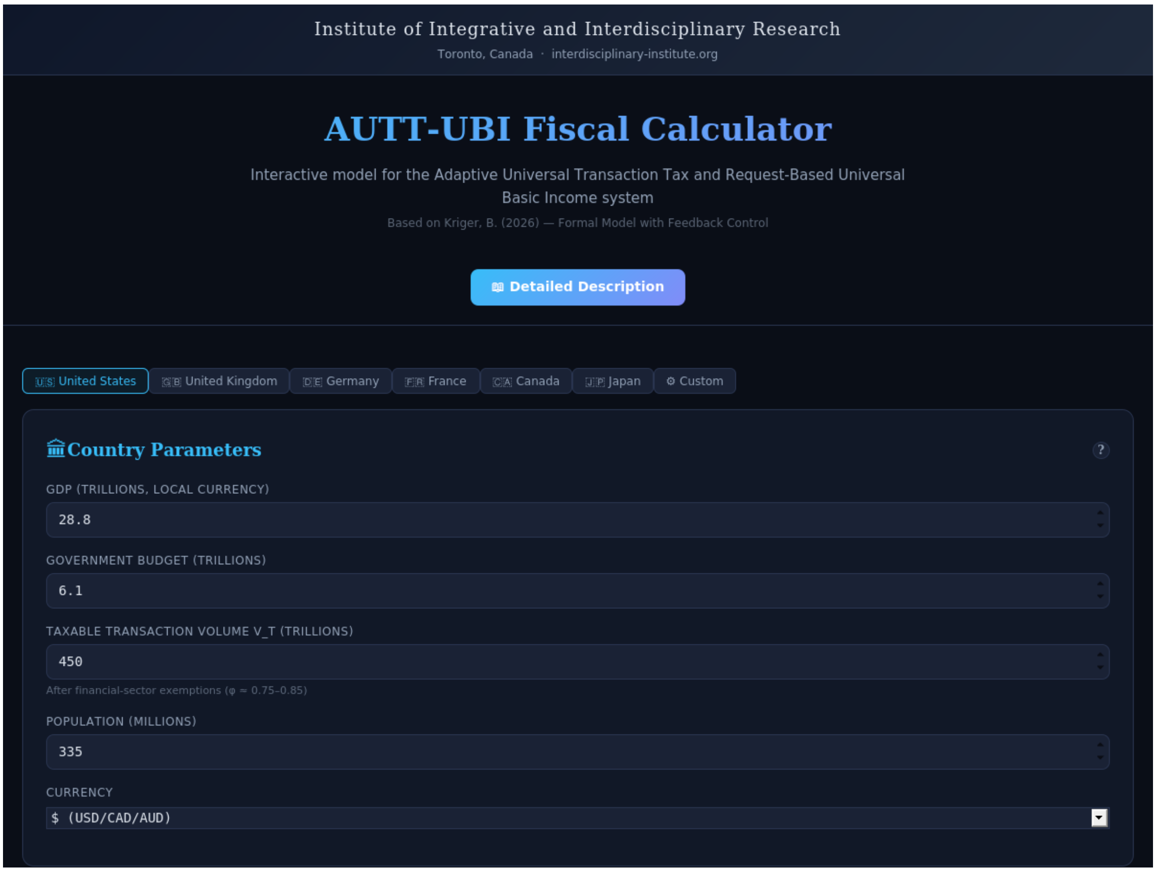

Figure 7: Calculator header with Institute branding, country presets, and input fields for country parameters (GDP, budget, taxable volume, population).

Figure 8: Model parameter sliders (cascade multiplier, UBI take-up, annual UBI, $\tau_{\max}$ ceiling, elasticity), household input panel, and core AUTT results showing required rate $\tau = 1.36\%$ for the United States.

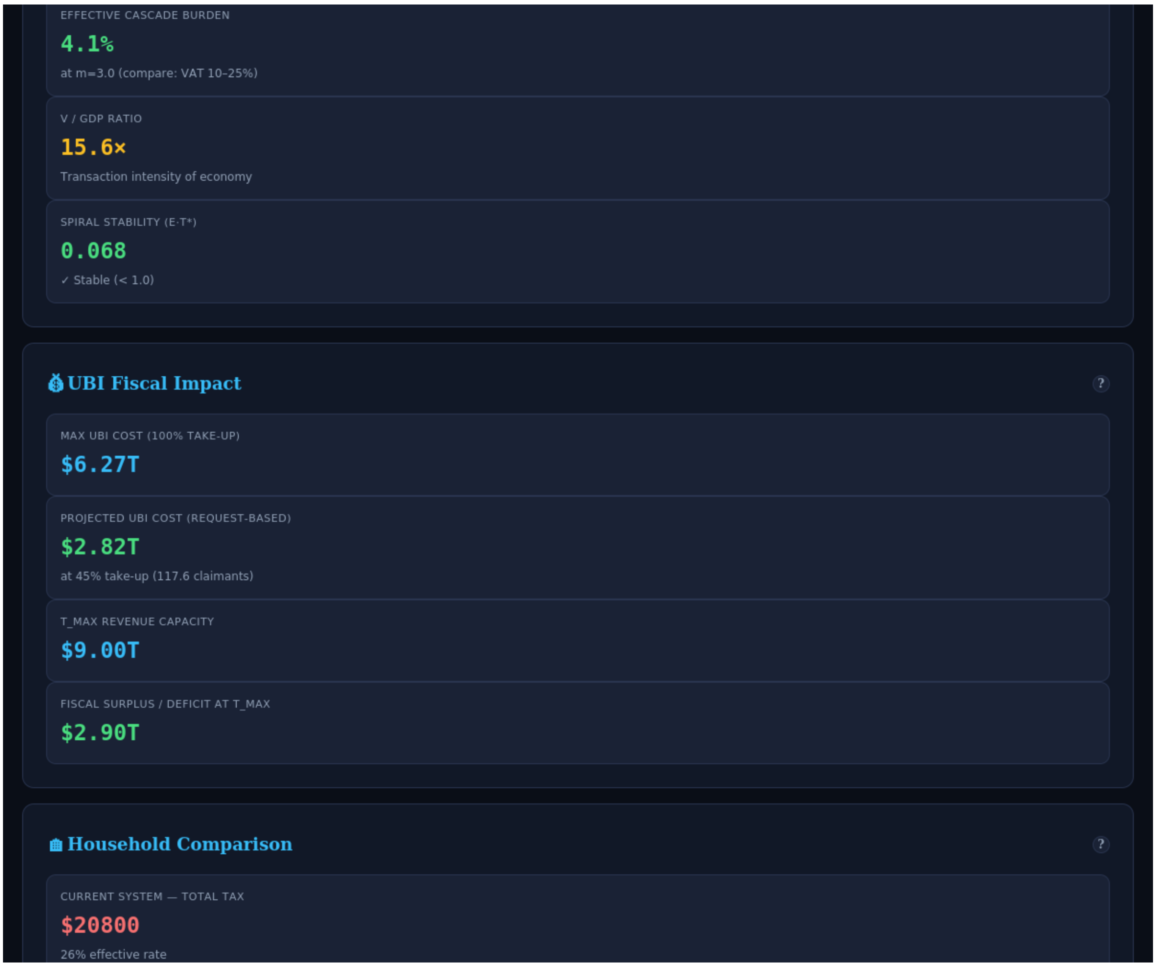

Figure 9: UBI fiscal impact panel ($6.27T maximum, $2.82T projected at 45% take-up, $2.90T surplus) and household comparison showing current tax $20,800 vs. AUTT burden $2,440, with capital liberation of +$42,360 (+72% including UBI).

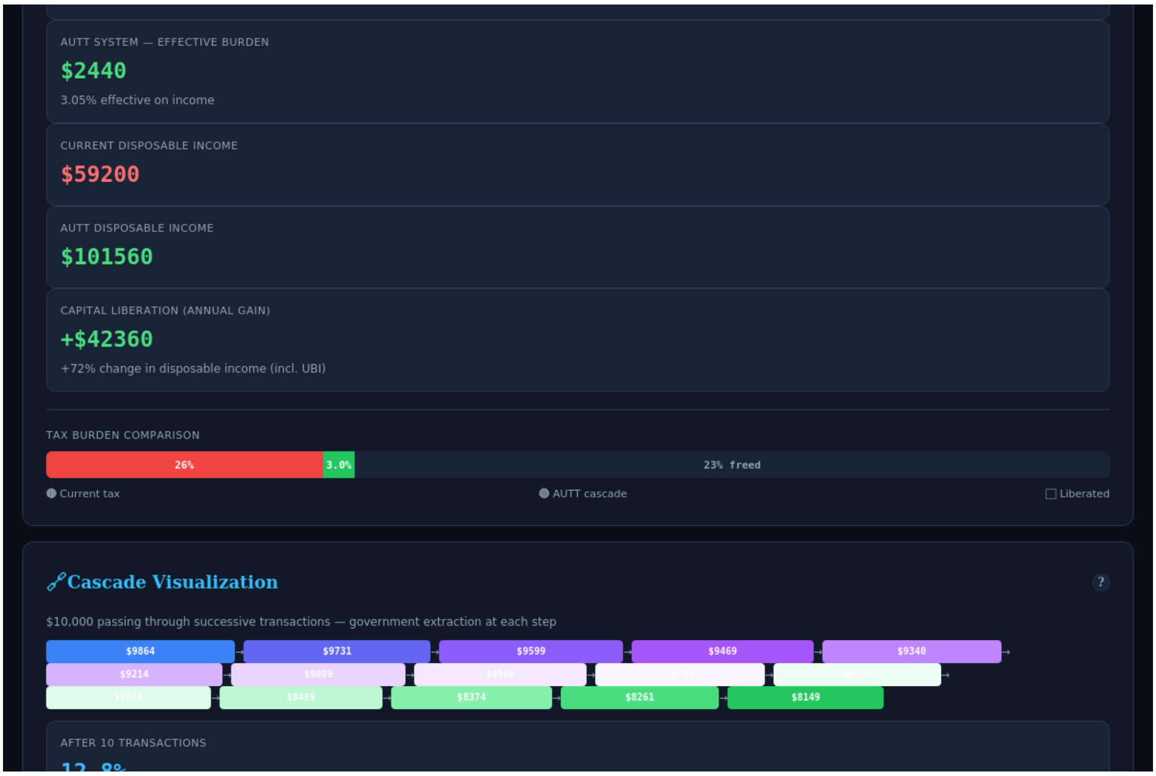

Figure 10: Tax burden comparison bar (26% current vs. 3.0% AUTT, 23% freed) and cascade visualization showing $10,000 passing through 15 successive transactions with cumulative government extraction.

B.2 Country Presets

The calculator provides one-click presets for six economies calibrated in Section 9: the United States, United Kingdom, Germany, France, Canada, and Japan. Each preset loads the country's GDP, government budget, estimated taxable transaction volume V_T (after financial-sector exemptions), population, and currency. Users may also enter fully custom parameters.

B.3 Input Parameters

The calculator accepts the following user-adjustable inputs:

Country parameters:

- GDP (trillions, local currency)
- Government budget target B (trillions)
- Taxable transaction volume V_T (trillions, after exemptions)
- Population (millions)
- Currency (USD, GBP, EUR, JPY)

Model parameters (continuous sliders):

- Cascade multiplier m (range: 1–10; default: 3). Controls the average supply chain length through which money passes before reaching the final consumer.
- UBI take-up rate (range: 10–100%; default: 45%). The fraction of the adult population that actively claims UBI. The CERB precedent suggests 35–45%.
- Annual UBI per adult (range: \$6,000–\$48,000; default: \$24,000). Adjustable for different jurisdictions and ambition levels.
- Constitutional ceiling $\tau_{\max}$ (range: 0.5–5.0%; default: 2.0%). The hard upper bound on the AUTT rate.
- Aggregate elasticity ε (range: 0.5–20; default: 5.0). Controls the behavioural response of transaction volume to the tax rate.

Household parameters:

- Household annual income
- Consumption rate (fraction of income spent rather than saved)
- Current effective tax rate (all taxes combined; default: 26% for US middle class)
- Number of adults claiming UBI (0, 1, or 2)

B.4 Computed Outputs

The calculator displays results in four panels, all updating in real time as any input is changed:

Panel 1: AUTT Core Results

- **Required AUTT rate** $\tau = B/V_T$, with a visual indicator of whether it falls below $\tau_{\max}$.
- **Effective cascade burden** $\tau_{\text{eff}} = \tau \cdot m$, with colour coding: green if below 10%, amber if 10–20%, red if above 20%.
- V/**GDP ratio**, indicating the transaction intensity of the economy.
- **Spiral stability parameter** $\varepsilon \cdot \tau^*$ (Proposition 18.1), with traffic-light indicator: green if < 0.25, amber if 0.25–1.0, red if ≥ 1.0.

Panel 2: UBI Fiscal Impact

- Maximum UBI cost at 100% take-up (fiscal worst case).

- Projected UBI cost at the user-specified take-up rate, with the number of claimants.
- $\tau_{\max}$ revenue capacity ($\tau_{\max} \times V_T$), i.e., the maximum possible AUTT revenue.
- Fiscal surplus or deficit at $\tau_{\max}$: the margin between maximum revenue and the budget target. Positive = the system is sustainable with headroom; negative = the system cannot meet the budget even at the ceiling rate.

Panel 3: Household Comparison

- Current system: total tax burden at the specified effective rate.
- AUTT system: effective burden = $\tau \times m \times c_k \times y_k$ (cascade on consumption).
- Disposable income under both systems, with and without UBI.
- **Capital liberation**: the annual gain in disposable income under the AUTT, expressed in absolute terms and as a percentage change. This implements the calculation from Section 20, Table 4.
- A visual comparison bar showing the fraction of income absorbed by current taxes (red), by AUTT cascade (green), and the freed portion (grey).

Panel 4: Cascade Visualization

- An animated flow diagram showing a notional $10,000 passing through 15 successive transactions, with the amount remaining after each AUTT deduction. This implements the worked example from Section 9 (Figure 4).
- Cumulative government extraction after 10, 50, 100, and 200 transactions, computed as $1 - (1 - \tau)^n$. At $\tau = 0.5\%$ and $n = 200$, cumulative extraction reaches approximately 63%—yet no single participant ever paid more than 0.5%.

B.5 Use Cases

For policymakers: Load a country preset, adjust the budget target to reflect desired spending levels, and observe the required rate and fiscal sustainability margin. Adjust the UBI amount and take-up rate to find the politically and fiscally optimal calibration.

For researchers: Test the sensitivity of the model to parameter variations. What happens when V_T is halved? When ε is doubled? When m increases to 8? The calculator provides immediate feedback, enabling rapid exploration of the parameter space without running simulations.

For the public: Enter your household income and current tax situation. See how much more disposable income you would have under the AUTT. See whether you would benefit from claiming UBI. Understand where the government's revenue comes from under the new system.

For journalists: Use the country presets to quickly generate comparative figures for articles. "Under the AUTT, the required rate for Germany is 2.38%, producing an effective cascade burden of 7.1%—compared to Germany's current VAT of 19%."

B.6 Technical Implementation

The calculator is a single-file HTML application with no external dependencies (all computation runs client-side in JavaScript). It can be hosted on any static web server, downloaded and run locally, or embedded in any web page. The source code is open and may be freely modified.

All formulas implemented in the calculator correspond directly to the formal model:

$$\tau = B/V_T \qquad \text{(Definition 3.1)} \tag{41}$$

$$\tau_{\text{eff}} = \tau \cdot m \qquad \text{(Section 6)} \tag{42}$$

$$\varepsilon\tau^* = \varepsilon \cdot B/V_T \qquad \text{(Proposition 18.1)} \tag{43}$$

$$\text{Extraction}(n) = 1 - (1-\tau)^n \qquad \text{(Section 6)} \tag{44}$$

$$\text{Burden}_k = \tau \cdot m \cdot c_k \cdot y_k \qquad \text{(Section 17)} \tag{45}$$